

Semialgebraic universal covers of normal projective varieties

OpenAI

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Abstract

We prove the Kollár–Pardon conjecture: the universal cover of a connected normal projective complex variety is biholomorphic to a semialgebraic open subset of a projective variety if and only if it is a product of a bounded symmetric domain, a complex affine space, and a simply connected normal projective variety.

1 Introduction

The universal cover of a projective variety remembers its fundamental group through the deck action and its complex geometry through the space on which that group acts. In complex dimension one, uniformization leaves only the projective line, the complex line and the unit disc. The question considered here asks whether an analogous decomposition survives in arbitrary dimension when the universal cover admits a finite real-algebraic description.

A subset of a complex algebraic variety is *semialgebraic* if, in algebraic affine charts, it is defined by finite Boolean combinations of real polynomial equalities and inequalities in the real and imaginary coordinates. We say that a complex space has a *semialgebraic projective-open presentation* if it is biholomorphic to a semialgebraic open subset of a projective variety, with openness taken in the ordinary complex topology. This is a condition on the complex space: the biholomorphism is not required to be algebraic, and the deck transformations of a covering are not required to preserve the chosen semialgebraic structure.

A *bounded symmetric domain* is a connected bounded domain in a complex vector space for which every point is an isolated fixed point of a holomorphic involution of the domain. We allow a point as a zero-dimensional bounded symmetric domain. The compact factor in our result may be singular. All universal covers and fundamental groups below are taken in the classical topology.

Theorem 1.1 (The Kollár–Pardon classification). *Let X be a connected normal projective variety over $\mathbb{C}$, and let $\tilde{X}$ be its universal cover. The following conditions are equivalent:*

- (i) $\tilde{X}$ is biholomorphic to a semialgebraic open subset of a projective variety.
- (ii) There are an integer $m \geq 0$, a bounded symmetric domain D , and a simply connected normal projective variety F such that

$$\tilde{X} \simeq D \times \mathbb{C}^m \times F.$$

A point is allowed as either D or F . No algebraicity of the deck transformations is assumed.

This proves Conjecture 2 of Kollár and Pardon [KP12, Conjecture 2] in its normal-projective scope. The three factors have different roles. The bounded factor records hyperbolic transverse geometry, the affine factor arises from Albanese varieties of compact fibres after finite covers, and

the remaining projective factor is simply connected. Retaining normal singularities is essential: the assertion does not replace the compact factor by a smooth resolution.

The group-theoretic input is the theorem of OpenAI that the entire ordinary fundamental group of a connected smooth special compact Kähler manifold is virtually abelian, meaning that it contains an abelian subgroup of finite index [Ope26b, Theorem 1.1]. We state its precise interface in Section 6 and apply it only to smooth compact special manifolds furnished by Campana’s core. That theorem is proved in the separately cited companion article. The geometric construction of the transverse quotient and the arguments passing from those compact fundamental groups to the deck action are proved here.

1.1 History and related results

The affine-space case belongs to Iitaka’s uniformization problem: one asks whether a smooth projective variety covered by $\mathbb{C}^n$ is a finite unramified quotient of an abelian variety. Nakayama proved the relevant structure theorem under semiampleness of the canonical divisor and derived the affine-space statement from abundance [Nak99, Theorem 1.4]. This is an important predecessor of the separation between a group-theoretic reduction and the geometry of the Albanese map.

Claudon, Höring and Kollár extended the question from affine to quasi-projective universal covers. Their classification and its general-cover extension were proved assuming smooth abundance [CHK13, Theorem 1.1, Conjecture 1.2 and Corollary 1.5]. The zero-boundary case of OpenAI’s log-abundance theorem [Ope26a, Theorem 1.1] supplies exactly that premise. Their Albanese bundle theorem itself was unconditional [CHK13, Theorem 1.4]. Semialgebraic openness admits the additional bounded symmetric factor, and Kollár and Pardon formulated the resulting product conjecture in [KP12, Conjecture 2].

Kollár and Pardon established a nonabelian lattice case of this conjecture. Let X be smooth and projective, and let $\pi_1(X) \twoheadrightarrow \Gamma$ be a quotient such that Γ acts freely, properly discontinuously and cocompactly on an irreducible bounded symmetric domain D of dimension at least two. Their Theorem 3 identifies a biholomorphic realization of the associated Γ -cover as a semialgebraic subset of a projective variety with a fibre-bundle structure $X \rightarrow D/\Gamma$ and a product presentation of that cover as $D \times F$, with F projective [KP12, Theorem 3]. Their proposed approach separates determining the fundamental group, constructing a morphism to the locally symmetric quotient, and proving the bundle assertion [KP12, §3]. In the present proof, the intrinsic quotient is constructed before its deck action is proved discrete and its base is identified as symmetric.

Kollár and Pardon also proved a fibre-bundle theorem that remains an essential input here: if a morphism from a normal projective variety to a smooth projective variety with contractible universal cover has an entire semialgebraic projective-open pullback, then it is a holomorphic fibre bundle [KP12, arXiv version 2, Theorem 20]. Their topological and Chow-space arguments in Sections 1–2, together with the control of invariant subvarieties in [CHK13, Section 2], are predecessors of our compact-factor argument. We use the bundle theorem after constructing the normal relative Albanese family, and adapt the invariant-subset method to the polarized fibre classes on its Stein universal base.

Compactifiable Kähler covers give a related analytic problem. Claudon and Höring, with Campana’s appendix, proved an Albanese bundle theorem after finite cover under a Kähler compactification and an almost abelian group hypothesis [CH13, Theorem 1.5]. Their minimal-counterexample analysis identifies specialness under the weaker compactification condition [CH13, Proposition 1.4]. Campana’s core [Cam04] and the compact special abelianity theorem [Ope26b] provide the group input in the present proof. The compactification conditions in these analytic

results and semialgebraic projective openness are kept distinct.

A recent approach imposes definability on geometric data as well as on the space. Rogov proves a metric form of the Kollár–Pardon statement for smooth aspherical projective varieties with definable lifted Kähler metric [Rog24, Theorem C]. That metric hypothesis supplies information beyond a semialgebraic presentation of the complex space. A different recent result shows that finite-CW homotopy type alone does not suffice for the bundle conclusion: Corrêa, Kollár, Schreieder and Wang construct examples with finite-CW-type universal covers whose Albanese maps are not differentiable fibre bundles [CKSW26, Corollary 30]. Our application of the Kollár–Pardon bundle theorem retains the entire semialgebraic pullback.

The analytic tools also have distinct roles. Real analytic uniformization and preparation [BM88, LR97] supply controlled paths and finite scaling data. The thin-set theorem of Vâjăitu [Vâj00] turns small pseudoconcave omissions into analytic holes. Bochner’s tube theorem and Berndtsson’s variation theorem [Nog20, Ber06] identify whole quadratic fibres. The Nadel–Frankel splitting theorem, in the form recalled by Farb–Weinberger [FW08, Theorem 1.10], identifies the transverse symmetric factor once the compact quotient has been constructed. Finally, Grauert’s Oka principle [Gra58, FP00] turns the resulting local compact-factor products over a contractible Stein base into a global product. The new constructions join these inputs through projective windows, bounded-path localization and an intrinsic quotient.

1.2 The proof and its main obstacles

We first explain the two geometric problems that prevent a direct application of bounded-domain uniformization. The given cover can contain compact positive-dimensional subvarieties, and its semialgebraic boundary can have several successive degeneracies. A scaling near one boundary face therefore need not produce the whole cover or a bounded domain. Moreover, convergence on compact subsets of a prospective limit gives only an interior inclusion: paths in the original domains could reach other components or escape through parts of the boundary lost under scaling. The proof constructs the compact directions and controls this additional boundary behaviour together.

Successive boundary faces. Write $U = \tilde{X}$. A functorial resolution of its projective presentation gives a smooth semialgebraic open set V and a proper modification $\mu : V \rightarrow U$. The same deck group acts on V , with smooth projective quotient. Compact Kähler extension makes V locally pseudoconvex. Near a regular real-algebraic boundary hypersurface, the directions on which the Levi form vanishes integrate into complex plaques. Their algebraic closures supply projective parameter families. Scaling in the directions transverse to a plaque produces a quadratic epigraph whose coefficients are meromorphic functions on the plaque’s projective model. Repeating on that lower-dimensional model terminates in a finite tower of positive-matrix inequalities. We denote this limit domain by P .

Two kinds of retention are needed in this induction. Projective hole propagation ensures that each selected fibre is a whole projective variety minus an analytic subset, and that these holes stay inside a divisor defined over a full parameter box. Triangular recentering maps control the parameters along bounded paths. Analytic discs attached to their real orbits show that the endpoint of every such limiting path lies in $\overline{P}$. Sections 2 and 3 establish these two tools; Section 4 applies them to the successive faces. The outcome includes contractibility and a bounded realization of P , together with complete real holomorphic vector fields that span its complex tangent spaces.

From a window to an intrinsic quotient. The projective families initially describe only local regions of V . For such a region, a bounded plurisubharmonic trace of its incident parameters is constant on every projective variety minus analytic holes. Strict plurisubharmonicity in the parameter then forces two intersecting fibres to coincide. Thus the fibres are intrinsic to V , even though their construction used a chosen boundary face. Holomorphic functions that decay at the moving chart boundaries control the exhaustion obtained by deck translations. We call these functions *guards*. They produce a global quotient $f : V \rightarrow M$, and a separate compactness argument for the inverse charts proves $M \simeq P$. This inverse argument is what upgrades limit maps to a biholomorphism.

The bounded realization makes f constant on the compact fibres of μ , so it descends to $f_X : U \rightarrow M$. To retain singular normal compact factors, we also prepare normal fibres, flatness, fibrewise resolutions and Whitney-stratified submersions before choosing the scales. The exhaustive translated charts propagate these properties to every fibre. Section 5 proves the quotient and these normal-source assertions.

The transverse action and its kernel. Put $\Gamma = \pi_1(X)$, let Λ be its image in $\text{Aut}(M)$, and put $K = \ker(\Gamma \rightarrow \Lambda)$. The induced action on M is not initially known to be discrete. Campana's core has smooth compact special general fibres. The abelianity theorem and a Liouville argument show that the lifts of each such fibre have countable image in M . Orbifold extension to the full proper projective parameter families turns these countable sets into locally finite sets. Their dense union forces the transverse action to be discrete. After a finite cover the action is free. The compact quotient has ample canonical bundle, and the Nadel–Frankel splitting theorem, combined with the spanning vector fields, identifies M as a bounded symmetric domain. A compact-fibre monodromy map proves that K is finitely generated; a second application of core extension and compact special abelianity proves that it is virtually abelian. These are the steps of Section 6; transverse symmetry is derived within this argument.

Separating the affine and compact factors. Choose a characteristic finite-index free-abelian subgroup $K' \subset K$. The normal projective fibres of $U/K' \rightarrow M$ have fundamental group K' . Their relative Albanese construction, carried out through the simultaneous resolution and descended by normality, gives a family of tori. The semialgebraic fibre-bundle theorem of Kollár–Pardon applies to the *entire* Albanese pullback. Its universal covering produces a proper map

$$j : U \longrightarrow T \simeq M \times \mathbb{C}^m$$

with simply connected normal projective fibres.

The remaining issue is variation of those compact fibres. In a projective Chow space the fibres, and each of their polarized isomorphism classes, form semialgebraic loci. A topological rigidity argument forces the closure of every invariant class to be all of T . Two disjoint semialgebraic subsets cannot both be dense, so there is only one class. Relative Hilbert embeddings give local holomorphic products, and the Oka principle over the contractible Stein base T gives the global product with F . Section 7 proves this splitting and the converse. Section 8 assembles the classification. Section 9 records the quasi-projective-cover consequences of smooth abundance. Figure 1 records the three maps constructed in the proof and the distinct group inputs.

1.3 Conventions

All projective varieties are reduced and irreducible unless stated otherwise; a connected normal variety has one irreducible component. A domain is connected and open. Local pseudoconvexity

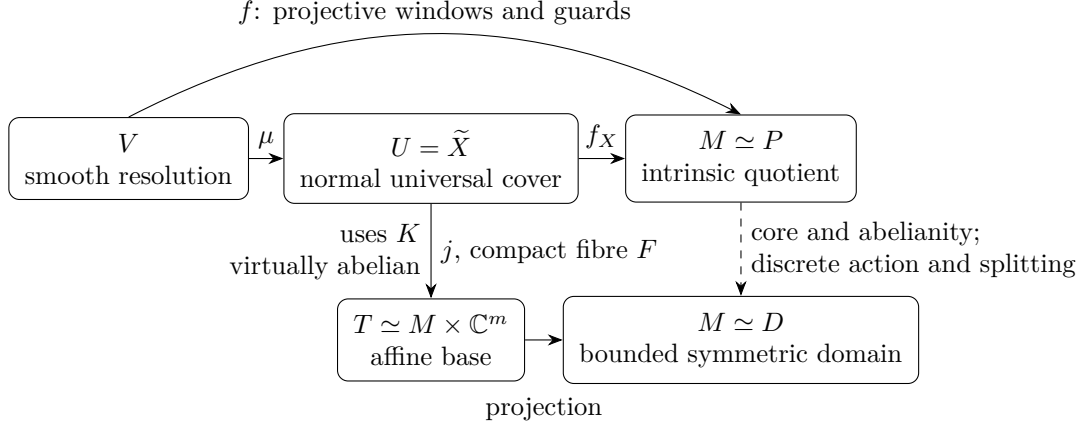


Figure 1: The geometric maps in the proof. The top row constructs the intrinsic quotient on a resolution and descends it to the normal cover. The bottom row separates its affine and compact directions. Solid arrows are geometric maps; the dashed arrow records the theorem identifying the same base as symmetric. After the quotient is constructed, compact special abelianity is used both to prove transverse discreteness and to control the kernel K needed for j .

means local Steinness at the boundary in the indicated ambient complex manifold. Families with projective fibres may be analytic pullbacks of algebraic families; bounded algebraic data in their projective variables are specified when needed. Generic real parameters are chosen on relatively compact boxes after removing the exceptional subanalytic sets of the particular construction. The precise convention, including countable families of tests, is Definition 2.1.

2 Preparation, kernels, and projective holes

This section supplies the two kinds of control needed for projective windows. Preparation makes the boundary limits finite and preserves their real parameters. The second half of the section proves that projective fibres which are present up to algebraic holes retain this property throughout a connected parameter domain.

All manifolds in this section are complex manifolds unless real coordinates are specified. A locally pseudoconvex open subset of a manifold means an open subset which is locally Stein at every boundary point. The open sets are allowed to be disconnected before a distinguished component is selected. Subanalytic data on a relatively compact coordinate box are always assumed to extend as globally subanalytic data to a slightly larger box. Rational powers of a positive scale variable are allowed. Polynomial descriptions in projective variables are understood in a finite collection of bounded algebraic charts, including charts at infinity. In this paper, *bounded algebraic data* means that one such finite atlas and uniform finite bounds for the number and degrees of the polynomial fibre equations and tests have been fixed. The coefficient arrays depend analytically, or globally subanalytically, on the retained parameter boxes as specified in each statement. For an algebraic map, the same convention is imposed on its graph. A *bounded-data semialgebraic family* has a fixed finite Boolean description of this kind in real and imaginary fibre coordinates. These are bounds on finite algebraic complexity; estimates for coefficients, scale matrices or their inverses are stated separately when required.

2.1 The preparation conventions

Definition 2.1 (Generic real parameters). An assertion at generic parameters in an open real box means that it holds outside a locally finite union of subanalytic subsets of smaller dimension. When countably many fixed compact boxes or positive accuracies are tested, a countable union of such exceptional sets is permitted. Each application involving finitely many data is subsequently restricted to a neighborhood of the chosen parameter on which those data have the asserted regularity. No uniform neighborhood for all members of a countable list is implicit.

We use the following standard forms of real analytic preparation. The underlying results are the uniformization and rectilinearization theorems and the distance inequality of [BM88, Theorems 0.1, 0.2, and 6.4], together with one-variable preparation with parameters [LR97, Theorem 1 and Sections 0.3–0.4]. Parusiński’s earlier preparation theorem [Par94, Theorem 7.5] is a predecessor of this method; the specified centre condition used below is the one in Lion–Rolin. For finite definable partitions, choice and components we use [Cos99, Theorems 2.10, 3.1 and 3.9].

Lemma 2.2 (Power bounds and generic Puiseux preparation). *Let A be globally subanalytic and let $f(v, t)$ be globally subanalytic for $v \in A$ and $0 < t < \epsilon(v)$, where ϵ is positive. If $f(v, t) > 0$, there is an integer N such that, for every fixed v ,*

$$t^N < f(v, t) < t^{-N} \quad \text{for all sufficiently small } t > 0.$$

The threshold may depend on v . For finitely many such functions one may use a common N .

If A is an open real box, then near a generic v_0 a given finite collection of functions admits convergent expansions

$$f(v, t) = \sum_{j \geq j_0} a_j(v) t^{j/q}.$$

with a common positive integer q and coefficients real analytic in v . After multiplication by a sufficiently high power of t , these are jointly real analytic functions of $(v, t^{1/q})$ through $t = 0$. Convergence and every fixed number of v -derivatives are uniform on smaller compact parameter boxes.

Proof. Apply one-variable preparation with t as the last variable. The finitely many preparation cylinders meeting $t = 0$ have, on their bottom bands, center zero: a nonzero center comparable to t cannot remain comparable as t tends to zero with v fixed. On each such band the prepared function is a rational power of t times a nonzero function of v times a unit bounded above and below. There are only finitely many rational exponents. Taking N larger than all their absolute values absorbs the positive factors depending on fixed v and proves the first assertion. Apply the same argument to $1/f$ when necessary.

For the second assertion, restrict to a full-dimensional base cell on which the finitely many coefficient functions in the preparation are analytic and their required nonzero denominators remain nonzero. In the analytic unit choose projective coordinate charts about the limiting arguments. An argument involving a negative power of t either has identically zero numerator on the chosen base cell or tends to infinity; in the latter case its reciprocal is a valid analytic coordinate and its nonzero coefficient can be inverted. In these charts the arguments are analytic in v and nonnegative rational powers of t . Clear their denominators. The unit is analytic on a neighborhood of its compact argument set, giving the asserted joint analytic extension. Ordinary convergence of analytic power series on a smaller box gives the derivative assertion. The discarded base cells and coefficient singularities have smaller dimension. $\square$

Remark 2.3. An estimate obtained with additional parameters held fixed does not permit those parameters to move during a correction. When a distance inequality is needed fibrewise, apply preparation to the definable fibrewise distance and residual functions. This gives a common power on finitely many parameter pieces; its positive constants and thresholds may still depend on the fixed parameters. Joint analytic preparation, when needed, requires restriction to a full-dimensional parameter piece.

Here is an explicit way to retain the fixed parameter. For compact closed fibres C_v and a nonnegative continuous definable residual $r(v, z)$ whose zero set is C_v , put

$$m(v, s) = \min\{r(v, z) : \text{dist}(z, C_v) \geq s\}$$

on a compact test box, assigning the value 1 when the set is empty. This is positive for $s > 0$. Lemma 2.2 bounds it below by s^N for sufficiently small s , with N independent of fixed v . Compactness excludes points of fixed positive distance when their residual tends to zero. Consequently, for small residuals,

$$\text{dist}(z, C_v) \leq r(v, z)^{1/N}.$$

All corrections therefore take place in the same fibre C_v . Empty C_v cause no spurious limit, since the residual then has a positive minimum on the compact test box.

Lemma 2.4 (Projective models with parameters). *Suppose an algebraic family in projective coordinates is pulled back by a local analytic parameter map. Suppose an irreducible member is specified by a marked irreducible germ. Near a generic point of a maximally totally real open parameter box, one may track that component on a local branch and replace the family and its evaluation maps by a smooth proper projective family of connected smooth models. The evaluation maps are holomorphic, and their restrictions to the projective fibres are algebraic of bounded data. Real analytic parameter dependence can first be complexified.*

Proof. Track irreducible components on a relative Hilbert or Chow parameter space; for their algebraic existence and projectivity see [Gro61, §3, Theorems 3.1–3.2] and [AN67, §2.4(b), Proposition 4]. On the locus of a fixed component type the tracking map is generically finite. Remove its branch and nonflat loci and choose a local branch. Resolve the resulting total family and the graphs of the finitely many evaluation maps by projective modifications. Generic smoothness removes a further proper analytic subset of the complex parameter space. A maximally totally real open box is not contained in a proper complex analytic subset, by the holomorphic identity theorem. Thus these deletions are legitimate at generic real parameters. The selected generic fibres are irreducible, and their smooth models are connected. After shrinking, properness and smoothness preserve connectedness. All constructions are made with finitely many projective equations and modifications; restricting to a relatively compact parameter box supplies the stated bounded data. $\square$

2.2 Open kernels and finite jets

Definition 2.5 (Open kernel). Let U_t be open subsets of a fixed manifold M , $0 < t < t_0$. Their open kernel is

$$\mathcal{K}(U_t) = \{z \in M : \text{some neighborhood } W \ni z \text{ satisfies } W \subset U_t \text{ for every sufficiently small } t\}.$$

Equivalently, if $F_t = M \setminus U_t$, then

$$\mathcal{K}(U_t) = M \setminus \limsup_{t \downarrow 0} F_t,$$

where the upper limit allows both t and the point of F_t to vary. In a varying family these tests are relative to the total space; smooth local trivializations identify the limiting fibre. A marked kernel is the connected component containing a specified point of the open kernel.

The neighborhood quantifier in this definition is essential. In particular, pointwise eventual membership is insufficient. The compact subsets of an open kernel are eventually contained in U_t , with a neighborhood, by a finite covering argument.

We use the Hartogs-figure criterion for local pseudoconvexity and the Euclidean Levi theorem in their standard forms; see [FG02, II.3.4 and II.3.7] and [Dem12, VIII.9.11(a)]. The exhaustion and directional-radius criteria used below are [Dem12, I.7.2].

Lemma 2.6 (Pseudoconvexity of open kernels). *An open kernel of locally pseudoconvex open sets is locally pseudoconvex. The assertion also holds in varying manifolds equipped with smoothly converging holomorphic coordinate charts.*

Proof. Use the local Hartogs-figure criterion in a coordinate polydisc. A relatively compact Hartogs figure with a neighborhood contained in the kernel is contained, together with that neighborhood, in every U_t for sufficiently small t . Pseudoconvexity fills its associated polydisc in each U_t . To obtain neighborhood inclusion for a compact subset of the filling, slightly enlarge the figure within the given neighborhood before applying the criterion. The resulting enlarged fillings supply a neighborhood of that compact subset in every small U_t . Thus the filling lies in the kernel. This is precisely the Hartogs-figure criterion for the kernel. The same argument uses the corresponding figures in smoothly varying holomorphic charts; strict compact containment persists under this change. $\square$

Lemma 2.7 (Finite-jet specialization). *In bounded projective charts let a family of sets be described by finitely many Boolean combinations of equations and inequalities polynomial in the real fibre coordinates y , with coefficients real analytic in real parameter coordinates b . Fix x and substitute*

$$b = x + R_t h,$$

where the entries of R_t have convergent Puiseux expansions, $R_t \rightarrow 0$, and both R_t and R_t^{-1} have power bounds. The upper limits of these sets, and hence the open kernels of their complements, are semialgebraic in (h, y) on the central fibre. Here h ranges over its whole affine space, while the substitution is tested only where b belongs to the original parameter box. When external parameters are present, assume that all data, including R_t , are jointly globally subanalytic and that the displayed analytic coefficients are jointly analytic. Then the resulting descriptions have analytic coefficients after generic restriction. If all the original data are merely globally subanalytic, the corresponding conclusions are subanalytic.

Proof. The subanalytic assertion follows immediately by writing the upper-limit quantifiers with a positive accuracy and a positive upper bound for t . We prove the stronger assertion in the stated polynomial-in- y situation. Work on one basic piece, written as

$$F_i(b, y) = 0, \quad H_j(b, y) \geq 0, \quad L_k(b, y) > 0.$$

Finite unions cause no difficulty. First require a common strict slack $L_k \geq t^A$. This does not change the upper limit when A is sufficiently large. Indeed, for a proposed limit point, a positive accuracy, and a fixed bound on h , take the supremum of the minimum strict slack among witnesses within that accuracy; cap this supremum by 1. Whenever witnesses exist for arbitrarily small t ,

they exist for every sufficiently small t by one-dimensional subanalyticity. The positive supremum is definable. Lemma 2.2 gives one exponent A for this family, allowing the threshold to depend on the point, accuracy, and bound. A witness with at least half the supremum gives the desired slack after increasing A .

For a second exponent $B > A$, allow residual errors of size t^B in the equations and weak tests. Include the slack as an extra variable. On a slightly larger compact original-coordinate box the distance inequality for the resulting closed semianalytic set gives

$$\text{dist}((b, y, \sigma), C) \leq c \text{residual}(b, y, \sigma)^\theta$$

for some $\theta > 0$. The same statement with external parameters fixed follows from the fibrewise version described in Remark 2.3. Choose B so large that $B\theta > A + 1$ and $\|R_t^{-1}\|t^{B\theta} \rightarrow 0$. A point satisfying the relaxed tests can then be corrected to C ; the correction is $o(t^A)$ in slack, $o(1)$ in y , and $o(1)$ in the normalized coordinate h . Strict tests still have, for example, half the original slack. Thus these relaxed tests have exactly the original upper limits.

Taylor-expand every coefficient at $b = x$. Since $R_t \rightarrow 0$ with a positive power bound, sufficiently high finite Taylor order makes all remainders $o(t^B)$ on each fixed bounded h -box. Truncate the Puiseux series at correspondingly high order. Changing the relaxed error by a fixed factor absorbs these remainders and leaves the same upper limit, by the preceding correction argument. The resulting tests are polynomial in h, y and in a common root of t , with finitely many negative powers removable by multiplication by a positive power of t . Real quantifier elimination therefore gives a semialgebraic upper limit.

The exponent choices can be made independently of the fixed bound on h : include its reciprocal as another bounded fixed parameter in the slack and distance tests. The constants and the small- t threshold may deteriorate with that bound; the finitely many power exponents do not. Consequently one finite polynomial description works on the full affine h -space. A finite projective atlas treats all fibre points. The finite Taylor coefficients and the finitely many choices used above are analytic in external parameters after generic restriction. In particular, apply Lemma 2.2 jointly to the entries of R_t and R_t^{-1} and discard the zero loci of their nonzero leading coefficients. Their relevant positive orders of decay and inverse power bounds are then fixed on the smaller parameter box. This supplies a common Taylor order there, rather than only a separate order for each fixed parameter. $\square$

Lemma 2.8 (Generic convergence of distance tests). *Let x range over an open real box U , let K be compact and globally subanalytic, and let $d_t(x, z)$ be a bounded globally subanalytic family of functions on K , for $0 < t < t_0$, uniformly Lipschitz in z . There is a relatively closed subanalytic subset $E \subset U$ of dimension smaller than $\dim U$ such that the extension*

$$\widehat{d}(x, z, t) = \begin{cases} d_t(x, z), & t > 0, \\ d_0(x, z), & t = 0, \end{cases} \quad d_0(x, z) = \lim_{t \downarrow 0} d_t(x, z)$$

is jointly continuous at every $(x, z, 0)$ with $x \in U \setminus E$ and $z \in K$. In particular, for such x , if $x_t \rightarrow x$ and $z_t \rightarrow z$, then

$$d_t(x_t, z_t) \rightarrow d_0(x, z).$$

For every compact $L \subset U \setminus E$, the convergence $d_t \rightarrow d_0$ is uniform on $L \times K$. One common exceptional set and one smaller open center box may be used for finitely many such tests. For countably many tests one may exclude the union of their exceptional sets, but no common open center box is asserted. These conclusions apply in particular to bounded, capped distance functions of definable closed sets.

Proof. Pointwise limits exist by one-dimensional definable monotonicity. They have the same Lipschitz bound in z . A finite net in K thus upgrades convergence at fixed x to uniform convergence on K . For each positive rational δ , the supremum of the positive numbers a for which

$$|d_t(x, z) - d_0(x, z)| < \delta \quad (0 < t < a, z \in K)$$

is a positive definable function of x , after capping it by 1. It is continuous on a dense open definable set, and hence locally bounded below there. This controls the error also at moving centers x_t near a chosen generic x . For each point of a countable dense subset of K , the definable function $d_0(x, z)$ is likewise continuous at generic x . A finite-net argument and the common Lipschitz constant give continuity in x uniformly on K at all points outside the resulting countable union of lower-dimensional subanalytic sets. Combining the two estimates proves that $\hat{d}$ is continuous at every point of $\{x\} \times K \times \{0\}$ for all such x .

To obtain one open good locus for this fixed compact test, define

$$\Sigma = \{x \in U : \text{for some } z \in K, \hat{d} \text{ is not continuous at } (x, z, 0)\}.$$

This is a subanalytic set: both d_0 and the extension are definable, and failure of continuity is expressed by the finite quantifiers

$$\exists z \in K \exists \epsilon > 0 \forall \delta > 0 \exists (x', z', t') : \begin{cases} x' \in U, z' \in K, t' \geq 0, \\ |x' - x| + |z' - z| + t' < \delta, \\ |\hat{d}(x', z', t') - d_0(x, z)| \geq \epsilon. \end{cases}$$

All variables are restricted to the domains of the given functions. The preceding argument puts Σ inside a countable union of lower-dimensional subanalytic sets. Their closures are nowhere dense, so the Baire theorem shows that Σ has no interior. Since Σ itself is subanalytic, it has dimension smaller than $\dim U$, as does its relative closure E .

On $U \setminus E$ the extension is therefore jointly continuous at every boundary point with $t = 0$. Compactness of $L \times K$ turns this into uniform convergence on that product. Taking a finite union of the exceptional sets proves the finite-family assertion. Taking a countable union gives precisely the weaker countable-family assertion stated above. $\square$

Corollary 2.9 (Freezing the real center). *Let a definable family be written using $\operatorname{Re} b$ and $\operatorname{Im} b$, and let $R_t \rightarrow 0$ be a fixed real power matrix, independent of the real center x . For the substitution*

$$b = x + R_t(u + iv)$$

its upper-limit and open-kernel tests at generic x are independent of u . Thus the kernels are invariant under real translations in the scaled coordinates. The tests may include remaining bounded fibre variables jointly. More generally, when a normalization $R_t(x)$ depends definably on x , Lemma 2.8 applies to the normalized closed-set tests themselves.

Proof. For the fixed matrix use $a = \operatorname{Re} b$ as an independent parameter and take the closed-set distance tests after setting $\operatorname{Im} b = R_t v$. Their generic limit at $a = x$ is unchanged by replacing a by $x + R_t u$, by Lemma 2.8. This applies to arbitrary bounded u, v and to their convergent perturbations, which are the quantifiers defining upper limits. Taking complements gives the kernel assertion. The final statement is the same lemma applied after, rather than before, the parameter-dependent normalization. $\square$

Lemma 2.10 (Power diagonalization). *Let $f(e, t, v)$ be a bounded globally subanalytic function with iterated limit $f_{00}(v) = \lim_{t \downarrow 0} \lim_{e \downarrow 0} f(e, t, v)$. There is an integer N_0 such that for every integer $N > N_0$,*

$$\lim_{t \downarrow 0} f(t^N, t, v) = f_{00}(v)$$

for each fixed v . The same choice works for finitely many tests; the parameter v may include fixed box sizes and positive accuracies. For bounded closed-set tests, apply this to the distance functions on slightly larger boxes, retaining a smaller box to avoid artificial edge effects.

Proof. Let $f_0(t, v) = \lim_{e \downarrow 0} f(e, t, v)$. For fixed positive δ , the supremum of those $a \in (0, 1)$ such that $|f(e, t, v) - f_0(t, v)| < \delta$ for every $0 < e < a$ is positive and definable. Lemma 2.2, with (v, δ) held fixed, bounds this threshold below by t^{N_0} for all sufficiently small t , with N_0 independent of (v, δ) . The asserted diagonal then lies below the threshold. Let t tend to zero and then let δ tend to zero. $\square$

2.3 Polynomial tests for compact holes

We now turn from limits of parameter sets to omissions inside compact fibres. The basic test measures a polynomial on the omitted set. Its plurisubharmonicity will turn containment along a parameter edge into containment throughout a neighbourhood.

Lemma 2.11 (Polynomial maximum over the holes). *Let B be a complex manifold and let $J \subset B \times \mathbb{C}$ be closed. Assume that $J \rightarrow B$ is proper, its fibres are locally bounded in the line coordinate, and $(B \times \mathbb{C}) \setminus J$ is locally pseudoconvex. If $g(b, t)$ is monic of positive degree in t with holomorphic coefficients in b , then*

$$\psi_g(b) = \log \max_{t \in J_b} |g(b, t)|$$

is plurisubharmonic, with value $-\infty$ for an empty fibre or a zero maximum. The assertion also holds for polynomials with nowhere-zero holomorphic leading coefficient, and for the constant test 1.

Proof. Work over a small Stein coordinate box. The map $F(b, t) = (b, g(b, t))$ is finite, proper, and open. Put $J' = F(J)$. Its complement Ω' is locally pseudoconvex. Here is the justification, including the ramification locus. Take a plurisubharmonic exhaustion of $\Omega = (B \times \mathbb{C}) \setminus J$, which is Stein over the small box by the local Levi criterion. On Ω' take the maximum of this exhaustion over all preimages under F . Away from branch values this is a finite maximum of plurisubharmonic functions. Local boundedness above and continuity of the finite fibres extend it plurisubharmonically across branch values. It is an exhaustion: a sequence approaching J' has at least one preimage approaching J , while properness of F handles escape to infinity. This proves the claim about Ω' .

Invert the image line coordinate and adjoin its point at infinity. In the inverse coordinate w , the resulting domain contains $w = 0$ over every base point. It is locally pseudoconvex there by local boundedness of J' . Its largest centered vertical disc has radius

$$r(b) = \left(\max_{v \in J'_b} |v| \right)^{-1},$$

where $r = +\infty$ when the maximum is zero or the fibre is empty. The Hartogs radius criterion says that $-\log r$ is plurisubharmonic, proving the assertion. Multiplication by a nowhere-zero holomorphic leading coefficient adds a pluriharmonic function.

For the constant test, note first that empty fibres form an open set by properness. Choose, locally on the base, a constant a outside a disc containing every J_b . The already proved test $t - a$ has a positive lower bound on every nonempty fibre and is $-\infty$ on empty fibres. If there is an empty fibre, it is $-\infty$ on an open base set and therefore identically $-\infty$ on the connected base box. Thus either every fibre is empty or none is. The test for 1 is accordingly constant $-\infty$ or constant zero. $\square$

In this proof the Hartogs radius criterion concerns the largest disc centered at a fixed holomorphic section; the whole vertical section need not be a disc. Equivalently, take its complete circular Hartogs subdomain. The criterion follows from the local Hartogs-figure test, and gives the negative logarithm of that radius.

Lemma 2.12 (Boundary trapping). *Let a proper holomorphic submersion be given near a relatively compact parameter interval I on the boundary of a one-sided complex disc. Let G be locally pseudoconvex on that side, and suppose every limit of its fibre holes as the transverse parameter tends to I belongs to a relative divisor D that extends holomorphically across I and is proper in each fibre. Then, after restriction near an interior point of I , the holes on that side belong to D . The conclusion holds with additional fixed parameters whenever these hypotheses hold locally uniformly in those parameters.*

Proof. At a possible limiting hole choose fibre coordinates (z, t) in which D is given by a Weierstrass polynomial $g(s, z, t)$. Choose the vertical disc with boundary disjoint from D and shrink all other variables. The limit hypothesis confines the holes to a strictly smaller vertical disc. After filling trivially outside this disc, they give a closed set J as in Lemma 2.11, with base (s, z) .

For fixed z , the subharmonic function $\psi_g(s, z)$ is bounded above near a compact subinterval of I and tends locally uniformly to $-\infty$ there. To see that it is identically $-\infty$, fix a smaller half-disc. Harmonic measure comparison bounds it by $C - M\omega(s)$ for every $M > 0$, where $\omega(s) > 0$ is the harmonic measure of a nontrivial subinterval. Let M tend to infinity. Thus g vanishes at every hole. Finitely many such cylinders cover the possible holes on compact fibre sets; outside them the limit hypothesis already excludes holes. This proves the assertion, including its locally uniform version. $\square$

Lemma 2.13 (Interior trapping). *Let G be open and locally pseudoconvex over a whole connected base box, with the preceding local confinement hypotheses. Suppose a relative divisor D is holomorphic in a whole base box and contains the holes on a nonpluripolar parameter slice inside the box. Then it contains the holes nearby. In particular this applies to an open piece of a real analytic maximally totally real slice.*

Proof. Use the same Weierstrass cylinders and polynomial maximum. For fixed remaining fibre coordinates, ψ_g is $-\infty$ on the specified nonpluripolar set. A plurisubharmonic function with this property is identically $-\infty$ on its connected box. For the last stated case this can also be checked without a capacity criterion: straighten the real analytic slice biholomorphically, and apply the one-variable uniqueness principle successively to the real intervals in a real coordinate box. The resulting vanishing on the complex box proves containment. $\square$

Lemma 2.14 (Tangency propagation in a connected fibre). *Let $q : Q \rightarrow \Delta$ be a holomorphic submersion, let $G \subset Q$ be open and locally pseudoconvex, and let A be a divisor in Q_0 . Suppose $Q_0 \setminus A$ is connected. Assume that on one side of a regular embedded real analytic arc through 0, every limit of holes belongs to A . If G_0 is nonempty, then*

$$Q_0 \setminus A \subset G_0.$$

It suffices to have these hypotheses on neighborhoods of compact paths in $Q_0 \setminus A$.

Proof. Since G_0 is open and A has no interior, it contains a point outside A . Connect that point to any prescribed point of $Q_0 \setminus A$ by a path and cover the path by finitely many submersion boxes avoiding A on the central fibre. On a sufficiently small fixed one-sided disc the limit hypothesis and compactness give all these boxes on the good side. In each box choose a disc $\Delta(c, r)$ in the transverse parameter internally tangent to the arc at 0. The full product of this disc with a smaller connected vertical patch belongs to G .

For a vertical point z let $R(z)$ be the largest radius of the transverse disc centered at c contained in G within this box. The Hartogs radius criterion gives that $-\log R$ is plurisubharmonic, and $R \geq r$ everywhere. At the starting patch, openness of G at the tangency point gives $R > r$ at some point: the rest of the closed tangent circle is compactly on the good side. If $R = r$ anywhere in the connected patch, $-\log R$ attains its maximum there and the strong maximum principle would force $R = r$ everywhere. Thus $R > r$ throughout the patch, so the central-fibre points of that patch are in G . Continue through successive overlapping boxes along the path. This reaches the prescribed point. $\square$

2.4 Thin holes and propagation in a projective family

The preceding lemmas propagate one specified container. We next show that bounded algebraic containers can be chosen through a boundary wall, then use this to propagate the existence of a Zariski open fibre. Analyticity of a sufficiently thin omitted set completes the argument across the remaining lower-dimensional parameter strata.

Theorem 2.15 (Analyticity of a thin pseudoconcave set). *Let M be a complex manifold of dimension n , and let $A \subset M$ be closed and subanalytic, with real dimension at most $2n - 2$. If $M \setminus A$ is locally pseudoconvex, then A is either empty or an analytic set of pure complex codimension one.*

Proof. The measure hypothesis in the standard analyticity theorem is automatic here. Indeed uniformize the closed subanalytic set near a compact coordinate box by a proper analytic map from a manifold of dimension at most $2n - 2$. The inverse image of the box is compact. A finite covering by coordinate patches makes the map Lipschitz on compact smaller patches, so its image has finite $(2n - 2)$ -dimensional Hausdorff measure there. The local Stein complement means exactly that A is 1-concave in the sense of [Vâj00, Definition 1]. For $n > 1$, apply [Vâj00, Theorem 1] with $q = 1$; it gives analyticity and pure dimension $n - 1$. For $n = 1$, a zero-dimensional closed subanalytic set is locally finite and hence is an analytic divisor directly. For smooth ambient spaces, the thin-pseudoconcave-set method already appears in [Hir73, Theorem 1.4]. $\square$

Lemma 2.16 (A relative divisor containing a proper subset). *Let $q : Q \rightarrow B$ be a smooth projective family with connected fibres, and let $A \subset Q$ be a relative algebraic subset with analytic coefficient data, proper in every fibre. Near a generic parameter there is a relative effective divisor D containing A , defined over the full smaller parameter box and proper in every fibre.*

Proof. First restrict near a generic parameter at which $A \rightarrow B$ is flat. Since q is smooth, the exact sequence of its ideal $\mathcal{I}_A$ shows that $\mathcal{I}_A$ is flat over this smaller base as well. In a relative projective embedding, choose a twist $\mathcal{I}_A(k)$ which is generated and has vanishing higher cohomology on the selected fibre [The26, Tag 0B5T], using the comparison in [Ser56, Theorems 1–3]. Upper semicontinuity preserves the vanishing after shrinking; constancy of the Euler characteristic in a

proper flat family makes h^0 constant. Grauert's constant-cohomology theorem therefore gives a locally free direct image and fibrewise base change [Gra60, §7, Sätze 3, 5 and 6]; see the correction [Gra63, p. 36] for the distinct general base-change statement. The evaluation map generates on the selected fibre. Nakayama's lemma and properness allow a further shrinking on which it generates throughout. We thus have local holomorphic sections of the twisted ideal whose restrictions generate every fibre ideal. Choose one whose restriction at the selected parameter is not identically zero; such a section exists because A is proper in that fibre. Extend it over a smaller base box. It stays nonzero at a local section of q through a point where its central restriction is nonzero. Thus its zero divisor is proper in every nearby fibre and contains A . The construction uses one fixed twist, hence has bounded algebraic fibre degree. Generic flatness supplies the initial parameter outside a proper analytic subset; the subsequent choices only shrink its neighbourhood. Such a parameter can be chosen in the maximally totally real box. This argument does not require the general bad base-change locus to be closed. $\square$

Lemma 2.17 (Bounded algebraic containers). *Consider a smooth proper projective family with connected fibres, and a bounded-data semialgebraic family of subsets of those fibres. There is a degree bound, depending only on the family description, such that every member contained in a proper complex algebraic subset is contained in an effective divisor of that degree or smaller. The parameters admitting such a container form a subanalytic set, and containers can be selected subanalytically. For a one-sided family approaching a generic real analytic parameter wall, its limiting containers can be chosen as proper effective divisors varying real analytically on that wall.*

Proof. Regard the finitely many real polynomial coefficients and the projective fibre equations as independent algebraic parameters. Real algebraic cell decomposition, with these parameters first, expresses each fibre as finitely many Nash pieces of uniformly bounded algebraic complexity. Their algebraic closures, and the complex Zariski closures of their projections into the complex fibre coordinates, consequently have bounded degree by elimination. The complex Zariski closure of the original set is the union of these finitely many closures. When it is proper, a homogeneous polynomial of bounded degree vanishes on it without vanishing identically on the ambient fibre. Its zero divisor is a container.

At a fixed degree the condition that a homogeneous equation vanish on every point of the semialgebraic set, but not on the whole fibre, is a quantified semialgebraic condition on its coefficients. Quantifier elimination and definable choice give the parameter locus and the selection. In a varying smooth projective family, record the chosen zero divisors as cycles in the finite union of relative Chow spaces of these degrees. Their closure is proper over a smaller parameter box. Its limiting cycles retain their fibre dimension, so their supports are proper divisors; a limiting homogeneous equation alone would not suffice, since it could vanish identically on a special fibre.

Across a generic real wall, one-sided subanalytic limits of this bounded cycle data exist and vary real analytically after a stratification and generic restriction. This follows equally by applying Lemma 2.2 to the normal wall variable in projective coordinate charts. More precisely, restrict to a full-dimensional real analytic stratum of the wall on which the cycle degree and these charts are fixed. Preparation in the one-sided normal variable gives an analytic limit map on that stratum and convergence uniform on each smaller compact wall box. Restrict this map to the real interval of a transverse analytic complex disc. Its real analytic coefficients complexify holomorphically in the disc variable, jointly real analytically in the other wall parameters. The equations defining the relative Chow space vanish on the real interval, so continue to vanish by the holomorphic identity theorem. Pulling back its universal cycle gives a relative divisor on the smooth pulled-back family. The analytic extensions exist on a common small disc after restriction to a compact smaller wall

box. This is the claimed locally uniform continued container; no holomorphic extension in all complex base variables at once has been asserted. $\square$

Proposition 2.18 (Projective-fibre propagation). *Let $q : Q \rightarrow B$ be smooth, proper, and projective, with connected smooth fibres. Let $G \subset Q$ be open, subanalytic, locally pseudoconvex, and fibrewise semialgebraic with bounded data. Put $S = q(G)$, and let S_0 be a connected component of S . If G_b contains a Zariski open dense subset of Q_b for every b in some nonempty open subset of S_0 , then*

$$J = q^{-1}(S_0) \setminus G$$

is an analytic divisor, possibly empty. Its intersection with each fibre is proper, so every G_b , $b \in S_0$, contains a Zariski open dense subset of Q_b .

Proof. Call a parameter good when its holes have proper complex Zariski closure. Lemma 2.17 makes the good locus subanalytic. Choose a finite local stratification of it and its complement. We first show that no generic real hypersurface wall can separate an open good region from an open bad region inside S_0 .

At a generic point of a proposed wall choose transverse analytic complex discs, with their real arcs in the wall. Concretely, choose a real analytic tangent vector field on the wall which is not in its complex tangent hyperplane, take its local real integral curves, and complexify those curves. Their imaginary directions are transverse to the wall, and the resulting family fills a neighborhood. Lemma 2.17 gives on each arc a limiting proper divisor from the good side and its holomorphic continuation on the disc. All limiting holes on that side are in its support. The central fibre has a point in G , since the wall lies in S . The complement of a proper divisor in a connected smooth projective variety is connected: a path between two points can be perturbed off its real-codimension-two strata. Lemma 2.14 therefore puts that entire complement in the wall fibre of G .

This also confines holes on both sides near the limiting divisor. Indeed, any compact subset of the wall fibre outside the divisor has a neighborhood in G ; finite coverings and properness make this uniform on smaller arc intervals. We can now choose Weierstrass cylinders for the continued divisor. On the real interval their holes belong to the divisor. Lemma 2.13, in the transverse disc, confines the holes to that divisor on both sides. Varying the transverse discs through generic wall points yields an open good region on the proposed bad side, a contradiction.

A connected open real manifold cannot be separated by a subanalytic subset of real codimension at least two: a compact path can be perturbed to meet only the hypersurface strata of a finite stratification. Thus the preceding wall argument shows that the open good locus is dense in S_0 . Apply the same argument to generic hypersurface strata whose two neighboring open regions are good. It makes their generic parameters good as well. What remains is a subanalytic subset $E \subset S_0$ of real codimension at least two.

Write $n = \dim_{\mathbb{C}} Q$ and $f = \dim_{\mathbb{C}} Q_b$. Outside E the fibre holes have real dimension at most $2f - 2$; over E they have dimension at most $2f$. The dimension theorem for subanalytic fibres therefore gives $\dim_{\mathbb{R}} J \leq 2n - 2$. Theorem 2.15 makes J an analytic divisor. No fibre is contained in J , because every parameter of S_0 has a point in G . Its intersection with a fibre is consequently a proper analytic subset of a projective variety, and is algebraic by Chow's theorem. This proves the final assertion. $\square$

Corollary 2.19 (Continuation of a fixed container). *In Proposition 2.18, let D be a relative analytic divisor defined on the full parameter box and proper in every fibre. If $J \subset D$ over a nonempty open subregion of S_0 , then $J \subset D$ over all of S_0 . The same conclusion holds for any relative proper analytic subset in place of D .*

Proof. Every irreducible component H of J dominates S_0 . To check this, $q|_H$ is proper, so its image is analytic. If that image were proper, dimension of H would force it to be a hypersurface in S_0 with full-dimensional generic fibres in Q_b . Those fibres would equal Q_b , contradicting $G_b \neq \emptyset$. Thus H meets the stated open subregion. The local equations of D vanish on a nonempty open subset of H , and hence on H by the analytic identity theorem. Apply this to every component. $\square$

Corollary 2.20 (A section detects the projection domain). *Suppose a full-box container D as in Corollary 2.19 is available. A local holomorphic section σ of q avoiding D lies in G at every parameter of S_0 . Near a boundary point of S_0 that lies in the parameter box, S_0 is locally pseudoconvex wherever such a section exists.*

Proof. The first assertion is immediate from containment of the holes. Since $S = q(G)$ is open, a boundary point of its component S_0 cannot belong to S : a small connected neighborhood within S would otherwise join it to S_0 . In submersion coordinates near $\sigma(b_0)$ choose a connected vertical patch disjoint from D . Over S_0 this whole patch is in G , while over the boundary it is disjoint from G .

More explicitly, $\sigma^{-1}(G)$ is locally pseudoconvex, since holomorphic pullback preserves the local Hartogs criterion, and it is contained in S . In a sufficiently small parameter ball, each component of S_0 intersected with that ball is a component of $\sigma^{-1}(G)$ there: a path in the latter is a path in S and hence cannot leave S_0 , while every point of S_0 belongs to the pullback. Thus the local pieces of S_0 satisfy the same criterion. This proves the assertion at b_0 . $\square$

3 Localization with triangular recenterings

An open kernel alone does not control the component reached by a bounded path. The additional structure used here is a family of real recenterings, available at every bounded real displacement. We prove the required upper containment, including the uniform path selection needed for the analytic-disc argument.

For open sets $D_t \subset \mathbb{C}^d$, $0 < t < t_0$, write $\mathcal{K}(D_t)$ for their open kernel in the sense of Definition 2.5. If a fixed point p belongs to this kernel, let P be its connected component. For $L > |p|$, let $A_t(L)$ be the component containing p in $D_t \cap B(0, L)$, whenever $p \in D_t$. Thus P is defined by eventual neighborhood inclusion, whereas $A_t(L)$ records access from the seed at one fixed scale and radius.

Throughout this section, subanalytic families are restricted to bounded coordinate boxes and are globally subanalytic there. In particular, quantification, closure, connected components in families, and definable selection are available. A generic conclusion permits deleting a countable union of subanalytic sets of dimension smaller than that of the real parameter box. A separate exceptional set is allowed for each integer L and each positive rational accuracy.

3.1 Normalized domains and recenterings

Fix a decomposition $\mathbb{R}^d = \mathbb{R}^{d_1} \times \cdots \times \mathbb{R}^{d_s}$, and write $u_{>j} = (u_{j+1}, \dots, u_s)$. Let $U \subset \mathbb{R}^d$ be a real box in a complex coordinate box Ω . Suppose $S \subset \Omega$ is a fixed subanalytic open set, locally pseudoconvex at its boundary in Ω . In particular S does not change when its real center changes. Let $R_t(x) \in \text{GL}_d(\mathbb{R})$ be subanalytic and tend to zero, locally uniformly at generic x . Define

$$D_t^x = \{Z : x + R_t(x)Z \in S\}, \quad C_t(x, u) = R_t(x)^{-1}R_t(x + R_t(x)u). \quad (1)$$

Only portions whose original coordinates lie in Ω are used. Assume that, locally at generic x , for every finite M and every integer $k \geq 0$,

$$\|C_t(x, \cdot) - G_x\|_{C^k(\{|u| \leq M\})} \longrightarrow 0, \quad G_x(u) = \text{diag}(G_{x,1}(u_{>1}), \dots, G_{x,s}(u_{>s})), \quad (2)$$

where all entries of G_x are real polynomials, $G_x(0) = I$, and $G_x(u)$ is invertible for every real u . The last condition means, in particular, that G_x^{-1} is bounded on each of the fixed finite boxes that will be used. Suppose a fixed $p \in \mathbb{C}^d$ belongs to $\mathcal{K}(D_t^x)$ throughout the real parameter box under consideration. Put P_x for its marked component.

Theorem 3.1 (Accessible upper containment). *Under these hypotheses, at generic x , for every finite $L > |p|$ and every $\eta > 0$ there is $t_0 = t_0(x, L, \eta) > 0$ such that*

$$0 < t < t_0 \implies A_t^x(L) \subset \{Z : \text{dist}(Z, \overline{P_x}) < \eta\}. \quad (3)$$

Consequently, if $t_n \downarrow 0$ and paths in $D_{t_n}^x$ start at p , stay in one fixed bounded box, and have endpoints tending to z , then $z \in \overline{P_x}$. The same holds for starting points tending to p inside a uniform initial ball. Fixed auxiliary parameters are permitted; generic restrictions are made in the real centers used for the recenterings.

The proof uses the recentering identity in (1) to turn paths at nearby real centers into real families of points in one normalized domain. In the limit these families have the form $u + G_x(u)F(a)$, where F is a limiting path. We will attach analytic discs to them, solving from the last block to the first, and obtain centers in the marked kernel approaching the path endpoint. The next two lemmas supply replacement paths whose dependence on the real center and on the path parameter permits this disc construction.

3.2 Controlled paths from ordinary uniformization

We use two precise forms of real-analytic resolution: a closed subanalytic set has a proper real-analytic uniformization by a manifold of the same dimension, and a nonzero analytic function becomes a monomial times a nonvanishing analytic function after a proper surjective real-analytic map. These are Theorem 0.1 and Corollary 4.9 of [BM88]. The relative estimates below are consequences proved here, rather than an additional relative resolution theorem.

Lemma 3.2 (Generic analytic power parametrization). *Let $U \subset \mathbb{R}^k$ be open and let $f : U \times (0, \epsilon) \rightarrow \mathbb{R}^m$ be bounded and globally subanalytic. Outside a closed subanalytic subset of U of dimension less than k , locally in x , there are an integer $q \geq 1$ and a real-analytic map $a(x, s)$, defined near $U' \times \{0\}$, such that*

$$f(x, t) = a(x, t^{1/q}) \quad (0 < t < \epsilon').$$

Here U' and ϵ' may be reduced. If a scalar coordinate of f is positive, it either has a nonzero limit or, after a further generic restriction, has the form

$$t^{\nu/q} a_0(x, t^{1/q}), \quad a_0(x, 0) > 0, \quad \nu \in \mathbb{N}.$$

On smaller parameter boxes the analytic maps have a common complex neighborhood and all their derivatives are bounded.

Proof. Apply Lemma 2.2 to the finitely many coordinate functions of f . On a generic parameter box they have joint convergent Puiseux expansions with a common denominator. Boundedness forces every negative-power coefficient to vanish, giving the analytic map $a(x, t^{1/q})$. For a positive coordinate, restrict further so that its first nonzero coefficient does not vanish. Positivity then makes that coefficient positive and gives the asserted factorization. The discarded preparation cells and coefficient zero loci come from finitely many data on the bounded boxes in use. Their lower-dimensional relative closures give the closed subanalytic exceptional set in the statement. Restricting the joint analytic extensions to smaller compact parameter boxes gives a common complex neighborhood and bounds for every derivative. $\square$

Lemma 3.3 (Uniformly controlled replacement paths). *Let $U \subset \mathbb{R}^k$ be open, let $D \subset U \times (0, \epsilon) \times \mathbb{R}^n$ be a bounded globally subanalytic family of open sets, and let v, w be bounded subanalytic sections that belong to the same component of $D_{x,t}$. Locally at generic $x_0 \in U$, there are a neighborhood U' of x_0 , $\epsilon' > 0$, and paths*

$$F_t : U' \times [0, 1] \longrightarrow \mathbb{R}^n$$

with endpoints $v(x, t), w(x, t)$ and image in $D_{x,t}$, with the following uniform properties. For each t , $F_t(\cdot, a)$ extends holomorphically to one fixed complex neighborhood $U_{\mathbb{C}}^t$, independently of t and a . On a smaller such neighborhood there is $C < \infty$ such that

$$\sup_{0 < t < \epsilon'} \sup_{x \in U_{\mathbb{C}}^t} (|F_t(x, a)| + |F_t(x, a) - F_t(x, b)|/|a - b|) \leq C \quad (a \neq b). \quad (4)$$

In particular, every fixed order of x -derivatives satisfies the same estimate, with its own constant. The replacement paths and their timing need not be subanalytic.

Proof. Apply definable Hardt triviality to the projection of D onto (x, t) , compatibly with the two marked endpoint graphs [Cos99, Theorem 5.22]. On each base cell the trivialization identifies these sections with fixed points in a model fiber. They lie in the same component; choose a fixed definable path between them and transport it by the continuous trivialization. This gives continuity jointly in (x, t, a) on that cell. The finite base walls have dimension at most k , and their limiting loci at $t = 0$ have dimension less than k . Deleting those loci puts a smaller product $U \times (0, \epsilon)$ in one cell. Retain the original path parameter a and let T be the closure of its graph in (x, t, a, z) . This graph has dimension $k + 2$. Over $t > 0$ and interior base parameters its closure is still that graph. Uniformize T properly, resolve t , and discard components supported at $t = 0$. The maps from the resulting manifold into (x, t, a, z) are analytic.

For every stratum of the normal-crossings divisor, delete the closure of its critical values in x . A stratum that cannot dominate the k -dimensional base is avoided altogether. At every remaining point over $(x_0, 0)$, the original x coordinates can be retained as tangential coordinates while all divisor coordinates are retained as normal coordinates. Indeed the differential of x is surjective on their common zero stratum, so the analytic inverse function theorem allows precisely this coordinate change. Absorbing the nonzero monomial unit gives, on each fixed sign piece,

$$t = y_1^{m_1} \cdots y_r^{m_r}, \quad y_i > 0. \quad (5)$$

The coordinates not displayed are free coordinates in an ordinary box. We use smaller closed boxes inside larger analytic chart boxes. Properness gives finitely many of them that cover every relevant graph point for x near x_0 and t small. Their maps have common complex neighborhoods in x after further shrinking.

Here is a precise gluing argument. At fixed (x, t) , the image of a smaller closed chart box with fixed signs and (5) is compact and connected: in logarithmic coordinates its monomial part is an affine hyperplane intersected with coordinate half-spaces, and the free part is a box. The positivity constraints do not spoil compactness: if $y_h \leq B_h$ and $t > 0$ is fixed, then $y_i \geq (t / \prod_{h \neq i} B_h^{m_h})^{1/m_i} > 0$. These finitely many sets are subsets of the actual continuous path graph for $t > 0$ and cover that graph. The intersection graph of the sets needed to connect its two endpoints therefore has a chain joining them; use a simple chain, whose length is bounded by the number of chart/sign pieces. The existence of each finite chain, its intersection points, and their lifts is a subanalytic condition. Selection followed by a generic restriction fixes one chain and bounded subanalytic endpoint lifts for all (x, t) in a smaller product. Adjacent endpoint images agree exactly. All endpoint lifts remain in the larger analytic boxes. This argument uses closed boxes to retain the interface points and does not assume that open chart images overlap uniformly.

By Lemma 3.2, all these finitely many endpoint coordinates have analytic power parametrizations with a common denominator. For two positive endpoint coordinates $A_i(x, t)$ and $B_i(x, t)$, join them by

$$y_i(x, t, s) = A_i(x, t)^{1-s} B_i(x, t)^s, \quad 0 \leq s \leq 1.$$

The monomial equality is retained, since it holds at both endpoints. The interpolation stays in the coordinate box. Free coordinates are interpolated linearly. If needed, insert one interior waypoint before making these interpolations; this also retains specified strict sign conditions. Every image point with $t > 0$ still belongs to the original graph, and hence its z coordinate belongs to $D_{x,t}$.

For completeness, the clock controlling the speed is independent of x . Write $y_i^0(s) = y_i(x_0, t, s)$. Positive endpoint coordinates have the form t^α times analytic positive units. On a fixed small complex neighborhood of x_0 , logarithms of the unit ratios to their values at x_0 are analytic and uniformly bounded, with bounded derivatives. It follows that

$$y_i(x, t, s) = y_i^0(s) E_i(x, t, s), \quad |E_i| + |\partial_s E_i| \leq C,$$

uniformly also for complex x . Thus

$$|\partial_s y_i(x, t, s)| \leq C(|(y_i^0)'(s)| + y_i^0(s)).$$

Each y_i^0 is monotone and bounded. Define the increasing clock

$$h_t(s) = \frac{s + \sum_i |y_i^0(s) - y_i^0(0)|}{1 + \sum_i |y_i^0(1) - y_i^0(0)|}.$$

After $s = h_t^{-1}(a)$, the displayed derivative estimate, the bounded total variations of the y_i^0 , and the elementary estimates for the free coordinates give a common Lipschitz bound. Shrinking the complex x neighborhood ensures that the complex interpolations remain in the complex analytic chart neighborhoods. Cauchy estimates give the same Lipschitz bounds for every x -derivative on a smaller neighborhood. Analytic chart maps preserve the estimates. Allocate fixed consecutive subintervals to the finitely many chart segments and concatenate them. The joints agree for real x , and hence also on the complex neighborhood by analytic uniqueness. This proves (4). $\square$

3.3 Attached discs and upper containment

Proof of Theorem 3.1. We first prove the analytic assertion for one controlled path family provided by Lemma 3.3. Fix its generic real center x and suppress x on R_t, G and P . Suppose $F_t(x, 0) = p$

and the paths are uniformly bounded. Passing to a subsequence, the uniform Lipschitz estimate gives

$$F_t(x, \cdot) \longrightarrow F(\cdot) \quad \text{in } C^\alpha([0, 1]) \quad (0 < \alpha < 1).$$

Here and below this convergence is along the selected subsequence. The derivative bounds in Lemma 3.3 and $R_t \rightarrow 0$ show that on every fixed finite real u box the actual graphs

$$\Phi_t(u, a) = u + C_t(x, u)F_t(x + R_t(x)u, a) \quad (6)$$

converge to

$$\Phi(u, a) = u + G(u)F(a). \quad (7)$$

This holds in C^α in a , with any fixed finite number of u derivatives, after lowering α if necessary. Every real point of (6) belongs to D_t^x : in original coordinates it is exactly

$$x' = x + R_t(x)u, \quad x' + R_t(x')F_t(x', a) \in S.$$

This identity is why all bounded real displacements, rather than a small fixed collection of orbits, must be retained.

The initial graph has room at every real displacement needed below. Indeed, the initial kernel condition, definable selection, and generic restriction give a real neighborhood U' and numbers $r, \epsilon > 0$ such that

$$B(p, r) \subset D_t^{x'} \quad (x' \in U', \quad 0 < t < \epsilon). \quad (8)$$

To see the uniformity, select positive radii and thresholds definably from the initial kernel condition, restrict to a cell on which they are continuous, and shrink to a relatively compact smaller box. For every fixed M , (2) and invertibility give $c_M > 0$ such that

$$B(u + C_t(x, u)p, c_M r) \subset D_t^x \quad (|u| \leq M, \quad 0 < t < \epsilon_M). \quad (9)$$

One may take c_M smaller than $(2 \sup_{|u| \leq M} \|G(u)^{-1}\|)^{-1}$. Thus the later real corrections are admissible even when they do not stay close to zero.

The triangular equation. We use Bishop's reduction of the attached-disc condition to a boundary conjugation equation [Bis65]; see also [HT78, §2, equation (2.1) and Proposition 2.1]. The triangular polynomial system and the endpoint estimates needed here are established below. Normalize circle measure to have total mass one, denote its mean by $\langle \cdot \rangle$, and let H be the real Hilbert transform with $H(1) = 0$ and $H(\cos n\theta) = \sin n\theta$. A complex boundary value $v + iw$ bounds a holomorphic disc precisely when $v - \langle v \rangle = -Hw$. The operator H is bounded on C^α for $0 < \alpha < 1$ by the conjugate-function estimate of [Pri16, pp. 100–102], and on L^q for $1 < q < \infty$ by the M. Riesz theorem in the form proved in [Ess84, equation (0.1) and §§1–2].

Choose a Lipschitz schedule $a : \partial\mathbb{D} \rightarrow [0, 1]$ that equals zero on a nonempty open arc I . Choose a smooth nonnegative function χ supported in I , with $\langle \chi \rangle > 0$. Add a complex parameter $\zeta_j \chi$ in each block. Write

$$V_j(u, \theta, \zeta) = G_j(u_{>j}(\theta))F_j(a(\theta)) + \chi(\theta)\zeta_j.$$

The real unknowns $u_j \in C^\alpha(\partial\mathbb{D}, \mathbb{R}^{d_j})$ are required to have mean zero. Their equations are exactly

$$u_j = -H \operatorname{Im} V_j - (I - \langle \cdot \rangle) \operatorname{Re} V_j, \quad j = s, s-1, \dots, 1. \quad (10)$$

Starting with block s , whose matrix is the identity, these formulas define a unique solution successively. They give a holomorphic disc $f_{a, \zeta}$ with boundary $u + G(u)F(a) + \chi\zeta$. Polynomial

composition is continuous on C^α , so every compact family of schedules and sufficiently small perturbations has a uniform C^α bound on its solutions. In particular it uses just one finite real u box. No estimate is asserted uniformly over schedules whose transition intervals subsequently shrink to zero.

Its center is

$$C_j(a, \zeta) = \langle G_j(u_{>j})F_j(a) \rangle + \langle \chi \rangle \zeta_j. \quad (11)$$

Later blocks do not depend on earlier perturbations. Hence the real differential of $\zeta \mapsto C(a, \zeta)$ is block triangular, with diagonal $\langle \chi \rangle I$ on each complex block. It is invertible at every solution. This proves full real rank $2d$, not only motion in the imaginary center directions.

Persistence and filling. Replace $G(u)F(a)$ in the equations by $C_t(x, u)F_t(x + R_t(x)u, a)$. On the Banach space of mean-zero real C^α functions, the differential of the limiting left-hand system is identity plus a strictly block triangular bounded operator. Its inverse is the finite sum for a nilpotent triangular operator. The convergence after (6) gives convergence of these systems and their first derivatives in operator norm near any compact family of limiting solutions. For example it follows directly from the C^2 bounds in u , the product estimate in C^α , and convergence of the a -dependent coefficients in a slightly stronger Hölder norm. The implicit function theorem therefore gives nearby solutions, uniformly along the compact family, and convergence of their centers and center differentials. Local uniqueness patches the solutions over the family. To justify a uniform center-image ball, let K bound the inverse differentials of the limiting center maps at zero perturbation. Compactness and C^1 convergence allow a fixed perturbation radius $r_0 > 0$ on which the variation of those differentials, including the small- t error, is at most $1/(4K)$, while the small- t inverse differentials at zero have norm at most $2K$. The contraction proof of the inverse function theorem then puts a ball of radius $r_0/(4K)$ about each unperturbed center in its center-map image. One first decreases r_0 further if the start-room restriction below requires it.

Use the homotopy of schedules $a_\sigma = \sigma a$, $0 \leq \sigma \leq 1$; all retain the start arc I . First use a preliminary closed perturbation ball to bound the limiting solutions in a finite u box. Enlarge that box slightly, compute its start-room constant in (9), and then decrease the perturbation ball so that $\|\chi\zeta\|_\infty$ is smaller than half that room. Only after these choices take t small. On the support of χ , (9) admits all the chosen small complex perturbations. Away from that support the boundary lies on the actual graphs (6). Thus the entire compact homotopy of disc boundaries lies in D_t^x . At $\sigma = 0$ and $\zeta = 0$ the limiting solution is $u = 0$ and the disc is constant p . For small t and small ζ , its actual disc lies in the initial ball. The continuity principle in the locally pseudoconvex domain now keeps every filled disc in D_t^x .

There is no issue at the exterior of the original coordinate box: for this fixed schedule the normalized boundary values, and hence the entire discs, are bounded by a fixed constant. Their original coordinates converge uniformly to $x \in \Omega$. Choose a fixed convex coordinate box Ω_0 with $x \in \Omega_0 \Subset \Omega$. The intersection $S \cap \Omega_0$ is locally pseudoconvex at every boundary point in $\mathbb{C}^d$, using the given condition on S and convexity at the artificial boundary. Its components are therefore Stein by the Euclidean Levi theorem. A plurisubharmonic exhaustion of the component in use bounds every filled disc by its boundary maximum; the compact boundary homotopy prevents a first exit. All the discs stay in Ω_0 by their already established coordinate bounds.

For each σ , the uniform center rank supplies a fixed ball $B(C(a_\sigma, 0), \rho)$ contained in D_t^x along the chosen subsequence, after slightly decreasing ρ . This already implies eventual inclusion for all small t : the set

$$\{t > 0 : B(C(a_\sigma, 0), \rho) \subset D_t^x\}$$

is a one-dimensional subanalytic set and has 0 in its closure. It therefore contains an interval

$(0, \epsilon_\sigma)$. Consequently $C(a_\sigma, 0)$ belongs to the full open kernel. The continuous curve of these centers starts at p , so all its points belong to P . This step does not require the replacement paths or the schedule homotopy themselves to be subanalytic.

Approaching the endpoint. For each $\delta > 0$, choose a Lipschitz loop a_δ that starts at 0, goes to 1 and returns, equals 0 on a nonempty arc, and equals 1 outside a set of measure at most δ . Apply the preceding argument at this fixed δ , with zero perturbation. It gives $C(a_\delta, 0) \in P$. Set $c = F(1)$. Since F is bounded,

$$F(a_\delta) \longrightarrow c \quad \text{in } L^q \quad (1 < q < \infty).$$

We claim that every block of u^δ from (10) converges to zero in every such L^q . For the last block this is the boundedness of H and subtraction of the mean. If it holds for the later blocks, polynomiality gives $G_j(u_{>j}^\delta) \rightarrow I$ in every finite L^q : each monomial is estimated by Hölder's inequality using the finitely larger exponents required by its degree. Multiplication by the uniformly bounded $F_j(a_\delta)$ then gives $V_j \rightarrow c_j$ in every finite L^q . Equation (10) proves the induction. Taking means in (11) yields

$$C(a_\delta, 0) \longrightarrow F(1).$$

Thus $F(1) \in \overline{P}$. The order of choices was: fix the loop and its nonempty start arc; solve and bound all corrections; choose their finite real box; let $t \downarrow 0$ and obtain a center in P ; only then shrink the exceptional arcs. No shrinking-arc estimate is used to justify the finite- t implicit function theorem.

The generic and uniform quantifiers. It remains to pass from controlled paths to all paths. The family of kernels $\mathcal{K}(D_t^x)$ is subanalytic, by its quantified ball definition; so is its component P_x marked by p . For fixed integer $L > |p|$ and positive rational η , let $E_{L,\eta}$ be the set of real centers for which arbitrarily small t admit a point of $A_t^x(L)$ at distance at least η from $\overline{P_x}$. This is a subanalytic set, using definability of components.

If $E_{L,\eta}$ had interior, choose within it a generic smaller real box. One-dimensional definability in t and selection give endpoints and paths to them for every sufficiently small t , after reducing the box generically. Use a slightly larger fixed spatial box for the paths. Lemma 3.3 supplies a controlled replacement family there. At a generic fixed center, a subsequence of its endpoints converges, and the analytic argument just proved puts its limit in $\overline{P_x}$. This contradicts their distance of at least η . Thus $\dim E_{L,\eta} < d$. Delete these exceptional sets for the countably many L, η . At every remaining center the negation of membership in $E_{L,\eta}$ is exactly (3). Increasing integer boxes and decreasing rational accuracies give the stated quantifiers for all finite boxes and all positive accuracies.

Finally, a starting point approaching p can be joined to p by a segment in the uniform initial ball. Enlarging the fixed path box by a fixed amount accommodates this segment and proves the last assertion. $\square$

Corollary 3.4 (Semidefinite outer tests). *Suppose, in addition, that the marked kernel is described recursively by strict positive-definiteness tests for finitely many continuous Hermitian matrix functions $M_j(Z)$, and that every point of $\overline{P}$ satisfies $M_j(Z) \geq 0$. Then for each finite L , at a generic fixed center,*

$$\inf_{Z \in A_t(L)} \lambda_{\min}(M_j(Z)) \geq -o(1) \quad (t \downarrow 0)$$

simultaneously for all j . The assertion is uniform over the accessible component in that fixed box.

Proof. If one inequality failed, a sequence in the bounded box would have a convergent subsequence on which the corresponding least eigenvalue is bounded above by a fixed negative number.

Theorem 3.1 puts its limit in $\overline{P}$, contradicting continuity and the semidefinite test there. There are only finitely many tests. $\square$

The corollary is applied first at a fixed radius. A diagonal choice of radii and scales may subsequently impose these outer estimates together with compact inner inclusion. In particular, an arbitrary joint limit of the radius and scale is not part of the assertion.

4 Successive faces and projective windows

Throughout this section let Y be a connected smooth projective variety and let $V \subset Y$ be a nonempty connected semialgebraic open subset that is locally pseudoconvex at its boundary. A domain is connected. A matrix inequality means positive definiteness of the indicated real symmetric matrix. Empty matrices are permitted; thus the test associated with an empty lower block is simply $\Im s > 0$. The open kernel always has the neighborhood meaning of Definition 2.5. In particular, its complement is the limiting accumulation set of the forbidden points, and not merely the set of pointwise failures of eventual membership.

We use the following precise consequences of the preceding preparatory results. Specializations with fixed rational weights preserve local pseudoconvexity and have semialgebraic open kernels; after restriction at generic real centers their tests are invariant under real translations. The references are Lemmas 2.6 and 2.7 and Corollary 2.9. We also use the projective hole tests with a container defined on a full parameter box, including its real edge: boundary trapping, interior trapping on a nonpluripolar slice, and propagation through a connected projection component, as in Lemmas 2.12 and 2.13 and Corollary 2.19. Whenever these results are applied below, the projective family is smooth and proper, its fibres are connected, its evaluation is holomorphic, and its open subset is the inverse image of the locally pseudoconvex domain under consideration. No conclusion about the boundary behavior of arbitrary holomorphic functions is part of these inputs.

4.1 Algebraic plaques and the contact quotient

Lemma 4.1 (Algebraicity of the null plaques). *Let M be a regular patch of a real algebraic hypersurface in a smooth complex algebraic n -fold. Suppose one side is locally pseudoconvex and the Levi form has constant rank r on M . Put $\ell = n - r - 1$. The Levi-null distribution integrates into complex ℓ -dimensional plaques. Each sufficiently small plaque is open in an irreducible complex algebraic variety of dimension ℓ . Locally at generic transverse parameters these varieties form a real analytic family of bounded degree. After complexifying the parameters, taking the relevant components, and resolving, they have smooth projective models in a smooth proper family.*

Proof. Choose a real analytic defining function ρ , and let θ be its characteristic one-form on M . Write $D = \ker \theta = TM \cap JTM$. Since the Levi form is semidefinite, its null space agrees with the characteristic distribution

$$K = \{v \in D : d\theta(v, D) = 0\}.$$

This distribution has real rank 2ℓ and is J -invariant. If X is a section of K and Y is a section of D , then $\theta([X, Y]) = -d\theta(X, Y) = 0$. Applying the Jacobi identity to two sections of K and one section of D shows that their bracket again has this property. Consequently K is involutive. The analytic Frobenius theorem gives analytic leaves, and their J -invariant tangent spaces make them complex submanifolds.

In affine algebraic coordinates write the real polynomial defining M as

$$\rho(z, \bar{z}) = v(z)^* C v(z), \quad C = C^*,$$

where v is a finite vector of holomorphic monomials; adjoining a constant monomial is harmless. For a parametrization f of a complex plaque, polarization of the identity $\rho(f(\zeta), \overline{f(\zeta)}) = 0$ gives

$$v(f(\eta))^* C v(f(\zeta)) = 0$$

for all η, ζ in a smaller plaque. Indeed the polarized expression is holomorphic in ζ and antiholomorphic in η , and its restriction to the diagonal determines every coefficient of its convergent series. The complex span W of the vectors $v(f(\zeta))$ is therefore totally isotropic for C .

The algebraic set $A = v^{-1}(W)$ contains the plaque and lies in $\{\rho = 0\}$. Every complex submanifold of M has tangent space in K : the restriction of ρ vanishes identically, so its Levi form vanishes there, and semidefiniteness identifies its zero vectors with its kernel. Applying this at regular points of each local component of A bounds that component's dimension by ℓ . The component containing the plaque thus has dimension exactly ℓ , and the plaque is open in it.

Only finitely many evaluation vectors are needed to span W . Choose them by fixed evaluations in analytic plaque coordinates, and restrict to a parameter neighborhood where their rank is maximal and constant. Their span is then an analytic map into a Grassmannian. The equations $v(z) \in W$ have uniformly bounded degrees, and their projective closures therefore have bounded degree. Track the component containing the marked plaque, using its irreducible germ. The family and its evaluation have the analytic algebraic presentation of Lemma 2.4. Complexification, a local branch in the component space, relative resolution, and generic smoothness give the stated smooth projective family. The generic restrictions here remove rank-drop and component-change loci; they do not assert uniformity over their closures. $\square$

Lemma 4.2 (The period matrix). *In the situation of Lemma 4.1, there is a real analytic $(r+1)$ -parameter family of plaques, with coordinates $(s, w) \in \mathbb{R} \times \mathbb{R}^r$, whose complexification evaluates with full complex rank near the original plaque. The remaining r real transverse coordinates can be retained as external parameters. At a fixed real center x_1 , let L be the smooth projective model of its plaque. There is a symmetric meromorphic $r \times r$ matrix H on L , holomorphic near the original plaque, such that $\Im H > 0$ there and the weighted tangent inequality is*

$$q_z(s, w) = \Im s - (\Im w)^\top (\Im H(z))^{-1} \Im w > 0. \quad (12)$$

The meromorphic data depend analytically on the generic external parameters.

Proof. The quotient of M by the characteristic plaques is a real analytic contact manifold of dimension $2r+1$. Indeed the hyperplane field D descends because $[K, D] \subset D$, and its induced exterior derivative is nondegenerate. Choose Darboux coordinates with contact form $da - \sum_j v_j du_j$. Holding v fixed and writing $a = s + v^\top w$, $u = w$, gives a family on which the pulled-back contact form is ds . Thus the w -directions form a Legendrian Lagrangian and the s -direction is transverse. The parameters v are the external parameters just mentioned.

At a point of a plaque, divide the complex tangent space by its null space. This gives a complex vector space of dimension r with a positive Levi form. The chosen real Lagrangian is complex independent: if its real span contained both a and $Ja \neq 0$, its symplectic form would vanish on (a, Ja) , contrary to Levi positivity. Its complexified evaluation, the leaf tangent, and the complexified transverse direction therefore span the ambient complex tangent space. This proves the full-rank assertion.

Let $e_1, \dots, e_r, f_1, \dots, f_r$ be a fixed real symplectic frame of the contact quotient, with the e_j the chosen Lagrangian. Normalize the characteristic form by its value on ∂_s . Its pullback is the fixed contact form, so its exterior derivative on the contact directions is the fixed symplectic form ω , including as the point moves along the plaque. The infinitesimal evaluations of the frame are holomorphic in the plaque variable. Use the evaluations of the e_j as a complex frame, and write those of the f_j as the columns of H . The complexified kernel of evaluation consists of $(-H\xi, \xi)$. It is isotropic for the complexification of ω , since ω has type $(1, 1)$. The equation

$$\omega((-H\xi, \xi), (-H\eta, \eta)) = \xi^\top (H - H^\top) \eta = 0$$

proves symmetry. Choose the sign of the second symplectic frame so that $B = \Im H$ is positive definite. For a vector a in the first real Lagrangian, its J -multiple in real symplectic coordinates is

$$Ja = (-\Re H B^{-1}a, B^{-1}a).$$

Consequently the metric $\omega(a, Ja')$ on that Lagrangian has matrix B^{-1} .

Here is the Taylor calculation, including the mixed terms. In the complexified family let $\tilde{\rho}(b, z)$ be the pulled-back defining function, positive on the selected side, and write $b = x + iy$. On the real parameter family it is identically zero. The Legendrian choice gives $\tilde{\rho}_{y_w}(x, z) = 0$ identically there, whereas $a(z) = \tilde{\rho}_{y_s}(x_1, z) > 0$ after orientation. Thus every pure x_w derivative and every $x_w y_w$ derivative occurring in the weight-two Taylor polynomial vanishes. Dividing by $a(z)e^2$ gives, uniformly on compact sets in (s, w, z) near the original plaque,

$$\frac{\tilde{\rho}(x_1 + (e^2 s, ew), z)}{a(z)e^2} \longrightarrow \Im s - (\Im w)^\top A(z) \Im w.$$

The positive matrix A is the normalized Levi metric on the chosen Lagrangian, up to the fixed positive convention in the Levi form and Taylor coefficient. Rescale the symplectic frames by that constant; the preceding computation then gives $A = (\Im H)^{-1}$ exactly.

Finally all infinitesimal evaluations come from a family algebraic in the projective fibre variable. Their entries are rational sections on the plaque variety. Taking the indicated frame ratios and pulling to its projective model makes H meromorphic on L . This also proves the asserted bounded algebraic and analytic parameter dependence. $\square$

4.2 The first kernel and its whole fibres

Complexify the family in Lemma 4.2, pull back V , and take the open kernel for $b_1 = x_1 + (e^2 s, ew)$. Denote by E the component containing $((i, 0), z_0)$ for a point z_0 of the original regular plaque. The fixed-weight specialization results show that E is semialgebraic and locally pseudoconvex, and that it is invariant under real translations in (s, w) and under $(s, w) \mapsto (a^2 s, aw)$, $a > 0$. These actions preserve its component because they are connected actions starting at the identity.

Lemma 4.3 (The whole seed fibre has a Bergman function). *At every z_0 in a sufficiently small original plaque patch, the component of the kernel fibre containing $(i, 0)$ is precisely*

$$T_A = \{(s, w) \in \mathbb{C}^{r+1} : \Im s > (\Im w)^\top A \Im w\}, \quad A = (\Im H(z_0))^{-1} > 0.$$

Its Bergman kernel is strictly positive at every point. This is a statement about the entire fibre component, not a relatively compact part of it.

Proof. Let K be any compact set in the entire scaled (s, w) -space. For sufficiently small e , the points $x_1 + (e^2 s, ew)$ for $(s, w) \in K$, and z in a fixed smaller plaque patch, lie in the original regular defining chart. The Taylor convergence in the preceding proof holds uniformly there. Strict positive points of q_{z_0} have neighborhoods eventually in the chosen side; strict negative points have neighborhoods eventually outside it. A zero of q_{z_0} cannot be in the open kernel, because every neighborhood contains a strict negative point. Since K was arbitrary, this identifies the whole positive component with T_A . It is connected. If $r = 0$ and both original sides occur, the two halfplanes are distinct fibre components: the real axis consists of the original boundary family and is still forbidden. The component through i is the upper halfplane in this case as well.

There is an explicit biholomorphism of T_A with a ball. First set

$$S = s + \frac{i}{2} w^\top A w, \quad W = \frac{1}{\sqrt{2}} A^{1/2} w.$$

An immediate expansion gives $\Im S - |W|^2 = \Im s - (\Im w)^\top A \Im w$. The map

$$(S, W) \mapsto \left(\frac{S - i}{S + i}, \frac{2W}{S + i} \right)$$

then maps the Siegel half-space onto the unit ball in $\mathbb{C}^{r+1}$. The complex Jacobian g of their composition is nowhere zero, and the change-of-variables formula yields

$$\int_{T_A} |g|^2 = \text{Vol}(\mathbb{B}^{r+1}) < \infty.$$

Thus every point evaluation on $A^2(T_A)$ is nonzero. If the fibre of E has other components, extend g by zero to them; this gives a holomorphic square-integrable function on that same whole fibre. No extension from a proper subdomain is being used. $\square$

Lemma 4.4 (Exact first epigraph). *There is a connected semialgebraic locally pseudoconvex domain $V_2 \subset L$ such that H is holomorphic on V_2 , $\Im H > 0$ there, and*

$$E = \{(s, w, z) : z \in V_2, \quad \Im s > (\Im w)^\top (\Im H(z))^{-1} \Im w\}. \quad (13)$$

In particular $V_2 = \{z : ((i, 0), z) \in E\}$.

Proof. We first classify the fibres away from a proper real algebraic subset of L . Refine a Boolean algebraic description of V using the irreducible real polynomial factors of its boundary equations, and take the factor defining the chosen regular hypersurface patch. Any other irreducible factor cannot vanish on every generic plaque: varying all $2r + 1$ real transverse parameters would make it vanish on an open part of the chosen irreducible hypersurface. The condition that its restriction to a plaque is identically zero is a finite algebraic condition on the bounded-degree family. Exclude its proper parameter locus. For the resulting generic plaque, discard the zeros of the remaining boundary equations, the singular locus of the chosen equation, and the zeros or poles of the frame determinants and normal derivative. These form a proper real algebraic subset of L : none contains the original regular plaque patch.

At any remaining point, the original Boolean description is locally empty, full, or consists of one or both sides of the chosen equation. The identities of the Taylor calculation and of $A = (\Im H)^{-1}$ continue as identities of real meromorphic functions on L . Accordingly the full kernel fibre there is empty, full, or one or both of the two strict quadratic sides. Each side is connected, being an

epigraph or a hypograph of a quadratic form. If such a side meets E , it is wholly in its fibre, since a path in that side stays in the same total kernel component.

Each fibre component is a pseudoconvex tube and hence is convex by Bochner's tube theorem [Nog20, Theorem 1.1]. Moreover a generic component of E cannot contain a complex affine line. Indeed, over a small coordinate polydisk $U \subset L$, the total domain $E \cap (\mathbb{C}^{r+1} \times U)$ is pseudoconvex in a Stein coordinate space. Berndtsson's theorem, with weight zero, says that

$$((s, w), z) \mapsto \log K_{E_z}(s, w)$$

is plurisubharmonic, allowing the value $-\infty$ [Ber06, Theorem 1.1]. The slices here are the actual unbounded slices; the fibre coordinates have not been truncated. These local assertions therefore agree on overlapping base boxes. If the restricted total domain is disconnected, apply the theorem to its components, or to an exhaustion componentwise. The Bergman kernel at a point only uses its fibre component, so this gives the same function.

A convex domain containing a complex affine line is invariant under translation in that line direction. To see this, take the convex hull of a small ball at any of its points and successively longer segments in the line; openness then gives each fixed translate of a smaller ball. In complex linear coordinates the domain is consequently a product with $\mathbb{C}$. Fubini and the fact that an entire square-integrable function on $\mathbb{C}$ is zero imply that its Bergman space is zero. An open chamber of such fibres would make the displayed plurisubharmonic function identically $-\infty$ on a nonempty open subset of connected E . This contradicts Lemma 4.3.

At a generic point the quadratic matrix is nonsingular. Its epigraph is convex precisely when it is positive semidefinite, and its hypograph is convex precisely when it is negative semidefinite. Hence the only generic nonempty components remaining are the positive definite epigraph, in $\{\Im s > 0\}$, and the negative definite hypograph, in $\{\Im s < 0\}$. No point of E can have $\Im s = 0$: an open neighborhood of such a point would contain a point over a generic base with that same imaginary scalar coordinate, contrary to this classification. Connectedness and the seed select the positive sign. Thus $\Im H > 0$ on a dense open subset of the projection V_2 of E .

The meromorphic Cayley matrix

$$C = (H - iI)(H + iI)^{-1}$$

has operator norm less than one on that dense subset. It is bounded where meromorphic and therefore extends holomorphically across its polar locus. Its norm is at most one throughout V_2 . Equality on some unit vector would, by the maximum principle applied to a suitable scalar matrix coefficient, give equality on the connected domain, contradicting strict contraction on the original plaque patch. Thus $\|C\| < 1$ everywhere, $I - C$ is invertible, and $H = i(I + C)(I - C)^{-1}$ is holomorphic with positive imaginary part. This defines the full epigraph F on the right of (13). The generic classification and openness imply $E \subset F$, since the non-strict limiting inequalities cannot have an interior zero, by variation of $\Im s$.

We give the remaining fullness argument, to keep it distinct from the Bergman argument. At a generic base point $E_z = F_z$. Stratify the exceptional base set. At a regular codimension-one wall point in V_2 , some point of F_z is in E by the definition of the projection. Every compact path in the connected slice F_z is covered by finitely many small coordinate boxes compactly contained in F . On a generic side of the wall all these boxes are available. The tangency propagation lemma, Lemma 2.14, applied successively to these boxes, propagates the one available central patch along the path. It uses only transverse parameter disks and relatively compact fibre patches, so compactness of the whole tube is not required. We conclude $E_z = F_z$ at the generic wall points.

The remaining omission $A = F \setminus E$ is relatively closed and subanalytic of real codimension at least two. Local pseudoconvexity and Theorem 2.15 make A complex analytic. It is invariant under all real translations in the fibre coordinates. For each such direction, analyticity continues this invariance to small complex translations: restrict its local defining functions to the complex translation parameter and apply the identity theorem. Iteration along paths in the connected slice F_z shows that if A_z is nonempty it is all of F_z . This is impossible for $z \in V_2 = q(E)$. Hence A is empty. Finally V_2 is the inverse image of E by $z \mapsto ((i, 0), z)$, proving its local pseudoconvexity and the last assertion. $\square$

4.3 Polynomial normalizations

The first epigraph reduces the boundary problem to its projective plaque model. When that model is scaled again, its period matrix varies with the second scale. The next two lemmas normalize this matrix while retaining polynomial limits on every bounded box; this is the compatibility needed for the induction.

We record the finite-dimensional argument used below. In the phrase “joint Puiseux expansion,” a single substitution $t = v^q$ is understood, after which a fixed power of v times each function is analytic on a neighborhood of $\{0\} \times U$. An expansion separately for every $x \in U$ does not have this meaning. The generic parameter statement of Lemma 2.2 is used in precisely this joint form.

Lemma 4.5 (Finite Taylor extraction). *Suppose $A_t(x)$ and $A_t(x)^{-1}$ have joint Puiseux expansions on a real parameter neighborhood, with polynomial bounds in t^{-1} , and $R_t(x)$ has the same property and tends to zero locally uniformly. If f is holomorphic on a fixed complex neighborhood, then any expression obtained by multiplying $f(x + R_t z)$ on either side by such Puiseux matrices, and subtracting a similarly bounded constant term, has a polynomial holomorphic limit on every bounded z -box whenever it is locally bounded on one nonempty complex open z -set. The limit and convergence are analytic in a generic real center and smooth to all orders in z on bounded boxes.*

In addition, if R_t is invertible and

$$C_t(x, u) = R_t(x)^{-1} R_t(x + R_t(x)u)$$

is bounded on real bounded u -boxes, it converges smoothly on those boxes to a polynomial $G_x(u)$. After generic restriction, $\det G_x(u) = 1$, and the limiting matrices have polynomial inverses.

Proof. Put $v = t^{1/q}$ with a common denominator for the finitely many Puiseux expansions. There are integers $a > 0$ and $b \geq 0$ such that $R_t = v^a A(v, x)$, all multiplying factors have norm $O(v^{-b})$, and their analytic parts extend to a common complex neighborhood. Taylor-expand $f(x + R_t z)$ through order m . On any fixed bounded complex z -box the remainder after multiplication is $O(v^{a(m+1)-c})$, where c is a fixed integer bounding the poles of all factors. Choose m with $a(m+1) > c$. The retained part has a convergent Laurent expansion in v , with finitely many negative terms and coefficients polynomial in z . Boundedness on a nonempty complex open set forces the coefficient of the lowest negative power to vanish there and hence identically as a polynomial. Repeating removes all negative powers. The coefficient of v^0 is the claimed polynomial, and the Taylor remainder proves convergence on every bounded complex box. Cauchy estimates give derivative convergence. The same argument with the coefficient parameter x retained gives the generic local analytic statement.

For the ratio, Taylor-expand the analytic part of $R_t(x + R_t(x)u)$ in the increment $R_t(x)u$, and then multiply by $R_t(x)^{-1}$. Again sufficiently many Taylor terms leave a vanishing remainder. Boundedness for real u on an open box kills every negative Laurent coefficient, because a polynomial

vanishing on a real open box vanishes identically. This proves even locally uniform holomorphic convergence on bounded complex u -boxes. Finally, at a generic x ,

$$\det R_t(x) = v^k(d(x) + O(v)), \quad d(x) \neq 0.$$

Since $R_t(x)u \rightarrow 0$, the quotient of these determinants at $x + R_t(x)u$ and x tends uniformly to one. Thus $\det G_x = 1$. The adjugate formula gives both its polynomial inverse and uniform boundedness of the inverses on fixed boxes. $\square$

Lemma 4.6 (Normalizing a varying Siegel matrix). *Let S_2 be a fixed parameter domain with normalization $R_t^2(x_2)$, open-kernel component P_2 , and a fixed interior seed p_2 . Let H_* be symmetric and holomorphic on a full neighborhood of x_2 , with $\Im H_* > 0$ on S_2 . Assume the normalizations have joint Puiseux expansions and polynomial inverse bounds. Define*

$$T_t = (\Im H_*(x_2 + R_t^2 p_2))^{1/2}, \quad (14)$$

$$H_t(z) = T_t^{-1}(H_*(x_2 + R_t^2 z) - \Re H_*(x_2 + R_t^2 p_2))T_t^{-1}. \quad (15)$$

Then H_t converges on all bounded complex boxes to a symmetric holomorphic polynomial matrix τ , with $\tau(p_2) = iI$ and $\Im \tau > 0$ on P_2 .

Proof. The positive square root and its inverse are subanalytic algebraic operations on positive matrices. Generic joint Puiseux preparation therefore applies to T_t and T_t^{-1} ; positivity of the seed matrix and the nonzero leading coefficients give their power bounds. Every compact subset of P_2 lies in the normalized S_2 for small t , so H_t maps there into Siegel space and $H_t(p_2) = iI$. Apply the matrix Cayley transform. Its values lie in the unit matrix ball and its seed value is zero. Montel's theorem and the maximum principle imply bounds for the inverse Cayley transform on compact subsets of connected P_2 : a limiting contraction cannot reach the unit sphere at an interior point while taking zero at p_2 . Thus H_t is locally bounded on a nonempty complex open set.

Lemma 4.5 now gives the polynomial limit on all bounded boxes, not merely within P_2 . Its imaginary part is nonnegative on P_2 . For every fixed nonzero complex vector v , the harmonic function $v^* \Im \tau v$ is nonnegative and is $|v|^2$ at p_2 . The strong minimum principle makes it positive everywhere. This proves strict positivity of the matrix. $\square$

4.4 The induction and its parameter tests

Theorem 4.7 (Projective windows). *Let Y be a connected smooth projective variety and let $V \subset Y$ be a nonempty connected semialgebraic open subset that is locally pseudoconvex at its boundary. The following data exist, locally at generic centers in totally real parameter boxes. The same assertion holds in smooth projective families with additional real parameters, analytic coefficients, and bounded algebraic fibre data, when the hypotheses hold fibrewise. The constructed data depend analytically on those parameters after generic restriction and may be complexified locally.*

There is a complex box $B \subset \mathbb{C}^d$, a smooth proper projective family $q : Q \rightarrow B$ with connected fibres of dimension $\dim Y - d$, and a holomorphic evaluation $\text{ev} : Q \rightarrow Y$, algebraic with bounded data in the projective variables and of full rank on a coordinate patch meeting every fibre. Its fibre images $Z_b = \text{ev}(Q_b)$ are irreducible projective varieties of dimension $\dim Y - d$, and $Q_b \rightarrow Z_b$ is proper and surjective. If $G = \text{ev}^{-1}(V)$, a component S of $q(G)$ and a relative proper algebraic subset $D \subset Q$ satisfy

$$Q_b \setminus D_b \subset G_b \quad (b \in S). \quad (16)$$

The container D is analytic on the full box and proper in every fibre, after shrinking the box. The actual holes $Q|_S \setminus G$ are analytic; their images give precisely $Z_b \setminus V$ and are algebraic in each fibre. The domain S is locally pseudoconvex at its parameter boundary. Its boundary inside B is disjoint from $q(G)$. All stated shrinkings of B may be made before the choice of scales and sequences.

There are a fixed interior seed p , real block diagonal matrices $R_t(x)$ tending to zero with polynomial inverse bounds, and a component P of the open kernel of

$$S_{t,x} = R_t(x)^{-1}(S - x)$$

at p . A fixed neighborhood of p is eventually in $S_{t,x}$. The domain P is a finite recursive tower, beginning at a point and adding a block $(s, w) \in \mathbb{C} \times \mathbb{C}^r$ over a previously constructed domain P_2 by the condition

$$z \in P_2, \quad \Im M(s, w, z) > 0, \quad M(s, w, z) = \begin{pmatrix} s & w^\top \\ w & \tau(z) \end{pmatrix}, \quad (17)$$

where τ is symmetric and polynomial holomorphic and $\Im \tau > 0$ on P_2 .

For each bounded real u -box the ratios

$$R_t(x)^{-1}R_t(x + R_t(x)u) \longrightarrow G_x(u) \quad (18)$$

converge with every derivative. The matrix G_x is polynomial and block diagonal, each block depends only on later block coordinates, $G_x(0) = I$, and every block has determinant one. Its inverse is polynomial. For all sufficiently small real u ,

$$Z \longmapsto u + G_x(u)Z \quad (19)$$

is an automorphism of P .

Finally the bounded-path localization conclusion holds: any limit of endpoints of paths in $S_{t,x}$ starting at the seed and staying in a fixed bounded box belongs to $\overline{P}$. Equivalently, on each fixed bounded box the accessible components have outer distance to $\overline{P}$ tending to zero.

Proof. We induct on $\dim Y$. If the boundary has no ordinary real hypersurface patch, semialgebraic stratification shows that its complement has real codimension at least two. Indeed an open exterior would be separated from the nonempty open set V by a stratum of codimension one. Theorem 2.15 then makes $Y \setminus V$ analytic, and hence algebraic by Chow's theorem. Set $d = 0$, $Q = Y$, $P = S$ a point, and take this complement as the container. In a family, Lemma 2.17 selects a containing proper algebraic set with bounded equations; generic analytic selection of their normalized coefficients gives a real analytic container. Complexify those coefficients and use Lemma 2.16, shrinking so that the resulting relative divisor is proper on every fibre of the full box. The actual complements need not themselves be selected as an analytic parameter family. The same construction is available as a terminal step as soon as the complement consists only of algebraic holes. All zero-dimensional assertions have their evident empty-matrix meaning.

Composing the projective families. Otherwise choose a regular algebraic boundary patch of constant Levi rank, orient its inward side, and apply Lemmas 4.1–4.4. This gives the geometric first kernel E over the smaller projective model L , with projection $V_2 \subset L$ and $\dim L = \dim Y - r - 1 < \dim Y$. Write b_1 for the complex first-family parameters and x_1 for their real center; e will be their scale. Apply the induction hypothesis to (L, V_2) , keeping x_1 and the unused real contact parameters as external parameters. Its output is

$$q_2 : Q_2 \rightarrow B_2, \quad D_2, \quad S_2, \quad (P_2, p_2, R_t^2).$$

Here $b_2 \in B_2$ is the lower-stage parameter, x_2 its real center, and t its scale. For now b_2 is left unscaled.

Replace D_2 , if necessary, by the full-box proper relative divisor of Lemma 2.16. This only reduces the available complement and preserves its containment in the open family. Complexifying the analytic dependence on x_1 composes the lower family with the first plaque family. We obtain

$$q : Q \longrightarrow B_1 \times B_2, \quad \text{ev} : Q \longrightarrow Y, \quad G = \text{ev}^{-1}(V),$$

where q is smooth, proper, and projective. The continued divisor, still denoted D_2 , is defined on this full box. All resolutions, component choices, and box shrinkings here precede the choice of scales. We must select one component S of $q(G)$ retaining the complements of D_2 , then identify its first kernel. This will allow the lower scale t to be introduced without changing the actual domain being normalized.

On a projective fibre of Q_2 over S_2 , the pullback of H is constant. It is meromorphic on that connected projective fibre and lies in Siegel space off D_2 . Its Cayley transform is bounded there, extends across D_2 , and is constant on a compact connected complex manifold. The resulting symmetric matrix is therefore a meromorphic function $H_*(b_2)$, with positive imaginary part on S_2 ; its dependence on the retained first parameters is understood. Constancy along fibres persists meromorphically on the full connected box by uniqueness. Evaluate on a local holomorphic section avoiding indeterminacy to make H_* holomorphic on a full smaller box about a generic real center x_2 . A proper complex analytic polar set cannot contain an open subset of the totally real center box, so this choice is available.

The fixed parameter domain and its whole fibres. We now pass from the limiting fibres of E to the actual fibres of G . Fix an open chamber compactly inside S_2 and let $b_1 = x_1 + (i\eta, 0)$, $\eta > 0$. Every point of a central projective fibre off D_2 evaluates into V_2 . By (13), a neighborhood of each such point is eventually in the inverse image of V . Properness gives the following useful closed-set formulation: every accumulation point, as $\eta \downarrow 0$, of a hole on a compact parameter subbox belongs to D_2 . Otherwise a convergent sequence of holes would end in one of the just-described eventually included neighborhoods.

Here the limiting statement holds for all approaches through the upper half-disk, not just for the vertical ray. Write its scalar coordinate as $u + i\eta$, recenter the real first parameter at u , and put $e = \sqrt{\eta}$. In normalized coordinates the point is again $(i, 0)$. On a compact fibre patch outside D_2 , this point has a positive distance from the limiting forbidden set. Apply Lemma 2.8 to the capped distances of the normalized forbidden sets, including the remaining real parameters and the compact fibre variables jointly. Use one fixed bounded seed box and finitely many compact projective coordinate charts. The finite-test part of that lemma supplies one open real center neighborhood on which convergence is uniform on compact parameter and test boxes. Now choose an accuracy smaller than half the positive separation of the patch under consideration. The same separation holds for all sufficiently small e and nearby real centers. The threshold may depend on the patch; no positive lower bound for the separation near D_2 is required. Every sequence of upper half-disk holes approaching a point off D_2 would violate this separation on a compact patch about its limit. Thus all such accumulation points belong to D_2 , as required by the boundary trapping hypothesis. The container has a holomorphic continuation on the full box. Lemma 2.12, applied to the scalar upper half-disk, therefore traps the holes in that continuation on the indicated side. For the remaining first parameters use Lemma 2.13: their real boxes are totally real and nonpluripolar. This gives actual containment on a nonempty complex open parameter region. Choose the component S of $q(G)$ reached there. Proposition 2.18 makes the actual holes analytic throughout S , and Corollary 2.19 keeps them inside this same full-box divisor D_2 . Their proper

images are precisely $Z_b \setminus V$; these are compact analytic and hence algebraic. The container is proper fibrewise after shrinking at a generic center. Corollary 2.20 now gives local pseudoconvexity of S at its boundary. Since $q(G)$ is open, a boundary point of one of its components cannot belong to any of its components; this proves the asserted exclusion from $q(G)$.

For the first-kernel comparison below, make one further full-box restriction. The total evaluation $Q_2 \rightarrow L$ is generically a submersion. Choose at the generic center a point off D_2 where its differential has full rank. Smoothness of the projective family gives a holomorphic section σ through this point. Shrink the box so that σ is everywhere off the continued container and the evaluation retains full rank on a fibre-coordinate patch around σ . Include the first parameters in this holomorphic choice. The differential includes both base and projective fibre variables; it need not have full rank in the fibre directions alone.

After these box restrictions, retain the selected connected components S and S_2 . Fix b_2^0 in the retained chamber of S_2 . For a compact path in S_2 starting at b_2^0 , choose local sections of Q_2 off D_2 along a finite covering of the path. Their evaluations lie in V_2 . The strict inequalities in (13) and neighborhood eventual inclusion put their small first-scale displacements in $q(G)$, with overlaps joining them to the region used to choose S . Thus every such displaced path lies in this same S for all sufficiently small first scales. The threshold may depend on the compact path, but S is fixed before either scale tends to zero.

The first kernel in parameter space. Let $\mathcal{K}_1$ be the full first parameter kernel of this fixed S , with $b_1 = x_1 + (e^2 s, ew)$ and b_2 unscaled. Its component A through $(i, 0, b_2^0)$ is exactly

$$A = \left\{ (s, w, b_2) : b_2 \in S_2, \quad \Im \begin{pmatrix} s & w^\top \\ w & H_*(b_2) \end{pmatrix} > 0 \right\}. \quad (20)$$

The compact-path argument and strict neighborhood inclusion in (13) put the entire displayed domain in this component. For the reverse inclusion, a parameter neighborhood in $\mathcal{K}_1$ must give a neighborhood in the geometric kernel E . Use the section σ and evaluation patch prepared above to make this an open evaluation test.

Suppose a neighborhood belongs eventually to the first normalized parameter domain. By (16) every point in this evaluation patch is then in the inverse image of V . The submersion theorem, on a smaller fixed patch, sends these neighborhoods to neighborhoods in the first plaque family. The openness here is uniform after normalization. In local coordinates the normalized evaluation has the form

$$(s, w, b_2, v) \longmapsto (s, w, F(x_1 + (e^2 s, ew), b_2, v)),$$

where v is a projective fibre coordinate. The first coordinates are unchanged, and $D_{(b_2, v)} F$ has a uniformly bounded right inverse on a smaller patch around σ . The remaining derivative block tends to zero because it contains the factors e^2, e . The block triangular differential consequently has a uniformly bounded right inverse on each compact normalized path neighborhood. The quantitative local submersion theorem gives the needed neighborhoods of a fixed positive size. In the limit this is a neighborhood test for membership in the full first geometric kernel. Along a path $(s(a), w(a), b_2(a))$ from $(i, 0, b_2^0)$ in the selected component, the tested points

$$(s(a), w(a), \text{ev}_2(\sigma(x_1, b_2(a))))$$

form one continuous path in that kernel. At the starting point, the section is off D_2 over S_2 , so its evaluation is in V_2 and the tested point is in E . The entire tested path therefore stays in E . There are no changes of section in this argument. Its base values stay in $q_2(\text{ev}_2^{-1}(V_2))$ and, by connection to the seed, in its component S_2 . Equation (13) gives precisely the strict matrix test in (20). This

proves the reverse inclusion. The same rank argument, followed by the first family's full-rank evaluation, proves full rank of the final evaluation on a patch meeting every fibre in a smaller box. Its differential is then injective also on the fibre tangent. The irreducible image therefore has dimension $\dim Q_b = \dim Y - d$; properness of Q_b gives the remaining fibre-image assertions.

The second specialization and its marked component. We have obtained a fixed domain S and identified its first marked kernel A . Now normalize H_* by (14)–(15). In the first block use $b_1 = x_1 + (e^2 s, eT_t w)$; in later blocks use $b_2 = x_2 + R_t^2 z$. Congruence of the matrix inequality by $\text{diag}(1, T_t^{-1})$ identifies its first-stage test with

$$\Im \begin{pmatrix} s & w^\top \\ w & H_t(z) \end{pmatrix} > 0.$$

Lemma 4.6 gives its limit (17). Apply the above t -dependent affine normalization to $\mathcal{K}_1$ and A to obtain $\mathcal{K}_{1,t}$ and A_t . The seed $(i, 0, p_2)$ lies in A_t for every small t , because $H_t(p_2) = iI$. Consider a compact path from this seed in the marked component of the open kernel of $\mathcal{K}_{1,t}$. A compact connected tube of small neighborhoods around this path is eventually contained in $\mathcal{K}_{1,t}$. Since that tube contains the seed, it is contained in its component A_t for all small t . Its projection supplies neighborhoods eventually in the normalized S_2 . The projected path consequently belongs to the second open kernel and, by connection to p_2 , to P_2 . The limit matrix is nonnegative on the tube. With its lower block positive on P_2 , varying $\Im s$ shows that a zero Schur complement cannot be an interior point. Thus the limiting matrix test is strict. Conversely, compact neighborhoods in the strict tower inequalities are eventually in A_t , by kernel inclusion for S_2 and locally uniform convergence of H_t ; their paths connect them to the seed. This proves the equality of marked iterated kernels. In particular, other components of the full first kernel cannot enter this marked component when the second scale tends to zero. Only compact-path neighborhood inclusion is used here, not the later accessible-path localization theorem.

A single scale. Apply the closed-set power diagonalization theorem, Lemma 2.10, to these fixed definable forbidden-set tests and their complements. It supplies an integer N such that replacing e by t^N preserves the iterated open kernel on every fixed bounded box, with the seed component just identified. Accuracy and box radius are fixed parameters in that theorem; the exponent is uniform in those parameters whereas the threshold in t is allowed to depend on them. Increasing N preserves this conclusion. We shall do so finitely many times below. The final normalization is

$$R_t = \text{diag}(t^{2N}, t^N T_t, R_t^2). \quad (21)$$

It tends to zero and has polynomial inverse bounds. A compact ball about $(i, 0, p_2)$ supplies the fixed seed neighborhood.

Recentering ratios and localization. We verify the ratio condition, including the square-root order. By induction the later-block ratio $C_t^2(u_2)$ is bounded and converges polynomially. Replacing the later center by $x'_2 = x_2 + R_t^2 u_2$ makes its seed evaluation point equal to

$$x_2 + R_t^2(u_2 + C_t^2(u_2)p_2).$$

This is a bounded normalized point for bounded u_2 . Polynomial bounded-box convergence of H_t therefore bounds

$$T_t^{-1}(T'_t)^2 T_t^{-1} = (T_t^{-1} T'_t)(T_t^{-1} T'_t)^\top.$$

It follows that $T_t^{-1} T'_t$ is bounded. No commutativity of the positive square roots is asserted or needed.

All functions involved have joint analytic Puiseux expansions near generic centers. Their derivatives of any fixed finite order have power bounds. The shift of x_1 is $O(t^N \|T_t\| + t^{2N})$ on bounded boxes. Choose N larger than the finitely many pole orders in the lower-scale data, their inverses, and the Taylor remainders used here. Its effect on those data and on the square-root ratio then tends to zero. The normal-coordinate ratio is identically one, and the tangential ratio has a limit depending only on u_2 . Apply Lemma 4.5 to obtain polynomial convergence with all derivatives. Applying its determinant argument separately to T_t and to each old block shows that every limiting block has determinant one. The later-variable dependence in older blocks is the induction hypothesis. This proves (18) and all its asserted moderation properties.

Finally use Lemma 2.8 for the closed forbidden-set tests in the normalized coordinates. At a generic real center the distance limits on each fixed box are unchanged by all centers tending to that center. A separate neighborhood and threshold are permitted for each fixed accuracy. The exact change of center in the original fixed set S is

$$x + R_t(x)(u + C_t(x, u)Z) = x + R_t(x)u + R_t(x + R_t(x)u)Z.$$

Taking neighborhood limit tests on both sides and using the bounded inverse ratios proves invariance of the full kernel under $Z \mapsto u + G_x(u)Z$. For small u , the image of the seed stays in the same component; hence this is an automorphism of P . The bounded-path conclusion is now Theorem 3.1, applied to the fixed pseudoconvex domain S , the seed ball, and the ratios just proved. This completes the induction. Here $D_t^x = S_{t,x}$ in the notation of the localization theorem. $\square$

4.5 Geometry of the limiting tower

Proposition 4.8. *The domain P of Theorem 4.7 is diffeomorphic to $\mathbb{R}^{2d}$, where $d = \dim_{\mathbb{C}} P$. In particular it is contractible. It is Stein and biholomorphic to a bounded domain of the same complex dimension. Complete real holomorphic vector fields from the identity component of its automorphism group span every complex tangent space over $\mathbb{C}$.*

Proof. At one stage of the tower put $B(z) = \Im \tau(z)$. The Schur complement rewrites its condition as

$$\Im s > (\Im w)^{\top} B(z)^{-1} \Im w.$$

The positive slack

$$\lambda = \Im s - (\Im w)^{\top} B(z)^{-1} \Im w$$

gives smooth coordinates (z, w, ξ, η) on the whole stage, where $\xi = \Re s$ and $\eta = \log \lambda$. Their inverse recovers $s = \xi + i(e^{\eta} + (\Im w)^{\top} B(z)^{-1} \Im w)$. Thus this stage is diffeomorphic to $P_2 \times \mathbb{C}^r \times \mathbb{R}^2$. Induction, starting from a point, proves the claimed diffeomorphism and contractibility.

For Steinness, let M_j be the polynomial matrices in all blocks. The function

$$\Phi(Z) = |Z|^2 - \sum_j \log \det \Im M_j(Z)$$

is strictly plurisubharmonic on P . The matrix entries have harmonic imaginary parts, and differentiating $-\log \det$ gives a nonnegative Levi form, equivalently the squared matrix norm of the derivative conjugated by the positive inverse square root. At a finite boundary point at least one positive matrix becomes singular, so $\Phi \rightarrow +\infty$. At infinity its negative terms are at worst logarithmic in $|Z|$, because the M_j are polynomial, whereas $|Z|^2$ dominates them. Thus Φ is an exhaustion and P is Stein.

For a bounded realization use one block at a time. Since $\Im M > 0$, the matrix $K = I - iM$ has Hermitian part greater than I , and hence $\|K^{-1}\| \leq 1$. Write its first column as $(a, b)^\top \in \mathbb{C} \times \mathbb{C}^r$. The lower-right block $D = I - i\tau(z)$ is invertible, and the Schur complement formula gives $a \neq 0$. The equations for that column recover the original coordinates holomorphically:

$$w = \frac{Db}{ia}, \quad s = \frac{i(1 - a + iw^\top b)}{a}.$$

Thus $(s, w, z) \mapsto (a, b, \psi_2(z))$, with an inductively obtained bounded realization ψ_2 , is injective, bounded, and has a holomorphic inverse on its image. It uses exactly $r + 1$ new coordinates and its image is open. This proves the same-dimensional bounded realization.

The automorphism group of a bounded domain is a real Lie group [Kob67, Proposition 3.1 and Theorem 6.2]. Differentiate the curves (19) at $u = 0$ to obtain complete real holomorphic vector fields. In the block order of the theorem their values have identity diagonal: differentiation in a coordinate of block j contributes its coordinate unit vector in that block, nothing in later blocks, and possibly vectors in earlier blocks, because G_k only depends on blocks later than k . This triangular matrix has determinant one at every point. The resulting fields therefore span the complex tangent space everywhere. $\square$

5 The intrinsic quotient

We construct a quotient whose fibres are the projective-open varieties found in the boundary windows. The first task is to show that these fibres are intrinsic: two of them which meet must coincide. Guards then make their translated charts exhaust the covering space, and compactness of the inverse charts identifies the quotient with the limiting tower. We begin by putting the given cover in the smooth setting required by the window construction.

5.1 The covering is locally pseudoconvex

Normalize the projective ambient variety along the component containing $\tilde{X}$, and use a functorial projective resolution [BM08, Theorem 1.1]. Compatibility with local analytic isomorphisms identifies its restriction with the pullback of a projective resolution $X^+ \rightarrow X$. Thus

$$\mu : V \longrightarrow \tilde{X}$$

is a proper modification, V is a smooth semialgebraic open subset of a smooth projective variety Y , and the lifted action of the deck group $\Gamma = \pi_1(X)$ is free and properly discontinuous with compact projective quotient X^+ . The inverse image V is connected: the fibres of a proper modification of a normal space are connected, and such a map cannot disconnect the inverse image of a connected open set.

Lemma 5.1. *The open set V is locally pseudoconvex in Y .*

Proof. This is precisely Corollary 2(a) of Ivashkovich [Iva08]. Here is also the Hartogs argument in the present projective case. On a Hartogs figure in a coordinate polydisc $B \subset Y$, the covering map $p : V \rightarrow X^+$ extends meromorphically to B , by the meromorphic Hartogs extension theorem of Ivashkovich [Iva92], in the form of Theorem 4 of [Iva08]. It agrees with p on the component of $B \cap V$ containing the figure.

If that component does not fill B , choose a straight segment from a point of the figure to a point outside V , and let a be its first exit. The complex line containing the segment is not

contained in the indeterminacy locus, since it meets the figure. Restriction of the meromorphic extension to this line extends holomorphically across its isolated indeterminacies: in projective homogeneous coordinates cancel the common vanishing order. Consequently p tends to a point $x \in X^+$ along the segment. An evenly covered neighbourhood of x contains the image of a connected final subsegment. That subsegment lies in one sheet and converges, by the inverse sheet map, to a point of V . Its limit in the Hausdorff space Y must also be a , a contradiction. Every such Hartogs figure therefore fills in V , which is the local pseudoconvexity criterion. Pulling back its local plurisubharmonic tests gives the same boundary property for holomorphic inverse images used in the construction. $\square$

5.2 Whole projective fibres in a window

The analytic-to-algebraic passages in this section use Remmert's proper-image theorem in the reduced-space form proved in [Gra60, §7, Satz 1], and Chow's theorem in [Ser56, Proposition 13]. We also use normalization and normal holomorphic removability as in [Dem12, II.7.3, II.7.12 and Remark II.7.14]. These apply to the possibly singular reduced incidence spaces as well as to their smooth models.

We may now apply Theorem 4.7. Write $n = \dim Y$, $r = n - d$, and denote the parameter family and its evaluation by

$$q : Q \longrightarrow B, \quad e : Q \longrightarrow Y.$$

The map q is smooth, proper, and projective, with connected r -dimensional fibres. Their images $Z_b = e(Q_b)$ are irreducible projective varieties of dimension r . The evaluation has generic rank n . The component S of $q(e^{-1}(V))$ used in the construction has a relative algebraic container for the complement of $G = e^{-1}(V)$ in $Q|_S$, furnished by that theorem. The same theorem makes the actual holes $J = (Q|_S) \setminus G$ analytic. The proper map $(q, e) : Q|_S \rightarrow S \times Y$ maps J to an analytic subset of the incidence family, whose fibre at b is exactly $Z_b \setminus V$: every preimage of a point outside V lies in J , and no preimage of a point in V does. Thus $Z_b \cap V$ itself is a Zariski open subset of Z_b , not merely an image of some chosen Zariski open set. All of these assertions concern whole fibres.

After normalization of the parameters, the accessible sets on successively larger balls have marked open kernel P . Every compact subset of P eventually belongs to the accessible set. Their upper limits on bounded sets belong to $\overline{P}$. Boundary points other than those on the artificial spheres lie outside $q(e^{-1}(V))$. The last assertion means that an incidence with a fixed point of V cannot leave the chosen parameter component across a nonartificial boundary. These are the interfaces with the construction and localization sections; no conclusion about a quotient is part of these interfaces.

If $d = 0$, the window construction says that the whole of V is a projective-open variety with analytic holes. Set $M = P$ to be a point and let $f : V \rightarrow M$ be the constant map. The normal-source assertions below follow from μ , and the conditional proper-family conclusions concern maps to a point. The guard, incidence, and inverse-map arguments that follow are needed only when $d > 0$, which we assume until the descent step.

5.3 The tower and its guards

Write the successive symmetric matrices defining P as

$$M_j(Z) = \begin{pmatrix} s_j & w_j^\top \\ w_j & \tau_j(Z_{>j}) \end{pmatrix}, \quad j = 1, \dots, k.$$

Here $Z_{>j}$ consists of the later blocks, τ_j is holomorphic and polynomial, and every coordinate of Z occurs as s_j or an entry of w_j . The defining conditions are $\text{Im } M_j(Z) > 0$ for all j .

Lemma 5.2. *The domain P is diffeomorphic to $\mathbb{R}^{2d}$, is Stein, and is biholomorphic to a bounded domain in $\mathbb{C}^d$. Its Kobayashi distance is a proper distance inducing its usual topology.*

Proof. The first three assertions are Proposition 4.8. For the distance assertion, the map $Z \mapsto (M_1(Z), \dots, M_k(Z))$ is a closed holomorphic embedding into the product of the corresponding Siegel upper half spaces. Closedness follows because the entries s_j, w_j recover Z , and a limit in that product still satisfies every strict matrix inequality. The Cayley transform identifies each Siegel upper half space with the unit ball for the operator norm in symmetric matrices. The Kobayashi distance from its origin is $\text{arctanh } \|W\|$: the radial disc gives the upper bound, and a norming complex linear functional gives the lower bound. Its automorphisms act transitively, so its distance balls are compact. Distance decrease therefore puts a Kobayashi ball of P inside a compact subset of P . Bounded realization implies nondegeneracy, and coordinate balls give the local upper estimates proving that the Kobayashi distance induces the usual topology. The balls are consequently closed in those compact sets, and hence compact. $\square$

Lemma 5.3. *There are normalized accessible parameter regions $A_i \subset \{|Z| < R_i\}$, where $R_i \rightarrow \infty$, and holomorphic functions h on their neighbourhoods, given by the same formula*

$$h(Z) = \exp \left(-\alpha \sum_j \text{tr} \log(I - iM_j(Z)) \right), \quad (22)$$

with the following properties. Their values lie in a fixed sector $\{\zeta : |\arg \zeta| < \theta\}$, $\theta < \pi/2$, and are uniformly bounded. On A_i , $|h(Z)| \leq C(1 + |Z|)^{-\alpha}$. Every compact subset of P is eventually contained in A_i . There are connected open subsets $H_i \subset A_i$, containing the seed and eventually every compact subset of P , such that

$$u_i = \log |h| + a_i |Z|^2 > m_i \quad \text{on } H_i, \quad a_i > 0,$$

where $a_i R_i^2 \leq 1$, $m_i \rightarrow -\infty$, and every boundary of H_i admitting an incidence with a point of V has $|h| \leq \delta_i$, with $\delta_i \rightarrow 0$.

Proof. Choose the scale on the ball of radius $R_i + 1$ so that localization gives $\text{Im } M_j \geq -\eta_i I$, with $\eta_i \downarrow 0$ and $\eta_i < 1/2$, throughout the accessible component. Require also the distance of its points to $\bar{P}$, on each previously chosen bounded ball, to tend to zero. This is a diagonal use of fixed-ball estimates.

For a matrix with $\text{Re } A_j \geq I/2$, the principal matrix logarithm is holomorphic and $|\text{Im tr log } A_j| < n_j \pi/2$, where n_j is its size. Choose $0 < \alpha < 1/(2 \sum_j n_j)$. This proves the sector bound. All singular values of A_j are at least $1/2$; hence

$$|\det A_j| \geq 2^{-n_j}, \quad \|\det A_j\| \geq 2^{-(n_j-1)} \|A_j\|.$$

Since the matrices contain every coordinate of Z , multiplying these inequalities proves the asserted decay estimate.

Let $r_i \downarrow 0$ bound $|h|$ near the artificial sphere. Take $a_i R_i^2 \leq 1$, and choose $m_i \rightarrow -\infty$ with $\log r_i + 1 < m_i$. Let H_i be the seed component of $\{u_i > m_i\}$ in A_i . Any connected compact subset of P containing the seed belongs to H_i eventually, since h has a positive minimum modulus there. At a level boundary $|h| \leq e^{m_i}$; the artificial boundary lies below the level; and the other boundaries admit no incidence with V . Take $\delta_i = e^{m_i}$. The function u_i is strictly plurisubharmonic because h is nowhere zero. $\square$

5.4 Incidence traces and whole fibres

After shrinking B at a generic real centre, the cycle map $b \mapsto Z_b$ is injective. To justify this, if its generic rank were less than d , the union of its r -dimensional cycles would have dimension less than $r + d = n$, contradicting the generic rank of evaluation. The constant-rank theorem then gives an injective restriction. Let

$$\mathcal{I} = \{(b, y) \in B \times Y : y \in Z_b\}.$$

We give it its reduced structure; it is a proper analytic family over B , of pure dimension n , obtained as the proper image of Q . The following argument is made separately at each fixed scale; its bounds need not be uniform in the scale.

Lemma 5.4. *The union $U_i = \bigcup_{b \in H_i} (Z_b \cap V)$ is open in V . Distinct fibres in this union are disjoint, and there is a holomorphic submersion*

$$\pi_i : U_i \longrightarrow H_i, \quad \pi_i^{-1}(b) = Z_b \cap V.$$

These fibres are connected. Moreover, if T is a smooth connected projective variety, $D \subset T$ is a proper analytic subset, and $g : T \setminus D \rightarrow V$ is holomorphic, then any such image meeting a fibre of π_i is contained in that fibre.

Proof. Fix $y \in V$. The set of its incident parameters is analytic. A positive-dimensional irreducible component meeting H_i would have a maximum of u_i greater than m_i : on the artificial boundary $u_i < m_i$, while the other parameter boundaries cannot contain an incidence with y . The part above any intermediate level between m_i and a value on that component is compact. The maximum principle for strictly plurisubharmonic functions on positive-dimensional analytic sets is a contradiction. Thus all incident parameters in H_i are isolated.

Near any one of these roots, the projection of $\mathcal{I}$ to V is proper and finite after choosing a small parameter neighbourhood whose boundary avoids the root. It is open: each local irreducible component of the finite source has dimension n , so its image germ is the full germ of the smooth n -dimensional target. Count the roots with their finite-map multiplicities. The function

$$\Phi_i(y) = \sum_{b: y \in Z_b, b \in H_i} \max(0, u_i(b) - m_i) \tag{23}$$

is bounded and plurisubharmonic on V , with value zero when there are no positive summands. Here and below one may first use a slightly larger high region and put the cutoff strictly inside it. This avoids any finiteness assertion exactly on a level seam. Away from a finite projection's discriminant the assertion is the sum rule for plurisubharmonic functions on holomorphic branches. Across the discriminant the roots remain in a compact parameter set, so the bounded removable extension gives the same trace, including multiplicities. Across a cutoff seam the summands tend to zero, and the plurisubharmonic pasting lemma applies. No upper part can appear without a limiting incidence. Finally, the subanalytic incidence family has a uniform finite bound on its number of isolated roots on this fixed ball. Multiplicity is bounded by the corresponding number of simple roots at nearby generic target points. This proves boundedness of the trace.

A bounded-above plurisubharmonic function on a Zariski open subset of a connected smooth projective variety extends across its analytic complement by [Dem12, I.5.24] and is constant by compactness. In particular Φ_i is constant after pullback to the projective model of any $Z_b \cap V$, $b \in S$, or by any map g in the statement. On a small disc in that model, lift the finite correspondence of

high roots to its normalization and make a finite ramified base change. All its branches then become holomorphic. The trace is a sum of subharmonic functions, and each high branch contributes u_i minus a constant. Since the sum is constant and u_i is strictly plurisubharmonic, each high parameter branch has zero derivative. This also proves constancy at a multiple root, by continuity.

If high fibres $Z_b \cap V$ and $Z_c \cap V$ meet, apply this observation on a projective model of the first fibre. A neighbourhood of the meeting point in that fibre lies in Z_c . Irreducibility and equal dimension give $Z_b = Z_c$, hence $b = c$. The finite incidence projection consequently has one point in each fibre over U_i . Its reduced graph is a finite bimeromorphic map onto a normal space, so it is an isomorphism. This proves holomorphy and openness. Since $\pi_i \circ e = q$ over the high region and q is a submersion, π_i is a submersion at every point: choose any preimage in Q_b . The complement of a proper analytic subset of the connected smooth Q_b is connected and maps onto $Z_b \cap V$; hence that fibre is connected. Finally the argument with g gives a constant parameter near a meeting point. The inverse image of the closed analytic subset $Z_b \cap V$ then contains an open set in the connected source $T \setminus D$, so is the whole source. $\square$

The induced guard $h \circ \pi_i$ has the boundary estimate

$$\limsup_{U_i \ni y' \rightarrow y} |h(\pi_i(y'))| \leq \delta_i \quad (y \in \partial U_i \cap V). \quad (24)$$

Indeed any offending sequence has a subsequence whose parameters converge in the closed artificial ball. Closedness of incidence gives an incidence with y . The limit cannot belong to H_i , which would put y in U_i , and cannot lie on a nonartificial boundary outside $q(G)$. It is therefore one of the low-guard boundaries in Lemma 5.3. This proves the estimate without a bound on tangent directions along the fibres.

5.5 A compactness lemma for cutoff guards

The next lemma allows the domains of the guards themselves to move. The small-boundary assumption is uniform: it applies at every boundary point internal to the test manifold.

Lemma 5.5. *Let D be a complex manifold, let $A_i \subset D$ be open, and let $h_i \in \mathcal{O}(A_i)$ satisfy $|h_i| \leq C$. Suppose that, for every $x \in \partial A_i \cap D$,*

$$\limsup_{A_i \ni z \rightarrow x} |h_i(z)| \leq \epsilon_i, \quad \epsilon_i \rightarrow 0.$$

Extend h_i by zero outside A_i . These extended functions have a subsequence converging uniformly on compact subsets of D to a holomorphic function. Suppose in addition that their nonzero values lie in $|\arg \zeta| \leq \theta < \pi/2$, that D is connected, and that $x_i \rightarrow x$, $h_i(x_i) \rightarrow a \neq 0$. Then every resulting limit is nowhere zero, and every compact subset of D lies in A_i eventually.

Proof. Increase ϵ_i by $1/i$ if necessary so it is positive. First work on a disc compactly contained in D , enlarging it slightly for estimates. Put $\eta_i = 2\epsilon_i$, and extend $v_i = \max(|h_i|, \eta_i)$ by the constant η_i . Pasting gives a uniformly bounded subharmonic function. If χ is a smooth cutoff equal to one on a smaller disc, $\int \chi \Delta v_i = \int v_i \Delta \chi$ gives a uniform bound for its Riesz mass there. On $|h_i| > \eta_i$,

$$\Delta |h_i| = \frac{|h'_i|^2}{|h_i|}$$

with the usual Euclidean Laplacian. It follows that the energy of h_i on this set is bounded, and that

$$\int_{2\eta_i < |h_i| < 4\eta_i} |h'_i|^2 = O(\eta_i).$$

Choose a smooth scalar cutoff λ , zero on $[0, 2]$ and one on $[4, \infty)$, and extend $F_i = \lambda(|h_i|/\eta_i)h_i$ by zero. The extension is locally in $W^{1,2}$: near every internal boundary the formula vanishes, and the preceding energy bounds control the remaining region. Moreover

$$\|F_i\|_{W^{1,2}(D')} \leq C_{D'}, \quad \|\bar{\partial}F_i\|_{L^2(D')}^2 = O(\eta_i), \quad |F_i - h_i| \leq 4\eta_i$$

for the zero extension in the last expression. Rellich compactness and the distributional Cauchy–Riemann equation show that a subsequence converges in L^2_{loc} to a holomorphic function h .

We verify uniform convergence, which does not follow just from this Sobolev bound. Fix an annulus compactly inside the test disc and first pass to a subsequence for which the squared L^2 errors $\|F_i - h\|^2$ on that annulus are summable. Fubini’s theorem gives L^2 convergence of the traces for almost every radius. The integrals in the radial variable of the tangential $W^{1,2}$ energies are uniformly bounded. Fatou’s lemma shows that their lower limit is finite for almost every radius. Choose a radius satisfying both conclusions and then a further subsequence with bounded tangential energies on that circle. Compactness of $W^{1,2}$ of a circle in the continuous functions gives uniform trace convergence; the difference between F_i and the zero extension of h_i tends uniformly to zero as well. If $h(z_0) \neq 0$, start with an arbitrarily small annulus about z_0 on which h takes values in a disc $B(a, r)$ with $0 \notin \overline{B(a, 2r)}$. The preceding radius choice then supplies such a surrounding circle inside this annulus. For large i , the original h_i is defined near that circle and its values belong to $B(a, 2r)$.

For completeness, the argument principle excludes hidden holes inside this circle. Choose a regular level $t_i \in (4\eta_i, 5\eta_i)$ and apply the argument principle to the part of $\{|h_i| > t_i\}$ inside the circle. This part has finitely many boundary components after a harmless choice of the level and outer circle: it is compact in A_i , and the level is real analytic and regular. On each inner boundary, the image has modulus t_i , so its winding number about a value w with $|w| > t_i$ is zero. For $w \notin \overline{B(a, 2r)}$, the outer winding number is also zero. Thus h_i cannot take such a value with $|w| > t_i$ inside. The component adjoining the outer circle cannot reach an inner level boundary: a continuous image from $B(a, 2r)$ to that small circle would pass through an excluded value. It therefore fills the disc, and all its values belong to $\overline{B(a, 2r)}$. Taking arbitrarily small circles and radii proves locally uniform convergence near z_0 .

At a zero of h , choose a surrounding circle with no zeros of h and small maximum modulus, and apply the maximum principle to the pasted v_i . This proves uniform convergence also there. If $h \equiv 0$, the sliced circles and the same maximum principle suffice directly. Hence every sequence has a uniformly convergent subsequence on a smaller disc.

The estimates were uniform for discs of fixed radius lying in a fixed coordinate box. They consequently apply to moving complex lines as well. They imply *asymptotic* equicontinuity of the zero extensions: if it failed, one could choose indices tending to infinity and two converging points with the same limit and a fixed positive difference of values. Apply the one-variable compactness result on the complex lines through these pairs, using fixed-radius parametrized discs in the box, to obtain a contradiction. The zero extensions can still have small jumps at each fixed index, so apply Arzela–Ascoli to the continuous functions $F_i = \lambda(|h_i|/\eta_i)h_i$, defined by the same cutoff formula and by zero in all dimensions. Their difference from the zero extensions is at most $4\eta_i$. Asymptotic equicontinuity, together with continuity of each of the finitely many initial F_i , gives ordinary equicontinuity of this family on compact subsets. Arzela–Ascoli and a countable exhaustion give

local uniform convergence of F_i , and hence of the zero extensions. The limit is holomorphic on every coordinate line, by the disc result, hence holomorphic.

Finally $\operatorname{Re} h \geq 0$, and the moving-point condition gives $h(x) = a \neq 0$. The strong minimum principle gives $\operatorname{Re} h > 0$ on connected D . Its modulus has a positive minimum on every compact set. Uniform convergence then excludes points where the zero extension was used. If compact containment failed for the original sequence, a violating subsequence would have a further uniformly convergent subsequence with the same nowhere-zero conclusion. This proves eventual compact containment for the original sequence as well. $\square$

Corollary 5.6. *Let D_i exhaust a connected complex manifold D , and let $g_i : D_i \rightarrow V$ be holomorphic. Suppose that at points $x_i \rightarrow x$ the images belong to U_i , and their normalized parameters $\pi_i(g_i(x_i))$ converge to the seed $p \in P$. Then every compact subset of D is eventually mapped into U_i ; after taking a subsequence, $\pi_i \circ g_i$ converges normally on D to a holomorphic map into P . The same assertion holds for maps into a manifold equipped with these projection charts and their guards.*

Proof. On each connected relatively compact test domain containing x , the maps g_i are defined for all sufficiently large i . Apply Lemma 5.5 there to $h \circ \pi_i \circ g_i$ on $g_i^{-1}(U_i)$. The boundary estimate is (24), including where the projection chart ceases to exist. At the moving seed the guard tends to $h(p) \neq 0$. A diagonal exhaustion of D therefore gives eventual inclusion in U_i and a nowhere-zero guard limit. Its positive minimum modulus on compact sets, together with the decay in Lemma 5.3, bounds all normalized parameters there. Montel's theorem gives a holomorphic parameter limit F .

To apply localization, join x to a prescribed point of D by a compact path, and precede it by a short path from x_i to x in a fixed coordinate ball. Include these paths in one relatively compact test domain. The preceding bounds hold on the whole paths, whose images eventually lie in U_i . Their parameter paths start at points tending to p and stay in one fixed bounded box. The moving-start assertion of Theorem 3.1 puts their limiting endpoints in $\bar{P}$. Hence $F(D) \subset \bar{P}$. For each constant vector v , $v^* \operatorname{Im} M_j(F)v$ is a nonnegative harmonic function, positive at the seed whenever $v \neq 0$. The minimum principle makes it positive everywhere. Thus all the strict matrix inequalities hold and $F(D) \subset P$. $\square$

5.6 Exhaustion and the inverse maps

Choose a point y_i over the normalized seed of π_i . Cocompactness supplies $\gamma_i \in \Gamma$ and, after passing to a subsequence, points $q_i \rightarrow q \in V$ with $\gamma_i q_i = y_i$. Corollary 5.6, applied to $\gamma_i : V \rightarrow V$, proves that

$$\widehat{U}_i = \gamma_i^{-1} U_i$$

eventually contains every compact subset of V . In particular these sets form a covering, although they need not be nested. Write $\widehat{\pi}_i = \pi_i \circ \gamma_i$.

Any two fibres of these charts, or of any of their deck translates, which meet are equal. Indeed each has a smooth projective model with an analytic complement, and the maximality assertion of Lemma 5.4 gives both inclusions. They therefore partition V into connected entire fibres. Give their set M the quotient topology. Each $\widehat{U}_i$ is saturated and its quotient is biholomorphic to H_i . On overlaps the coordinate change is holomorphic: use a local holomorphic section of the submersion $\widehat{\pi}_i$ and compose with the other projection. Equality of whole fibres makes the result independent of the section. The inverse change is constructed in the same way.

The quotient is Hausdorff. Given two distinct leaves, choose one point on each. They lie together in one sufficiently late $\widehat{U}_i$, which contains both entire leaves. Disjoint parameter neighbourhoods in

that chart have disjoint saturated inverse images, open also in V . The quotient is second countable because the quotient map is open and V is second countable. It is connected and is therefore a complex manifold. We obtain a surjective submersion

$$f : V \longrightarrow M.$$

The deck action descends to M . Let $M_i = f(\widehat{U}_i)$, let $\varphi_i : M_i \xrightarrow{\sim} H_i$ be its coordinate map, and let $\psi_i : H_i \rightarrow M_i$ be its inverse. In particular, for any connected parameter box $B_0 \Subset H_i$, the translated evaluation from $G \cap q^{-1}(B_0)$ maps onto the *entire* saturated set $f^{-1}(\varphi_i^{-1}(B_0))$. It is generically of full rank and comes from the smooth proper projective family $Q|_{B_0}$. Its actual omitted set is analytic and is contained in the specified relative algebraic divisor. The next section applies meromorphic extension to this proper family and uses surjectivity onto each entire leaf.

Every compact subset of M eventually belongs to M_i : finitely many local sections of f lift that compact set to a compact subset of V . Corollary 5.6 and local sections give a normally convergent subsequence

$$\varphi_i \longrightarrow \varphi : M \longrightarrow P.$$

If $b_i = f(q_i)$ and $b_0 = f(q)$, then $b_i \rightarrow b_0$, $\varphi_i(b_i) = p$, and $\varphi_i(b_0) \rightarrow p$.

Lemma 5.7. *For every compact $K \subset P$, the images $\psi_i(K)$ eventually lie in a compact subset of one fixed chart M_j .*

Proof. Enlarge K to a compact set contained in a connected relatively compact domain of P containing p . Suppose the assertion fails. Choose $j_\nu \rightarrow \infty$ and compact sets $L_\nu \subset H_{j_\nu} \cap P$ exhausting P : each fixed compact subset of P belongs to L_ν eventually. Failure means that, for each fixed j_ν, L_ν , there are arbitrarily large i for which $\psi_i(K) \not\subset \varphi_{j_\nu}^{-1}(L_\nu)$. Choose one such i_ν so large that H_{i_ν} contains the ν -th member of a fixed exhaustion of P , that $b_{i_\nu} \in M_{j_\nu}$, and that

$$|\varphi_{j_\nu}(b_{i_\nu}) - \varphi_{j_\nu}(b_0)| < 1/\nu.$$

These last conditions are possible because j_ν is fixed when i_ν is chosen. The displayed starting parameters therefore tend to p .

Test ψ_{i_ν} against the chart M_{j_ν} and its guard. Corollary 5.6 applies on the expanding domains in P . It says that $\psi_{i_\nu}(K) \subset M_{j_\nu}$ eventually, and that their tested parameters lie in one fixed compact subset $L \subset P$, after a subsequence. For large ν , $L \subset L_\nu$, contradicting the choice of i_ν . This proves the fixed-chart assertion; control in merely changing charts would not have sufficed. $\square$

By this lemma and Montel in the fixed charts, a diagonal subsequence of ψ_i converges normally to $\psi : P \rightarrow M$. Passing to the limits in $\varphi_i \psi_i = \text{id}$ and $\psi_i \varphi_i = \text{id}$ is legitimate on compact sets, by both normal convergences. Thus φ and ψ are inverse biholomorphisms. In particular $M \simeq P$, so the complete real holomorphic vector fields of Proposition 4.8 transfer to M and span every tangent space over $\mathbb{C}$.

5.7 Normal-source descent and regularity

Lemma 5.8. *The map f descends uniquely to an equivariant holomorphic map $f_X : \widetilde{X} \rightarrow M$. Its fibres are connected and, locally on M , have simultaneous projective compactifications with analytic holes. Each fibre is biholomorphic to a semialgebraic open subset of a projective variety.*

Proof. Realize M as a bounded domain. Each coordinate of f is constant on every compact connected fibre of μ : a holomorphic function on a compact irreducible complex space is constant, and connectedness makes the constants on its components agree. For completeness, a holomorphic function on the inverse image of a small open set under a proper modification descends on its isomorphism locus and is locally bounded by properness. Normal removability extends it across the omitted analytic subset. Consequently $\mu_*\mathcal{O}_V = \mathcal{O}_{\tilde{X}}$, which gives the unique holomorphic descent. Its values lie in M , and uniqueness gives equivariance. Also $\mu^{-1}(f_X^{-1}(m)) = f^{-1}(m)$, which proves connectedness.

On an original high chart, apply the projective ambient resolution map to the compact incidence family. Its proper image in parameter times the normal ambient compactification is analytic and proper over parameters. Its restriction to $\tilde{X}$ is exactly the graph of the single-valued descended parameter: proper surjectivity of μ and the preceding fibre identity prove both inclusions. Normalize this compactifying family if necessary; normalization does not change the normal graph portion. The graph portion is open, and its fibre complement is analytic and hence algebraic in each projective fibre. Equivalently, before a deck translation each fibre is the intersection of an algebraic image cycle with the original semialgebraic $\tilde{X}$. This also proves the semialgebraic assertion. Translated charts preserve the biholomorphic assertion, without requiring algebraic deck transformations. $\square$

5.8 Preparing the normal compactifying family

To use the normal-source fibres in the Albanese construction, we need normality and flatness at every parameter, a resolution on every fibre, and topological local triviality once the deck action on the base is free and discrete. All three properties will come from the original compact incidence families. We have so far described the quotient for a window at generic real centres. Before making the final choice of centre, scales, and deck translations, impose the following conditions on its full parameter box. The window construction permits these generic shrinkings, and the preceding quotient arguments then apply to the resulting choices.

Let $\mathcal{N} \rightarrow B$ be the normalization of the proper image of $\mathcal{I} \subset B \times Y$ in parameter times the normal ambient compactification, as in the descent proof. Its open graph portion is the corresponding open subset of $\tilde{X}$, and normalization leaves that portion unchanged. The source to be resolved is the reduced incidence space $\mathcal{I}$, rather than its parametrizing space Q . Lemma 5.4 identifies its entire high graph portion with an open subset of V , so this portion is smooth. On these graph portions the ambient resolution induces exactly μ .

Normal fibres and flatness. Shrink the full box so that $\mathcal{N} \rightarrow B$ is flat with normal fibres. The analytic result giving this shrinking is [Gre17, Theorem 6.2(2)(ii), including the properness clause]: for a proper morphism from a normal complex space to a smooth complex space, the image of the locus where the morphism is not normal is a nowhere dense closed analytic subset. Here a normal morphism means flat with normal fibres. Thus this theorem applies directly to $\mathcal{N}$ over the polydisc; no algebraization of the analytic coefficient functions is needed. A proper complex analytic subset cannot contain an open part of the full-dimensional totally real box, so a generic real centre and a neighbourhood of it avoid that image.

A resolution on every fibre. The correspondence from $\mathcal{I}$ to $\mathcal{N}$ induced by the ambient resolution is bimeromorphic: the full-rank evaluation meets the locus where that resolution is an isomorphism. Resolve its source over the full box, leaving the smooth graph portion unchanged.

Normality of the resolved source lifts the map to the normalized target, giving a proper projective map

$$\mathcal{R} \longrightarrow \mathcal{N} \quad \text{over } B.$$

Shrink generically so that $\mathcal{R} \rightarrow B$ is smooth. The exceptional sets of the map and its inverse birational correspondence have smaller dimension than the total spaces. The fibre-dimension theorem and properness allow another generic shrinking so their intersections with every fibre have smaller dimension than that fibre. The map on every fibre is consequently bimeromorphic. On the graph portions it is precisely the original map μ .

Submersivity on the intrinsic strata. Give the absolute space $\mathcal{N}$ its canonical minimal complex Whitney stratification. This stratification is intrinsic and restricts to open subsets; this is Teissier's canonical complex-analytic Whitney stratification [Tei82, VI, §3, Propositions 3.1–3.2]; see also [GFT18, §4.3, following Theorem 4.13]. After a further generic shrinking, the parameter projection is a submersion on every stratum. Here is why the shrinking can be made simultaneously. Over a relatively compact parameter box, properness leaves only finitely many strata to consider. The closure of their critical loci is stratified-critical: if critical tangent spaces approach a lower stratum, Whitney condition (a) puts its tangent space inside their limit, and the limiting rank is still less than the base dimension. Their image is closed by properness. It is a subanalytic set of real dimension less than $2d$, by the rank theorem on the complex analytic strata. In fact the critical image is complex analytic, by the proper-image theorem applied to the analytic closures of the critical loci. To make this last point precise, the canonical complex stratification has analytic stratum closures and frontiers. Form the Nash modification of the analytic closure of each stratum, which records the limiting tangent planes in a Grassmannian bundle. The rank-drop condition on its tautological tangent bundle is analytic, and the Nash projection is proper. Whitney condition (a) gives the asserted containment at lower strata after projecting back. Thus the critical image's intersection with the totally real box has empty interior. Remove that image.

From the prepared family to every fibre. Make these finitely many shrinkings before choosing the diagonal scales in Lemma 5.3 and the deck translations used for exhaustion. On each high chart, the graph portions of $\mathcal{R} \rightarrow \mathcal{N}$ now give the flat normal family, its fibrewise resolution, and the stratum-submersion property. Flatness and normality are invariant under local analytic isomorphisms, as is the absolute Whitney stratification. Deck translation therefore preserves all these properties. The translated high charts exhaust the source and contain entire fibres. Consequently the properties hold at *every* fibre and point of f_X , not only over a generic part of its base. The map f on V was already a submersion; its induced maps after proper quotient are smooth families. The same exhaustion makes μ bimeromorphic on every fibre.

The compactifications used to verify these assertions do not replace the resolution on overlaps. Write $U = \tilde{X}$ in the following equivariance identity. The lifted deck transformations satisfy $\mu \circ \gamma_V = \gamma_U \circ \mu$, so each translated graph portion still carries the same global map $\mu : V \rightarrow U$. In particular, for every subgroup K' used in a finite covering family below, the resulting resolution morphism is the actual quotient map $V/K' \rightarrow U/K'$. Its maps on first cohomology are induced by this single proper morphism, independently of the incidence chart used to verify fibrewise regularity.

Theorem 5.9 (Intrinsic projection). *Under the hypotheses of condition (i) of Theorem 1.1 and the boundary construction of Theorem 4.7, there are a contractible Stein manifold M , biholomorphic to the boundedly realizable tower P , an equivariant surjective submersion $f : V \rightarrow M$, and an*

equivariant holomorphic map $f_X : \tilde{X} \rightarrow M$ with $f = f_X \mu$. Both maps have connected fibres. Their fibres have simultaneous projective compactifications with analytic holes locally on M , after a deck translation. The normal-source fibres are normal and semialgebraically compactifiable up to biholomorphism, and f_X is flat.

Suppose additionally that the image Λ of the deck action on M is discrete and acts freely. Then the descended proper map $X \rightarrow M/\Lambda$, its pullback to M , and every connected finite covering family of that pullback are flat and topologically locally trivial, with connected normal projective fibres. Their induced resolution families are smooth proper projective families; the resolution maps are proper, have connected point fibres, and are bimeromorphic on every fibre. The induced maps on integral degree-one cohomology are morphisms of local systems.

Proof. Only the last paragraph remains. A discrete group acting on a bounded domain acts properly; freeness makes the quotient map a covering. Equivariance descends f_X to a holomorphic map from compact X , hence a proper map. Locally it has the normal graph descriptions above. Flatness and normal fibres follow from those descriptions. The intrinsic Whitney strata downstairs are submersive over the base, since this is true on the covering charts. Thom's first isotopy lemma, in the exact proper stratified-submersion form of [Mat70, Proposition 11.1], gives topological local triviality. Its fibres are connected because upstairs fibres are connected, and projective because they are compact analytic subvarieties of projective X .

Base change to M preserves these conclusions. A finite covering of its total space is locally analytically isomorphic to it; hence flatness, normality, and stratified submersivity persist, and properness follows from finiteness. Apply the same isotopy lemma. Its fibres are connected if the covering total space is connected: the component covering of the simply connected base M is trivial. The fibres are finite covers of projective varieties and are projective. The resolution families are proper submersions, so Ehresmann's theorem makes their integral cohomology locally constant. The normal families are topologically locally trivial by the preceding argument; functoriality of cohomology for the proper fibre maps gives the asserted morphism of local systems. Connectedness of the point fibres and fibrewise bimeromorphicity follow from μ and the generic preparation already propagated through all charts. $\square$

6 The transverse action and the fibre group

We use the intrinsic projection and the full incidence charts supplied by Theorem 5.9. Write

$$p : V \longrightarrow X^+, \quad f : V \longrightarrow M, \quad \rho : \Gamma \longrightarrow \text{Aut}(M), \quad \Lambda = \rho(\Gamma).$$

Here p is a connected regular covering of a smooth projective manifold, with group Γ , and f is an equivariant surjective holomorphic submersion with connected fibres. As in the construction, $\Gamma = \pi_1(X)$: the covering of X^+ is pulled back from the universal covering of the original normal projective variety X . The manifold M is contractible and biholomorphic to a bounded domain. The complete real holomorphic vector fields belonging to $\text{Aut}(M)^0$ have evaluations spanning $T_m M$ over $\mathbb{C}$, for every $m \in M$.

We recall precisely the compactification property used below. Locally in M , there is a smooth projective family $q : Q \rightarrow B$ with connected fibres over a connected box B , a proper analytic subset $D \subset Q$ contained in a relative divisor, and an open incidence locus $G \subset Q$ with

$$Q \setminus D \subset G, \quad e : G \longrightarrow f^{-1}(B), \quad f \circ e = q|_G.$$

After identifying the parameter box with its image in M , the map e is surjective onto the whole of $f^{-1}(B)$ and is generically of full rank. The inclusion $Q \setminus D \subset G$ does not assert equality: the points of $G \cap D$ will be retained. Each individual fibre of f also has a smooth projective compactification with analytic complement. All assertions in this section depend on this full incidence property, not only on a parametrization of a dense open subset of each leaf.

6.1 The core and the external inputs

A smooth compact Kähler manifold Z is *special* if it has no Bogomolov sheaf: there is no line bundle L with a nonzero morphism $L \rightarrow \Omega_Z^r$ and $\kappa(Z, L) = r > 0$. We use the following two consequences of Campana’s core theory [Cam04]. There is a projective modification $r : Z' \rightarrow Z$ and a holomorphic core $c : Z' \rightarrow C$ whose very general smooth connected fibre is special. Its orbifold base is of general type unless C is a point. Moreover, if T is a connected complex manifold, E is a reduced divisor, and $a : T \setminus E \dashrightarrow Z$ is meromorphic, then $c \circ r^{-1} \circ a$ extends meromorphically across E whenever this composition has maximal rank $\dim C$ somewhere. This last statement is the factorized form of the orbifold Kobayashi–Ochiai theorem, namely [Cam04, Theorem 8.2]. It concerns the orbifold core; the underlying variety C need not itself be of general type. If C is a point the extension assertion is immediate.

The other input is the compact special abelianity theorem of OpenAI, *The abelianity conjecture for special compact Kähler manifolds* [Ope26b, Theorem 1.1]:

$$Z \text{ smooth, compact Kähler and special} \implies \pi_1(Z) \text{ virtually abelian.} \quad (25)$$

The conclusion concerns the entire ordinary fundamental group and has no linearity or residual-finiteness hypothesis. We apply it to smooth connected compact special core fibres, and to the smooth compact manifold itself when its core is a point. All further conclusions about the transverse action and its kernel are proved below.

6.2 A Liouville lemma for abelian covers

The following bounded-holomorphic conclusion is a consequence of Lyons–Sullivan’s theorem for nilpotent covers [LS84, Theorem 1, p. 300], after passing to a finite cover of the compact base. We retain a self-contained proof for the abelian case needed here.

Lemma 6.1. *Let $a : W \rightarrow Z$ be a connected regular covering of a compact connected Kähler manifold. If its deck group is virtually abelian, every bounded holomorphic function on W is constant.*

Proof. Choose an abelian subgroup A of finite index in the deck group. The quotient $Z_A = W/A$ is a compact Kähler manifold. Pull a Kähler metric on Z_A back to W , and let $\Delta = \operatorname{div} \nabla$. The pullback metric is complete. Fix $o \in W$ and consider the convex set

$$\mathcal{H} = \{h \in C^\infty(W) : h > 0, \Delta h = 0, h(o) = 1\}.$$

Local Harnack inequalities [Ser64, Theorems 5–7] and interior elliptic estimates [Nir59, Lecture IV, equation (4.2)’ and its consequences] make $\mathcal{H}$ compact in the topology of smooth convergence on compact subsets. In relatively compact metric charts the scalar Laplacian has smooth uniformly elliptic coefficients; its lower-order terms in divergence form vanish, so these estimates apply without an additive Harnack term. Limits are still strictly positive: they are nonnegative harmonic functions equal to one at o , so the strong maximum principle applies.

Let h be an extreme point of $\mathcal{H}$. Its ray is minimal in the cone of positive harmonic functions. Indeed, if $0 \leq v \leq h$ is harmonic and $0 < v(o) < 1$, then

$$h = v(o) \frac{v}{v(o)} + (1 - v(o)) \frac{h - v}{1 - v(o)},$$

so extremality makes v a multiple of h . The endpoint cases follow from the maximum principle.

For each $\alpha \in A$ there is a number C_α such that

$$h(\alpha x) \leq C_\alpha h(x) \quad (x \in W) \quad (26)$$

for every positive harmonic function h . To see the uniformity, choose a compact set $K \subset W$ with $AK = W$. Harnack chains between the two compact sets K and αK give one constant for all positive harmonic functions. If $x = \beta k$, where $\beta \in A$ and $k \in K$, apply this inequality to $h \circ \beta$ and use $\alpha\beta = \beta\alpha$. This proves (26). Minimality of the ray of h now gives $h \circ \alpha = c_\alpha h$ for some $c_\alpha > 0$.

Consequently $\eta = d \log h$ is A -invariant and descends to a smooth one-form on Z_A . Its metric dual satisfies

$$\operatorname{div}(\eta^\sharp) = \Delta \log h = -|\eta|^2.$$

Integrating on Z_A shows that $\eta = 0$. Notice that the differential descends; a single-valued function $\log h$ on Z_A is not required. Thus every extreme point of $\mathcal{H}$ is the constant function one. The Krein–Milman theorem implies $\mathcal{H} = \{1\}$. Finally, the real and imaginary parts of a holomorphic function are harmonic for a Kähler metric. Adding a constant to each bounded real part makes it positive, so both parts are constant. $\square$

6.3 From countable levels to locally finite parameters

The following elementary analytic observation records the exact role of properness. Countability by itself would not give discreteness.

Lemma 6.2. *Let $q : Q \rightarrow B$ be a proper holomorphic map from a complex manifold, let $h : Q \dashrightarrow C$ be a meromorphic map to a projective variety, and let $G_0 \subset Q$ be open with $h|_{G_0}$ holomorphic. Fix $s \in C$. If*

$$q(G_0 \cap h^{-1}(s))$$

is countable, it is locally finite in B .

Proof. Resolve the graph of h , obtaining a proper modification $\tau : \widehat{Q} \rightarrow Q$ and a holomorphic map $\widehat{h} : \widehat{Q} \rightarrow C$. Put $\widehat{q} = q\tau$ and $A = \widehat{h}^{-1}(s)$, with its reduced analytic structure. Over G_0 , equality of meromorphic maps gives $\widehat{h} = h\tau$. In particular every point of $G_0 \cap h^{-1}(s)$ has preimages in $A \cap \tau^{-1}(G_0)$.

Let A_i be an irreducible component of A meeting $\tau^{-1}(G_0)$. On its nonempty relatively open part in this set, $\widehat{q}$ has countable image. A holomorphic map of positive rank on a smooth patch has uncountable image by the holomorphic rank theorem. Thus $\widehat{q}$ has rank zero on the regular locus of that open part and is constant on some nonempty relatively open subset of A_i . The identity theorem on an irreducible reduced complex space then makes $\widehat{q}$ constant on all of A_i . Equivalently, a fibre of $\widehat{q}|_{A_i}$ contains an open subset and hence equals A_i .

For a relatively compact smaller box $B' \Subset B$, $\widehat{q}^{-1}(B')$ is compact. The irreducible components of a complex analytic space are locally finite, so only finitely many A_i meet this compact set. Every attained parameter in B' is the constant value of one of the components meeting $\tau^{-1}(G_0)$. There are therefore only finitely many such parameters. Components lying entirely outside the

valid open set contribute no parameter and need not be excluded from the compact component count. $\square$

Apply the core construction to X^+ . Choose a smooth projective modification $r : Z \rightarrow X^+$ and a holomorphic core $c : Z \rightarrow C$. There is a Zariski open subset $X^\circ \subset X^+$ over which r is an isomorphism, the induced core map is holomorphic, and its restriction has full rank. Shrink this open set if necessary so that very general fibres meet it. For a very general $s \in C$ put

$$F_s = c^{-1}(s), \quad F_s^\circ = F_s \cap r^{-1}(X^\circ), \quad D_s = f(p^{-1}(r(F_s^\circ))) \subset M.$$

The variety F_s is smooth, connected, projective and special.

Lemma 6.3. *For every such s , the set D_s is countable and locally finite in M . The union of these sets, as s runs through very general values, is dense in M .*

Proof. Pull the regular covering p back along $F_s \rightarrow X^+$. Each connected component of the pullback is a regular cover of F_s with deck group the image of $\pi_1(F_s)$ in Γ , up to the basepoint identification. That image is virtually abelian by (25). The composite of the lifted map to V with each bounded coordinate of f is constant by Lemma 6.1. Hence f is constant on each connected component of the pullback. There are at most countably many components, since the fibre of p is countable. This proves countability of D_s .

Fix a full incidence chart $q : Q \rightarrow B$, $e : G \rightarrow f^{-1}(B)$. On $Q \setminus D$ consider

$$a = p \circ e : Q \setminus D \longrightarrow X^+.$$

Replace the containing analytic subset by a reduced divisor if needed. The total space Q is connected. Its full-rank evaluation locus is a nonempty open set; removing the containing divisor leaves a nonempty open subset on which a is submersive. Its image contains an open subset of X^+ and thus meets the dense ordinary core locus X° . At a preimage of such a point, the composition with the core has rank $\dim C$. The extension theorem therefore gives a meromorphic map

$$h : Q \dashrightarrow C, \quad h|_{Q \setminus D} = c \circ r^{-1} \circ p \circ e.$$

On the *full* open incidence locus G , uniqueness of meromorphic continuation identifies this extension with the same meromorphic composition. In particular it is holomorphic and agrees with that composition on

$$G_0 = e^{-1}(p^{-1}(X^\circ)) \subset G.$$

This identity also holds at points of $G_0 \cap D$: they have not been removed from the argument.

Surjectivity of e onto entire leaves gives the exact equality

$$D_s \cap B = q(G_0 \cap h^{-1}(s)). \tag{27}$$

Indeed a point on either side has a representative in the full incidence locus whose image in X° belongs to the ordinary core fibre at s . Lemma 6.2, applied to (27), proves local finiteness. Its proof does not require that additional exceptional sets in G_0 extend to the compactification: G_0 is an open set, and every compactified level component meeting it already has constant parameter. Nor does it require evaluation to be submersive at every representative of an attained parameter.

Very general values omit a countable union of proper analytic subsets of C . On the ordinary submersion locus their inverse images have dense complement, by the Baire theorem in coordinate boxes. The ordinary locus is dense in X^+ ; the covering p and the submersion f are open. Their images therefore show that the union of the D_s is dense in M . When C is a point use the unique level; the countability argument and connectedness already force M to be a point. $\square$

Proposition 6.4. *The image Λ is discrete in $\text{Aut}(M)$ and acts cocompactly on M .*

Proof. The group of automorphisms of a bounded domain is a real Lie group [Kob67, Proposition 3.1 and Theorem 6.2] and acts properly [BC54, §2, Propositions 2–3]. Let H be the closure of Λ in this group. For every very general s , the set D_s is Λ -invariant: deck transformations preserve p , and f is equivariant. A locally finite subset of a manifold is closed, so D_s is also H -invariant by continuity. The orbit under the connected group H^0 of any point of D_s is connected and lies in a discrete set. It is a point. Thus H^0 fixes the dense union of the D_s pointwise and consequently fixes M pointwise. It is the trivial subgroup. As a closed subgroup of a Lie group, H is a Lie group [Car52a, Chapter II, §III, no. 27]; zero-dimensional Lie groups are discrete. This proves the assertion for Λ .

The equivariant surjection f induces a continuous surjection $X^+ = V/\Gamma \rightarrow M/\Lambda$. Its source is compact, so the quotient is compact. $\square$

6.4 Removing stabilizers and identifying the domain

Lemma 6.5. *There is a torsion-free finite-index subgroup $\Lambda_1 \subset \Lambda$ acting freely on M . Its inverse image in Γ corresponds to a connected finite cover of the original normal projective variety X .*

Proof. The group Γ is finitely generated, being a quotient of $\pi_1(X^+)$, and hence so is Λ . Let $\mathfrak{g}$ be the real Lie algebra of $\text{Aut}(M)$. Selberg’s lemma [Sel60, Lemma 8], in the matrix-group form recalled in [Wol68, p. 435], applied to the finitely generated linear group $\text{Ad}(\Lambda) \subset \text{GL}(\mathfrak{g})$ gives a torsion-free subgroup L of finite index. Set

$$\Lambda_1 = (\text{Ad}|_{\Lambda})^{-1}(L).$$

For $m \in M$, properness and discreteness make $(\Lambda_1)_m$ finite. Its image in the torsion-free group L is trivial. If $\lambda \in (\Lambda_1)_m$, therefore, $\lambda_*\xi = \xi$ for every complete vector field $\xi \in \mathfrak{g}$. Evaluating at the fixed point gives

$$d\lambda_m(\xi(m)) = \xi(m).$$

The complex spanning property and complex linearity of the derivative give $d\lambda_m = 1$. A finite-order holomorphic automorphism fixing a point with identity derivative is the identity: if its order is k and z is a local coordinate system centered there, then $k^{-1} \sum_{j=0}^{k-1} z \circ \lambda^j$ is an invariant coordinate system, since its derivative is the identity. The inverse function theorem and analytic continuation prove $\lambda = 1$ on M . Thus all stabilizers are trivial. No faithfulness of the full adjoint representation has been assumed.

The subgroup Λ_1 is also torsion-free. Indeed, lift a finite triangulation of the compact smooth quotient M/Λ_1 to its contractible universal cover. Its cellular chains form a finite-length free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\Lambda_1]$. Restriction to a subgroup of prime order would remain a free resolution of finite length, contrary to the nonzero cohomology in arbitrarily high even degrees of a finite cyclic group. Every nontrivial finite-order element has a power of prime order, so no such element exists.

The subgroup $\rho^{-1}(\Lambda_1)$ has finite index in Γ . The corresponding connected topological covering of X is a finite unramified analytic covering. The proper-base Riemann existence theorem [Gro03, XII, Corollary 4.6] algebraizes it. Pullback of an ample bundle under the resulting finite morphism is ample [The26, Tag 0B5V], so the cover is projective; its analytic covering charts make it normal. This justifies the simultaneous finite-cover replacement. $\square$

Theorem 6.6. *Under the intrinsic-projection hypotheses recalled above, the transverse image is discrete and cocompact. After a finite cover it acts freely, and M is biholomorphic to a bounded symmetric domain, with a point allowed.*

Proof. Discreteness and the free finite-index action were just proved. Replace Λ by the subgroup supplied by Lemma 6.5 and put $B = M/\Lambda$. The intrinsic Bergman kernel of square-integrable canonical forms on the bounded realization of M is everywhere positive, and its logarithm has positive definite complex Hessian. The reciprocal kernel is a Hermitian metric on K_M ; invariance under biholomorphisms makes it descend to a metric on K_B with positive curvature. Consequently K_B is ample by the Kodaira embedding theorem [Kod54, §§1–2; see also §4], B is projective, and $c_1(B) < 0$. In complex dimension one the same positivity-to-ameness step follows from Riemann–Roch; the point case is immediate.

The manifold B is compact and aspherical because its universal cover M is contractible. The Nadel–Frankel splitting theorem in the aspherical form of [FW08, Theorem 1.10] applies. It gives, on universal covers, a holomorphic decomposition

$$M \simeq D_{\text{sym}} \times R,$$

where D_{sym} is a bounded symmetric domain and $\text{Aut}(R)$ is discrete. The cited statement is obtained from a product decomposition of a finite cover of B ; this does not change its universal cover. Its Hermitian locally symmetric factor is of noncompact type, as its canonical bundle is ample, so its simply connected cover has the bounded symmetric realization used here.

We explain why the identity automorphisms of M cannot mix these two factors. The Bergman metric of M descends to compact B , so is complete. On a product, Fubini’s theorem identifies the Hilbert space of square-integrable canonical forms with the Hilbert tensor product of the two factor spaces. Taking product orthonormal bases gives the product formula for the kernels. It follows that

$$g_M = g_{D_{\text{sym}}} \oplus g_R.$$

Both factor metrics are positive definite, since g_M is, and complete, since a factor slice is a closed totally geodesic submanifold of this complete product. Both factors are simply connected, since M is.

There is no Euclidean de Rham factor in M . Otherwise its parallel vector fields span a nonzero parallel real subspace preserved by the parallel complex structure, hence yield a holomorphic Euclidean factor $\mathbb{C}^a$, $a > 0$. Restricting a bounded realization of M to a complex line in this factor contradicts Liouville’s theorem. Uniqueness of the non-Euclidean irreducible de Rham factors [dR52, §7, Théorème III] now implies that every isometry in the identity component preserves each factor distribution, and in particular the two displayed factor foliations. Thus every element of $\text{Aut}(M)^0$ acts as a product automorphism. Its component on R is constant as a function of the group element, because $\text{Aut}(R)$ is discrete; it is the identity. Every vector field from this identity group therefore has zero R -component. The complex spanning property forces $\dim R = 0$. Hence $M \simeq D_{\text{sym}}$. $\square$

6.5 Compactifiable covers and finite-index fibre images

Lemma 6.7. *Let $a : U \rightarrow N$ be a connected regular covering of a smooth connected projective manifold, with deck group K . Suppose U is a Zariski open subset, in the analytic sense, of a smooth projective manifold $\bar{U}$. Then, using (25), the group K is virtually abelian.*

Proof. Choose a smooth projective core model, with C smooth,

$$r : N' \longrightarrow N, \quad c : N' \longrightarrow C.$$

Let $N^\circ \subset N$ be a Zariski open set on which r is an isomorphism and the ordinary core is holomorphic and submersive. The composition $c \circ r^{-1} \circ a$ extends meromorphically to $\overline{U}$ by the core extension theorem: the boundary is analytic, it can be included in a reduced divisor, and a is locally an isomorphism. Resolve this meromorphic map to obtain

$$\tau : \widehat{U} \longrightarrow \overline{U}, \quad h : \widehat{U} \longrightarrow C.$$

The modification can be chosen to be an isomorphism above $a^{-1}(N^\circ)$. Put $\widehat{U}_U = \tau^{-1}(U)$. The complement of $\widehat{U}_U$ is an analytic subset of the compact smooth projective manifold $\widehat{U}$.

Take $s \in C$ very general, also outside the proper analytic sets required for generic smoothness of h and c . The compact level $L = h^{-1}(s)$ has finitely many connected components, each a smooth compact complex manifold. For every component L_i meeting $\widehat{U}_U$, its complement of the analytic boundary is connected. Within this connected complex manifold, the further excluded set

$$(a \circ \tau)^{-1}(N \setminus N^\circ)$$

is analytic. If the component meets the retained ordinary locus, this is a proper analytic subset and its removal is still connected. Here we use the elementary fact that a proper analytic subset of a connected complex manifold does not disconnect it: a path can be perturbed to avoid its real-codimension-at-least-two stratification. The excluded subset need not extend to all of L_i for this argument. Components contained in the excluded set are simply discarded.

It follows that the retained ordinary level

$$L \cap \tau^{-1}(a^{-1}(N^\circ)) \simeq a^{-1}(r(F_s^\circ)), \quad F_s = c^{-1}(s), \quad F_s^\circ = F_s \cap \tau^{-1}(N^\circ), \quad (28)$$

has finitely many connected components. The isomorphism uses the choice that τ is an isomorphism on the ordinary locus. The fibre F_s is smooth, connected, projective and special, and F_s° is the complement of a proper analytic subset of it.

Fix a point of F_s° and a lift to U . The monodromy representation of the regular covering a identifies its fibre with K . If

$$H = \text{im}(\pi_1(F_s^\circ) \longrightarrow K),$$

the connected components of the restricted covering in (28) are the H -orbits in this fibre, equivalently the cosets of H in K . Thus

$$[K : H] = \#\pi_0(a^{-1}(r(F_s^\circ))) < \infty. \quad (29)$$

There is no increase of the relevant monodromy when the compact fibre is replaced by F_s° . Indeed the map to N extends as $r|_{F_s} : F_s \rightarrow N$. Moreover $\pi_1(F_s^\circ) \rightarrow \pi_1(F_s)$ is surjective, by perturbing loops off the proper analytic complement. Hence

$$H = \text{im}(\pi_1(F_s) \longrightarrow \pi_1(N) \longrightarrow K).$$

The group $\pi_1(F_s)$ is virtually abelian by (25); so is its quotient H . An abelian subgroup of finite index in H also has finite index in K , by (29). This proves the lemma.

If the core is a point, one may instead apply (25) directly to N , since N is then special and K is a quotient of $\pi_1(N)$. This also covers dimension zero. $\square$

Theorem 6.8. *After the finite-cover replacement of Lemma 6.5, let $K = \ker(\Gamma \rightarrow \Lambda)$. The group K is finitely generated and virtually abelian. For each $m \in M$, it acts on $f^{-1}(m)$ as the deck group of a connected regular covering of a smooth projective fibre.*

Proof. Since Λ acts freely, the stabilizer in Γ of m is exactly K . The equivariant submersion descends to a holomorphic submersion

$$g : X^+ \longrightarrow B = M/\Lambda.$$

It is proper because X^+ is compact. Its fibre N_b , for the image b of m , is a smooth connected projective manifold. Restricting the covering p gives

$$f^{-1}(m) \longrightarrow N_b$$

as a connected regular covering with group K ; all identifications follow by lifting to a small evenly covered neighborhood in B . The fibre compactifications in Theorem 5.9 give the hypothesis of Lemma 6.7. If the initial compactifying fibre is singular, it is smooth on the open leaf; resolve its boundary while leaving that leaf unchanged. The lemma therefore proves that K is virtually abelian. Finally the monodromy map $\pi_1(N_b) \rightarrow K$ is surjective, because the restricted covering is connected and regular. A compact manifold has finitely generated fundamental group, so K is finitely generated. $\square$

7 Compact fibres and the affine factor

We retain the notation of Theorem 5.9 and Theorem 6.8. Thus $U = \tilde{X}$, the base M is a bounded symmetric domain, and the image Λ of Γ acts freely and properly discontinuously on M , with compact quotient. The kernel K is finitely generated and virtually abelian. Passing to the finite cover used to obtain this free action changes neither U nor the assertion to be proved. Here and below Γ and Λ denote the groups after this replacement, so U/Γ is the chosen finite cover of X .

We first construct, for some integer $a \geq 0$, a proper map from U to $M \times \mathbb{C}^a$ whose fibres are simply connected normal projective varieties. This step passes through relative Albanese tori, which need not be isotrivial. We then prove that the compact fibres are all isomorphic and that their family is a holomorphic product.

7.1 The fibre cover and relative Albanese map

We record precisely which regularity conclusions of the projection construction are needed here. Equivariance identifies the pullback of $U/\Gamma \rightarrow M/\Lambda$ to M with

$$U/K \simeq (U/\Gamma) \times_{M/\Lambda} M \longrightarrow M.$$

This proper family and its connected finite covering families are flat and topologically locally trivial. Every fibre is connected, normal and projective. The simultaneous resolution gives a smooth proper projective family over M mapping to each such covering family, with connected point inverses and fibrewise birational maps. These maps induce the pullback maps of the integral first-cohomology local systems. Finally, the universal cover of every fibre has a semialgebraic projective-open presentation, up to biholomorphism. These are conclusions at *every* point of M . Normality of the total space, or generic normality alone, would not suffice for the argument below.

Lemma 7.1 (The characteristic fibre cover). *There is a characteristic finite-index subgroup $K' \subset K$ which is free abelian. The quotient $\mathcal{X} = U/K'$ is normal and its map $p : \mathcal{X} \rightarrow M$ is proper and flat. It is topologically locally trivial with connected normal projective fibres N_m , and*

$$\pi_1(N_m) \xrightarrow{\sim} \pi_1(\mathcal{X}) = K'.$$

The restriction of $U \rightarrow \mathcal{X}$ to N_m is its connected universal cover.

Proof. Choose a finite-index abelian subgroup of K . It is finitely generated, and hence contains a finite-index torsion-free subgroup B . Put $d = [K : B]$, and intersect all subgroups of K of index at most d . There are finitely many such subgroups, since homomorphisms from a finitely generated group to each of the finite symmetric groups form a finite set. Their intersection K' is characteristic, has finite index, and is contained in B . It is therefore free abelian. Since K is normal in Γ , so is K' .

The map $\mathcal{X} \rightarrow U/K$ is a finite covering. Thus the stated properness and regularity follow from the projection contract. As U is simply connected, its quotient by K' has fundamental group K' . A locally trivial bundle over the contractible manifold M is a homotopy equivalence on each fibre; alternatively, its homotopy exact sequence gives the displayed isomorphism. The covering induced on a fibre consequently corresponds to the zero subgroup of its fundamental group, and is connected. $\square$

Lemma 7.2 (Relative Albanese descent). *There is an integer $a \geq 0$ with $K' \simeq \mathbb{Z}^{2a}$ and a smooth proper family of Albanese torsors $q : \mathcal{A} \rightarrow M$ and a proper morphism*

$$\alpha : \mathcal{X} \longrightarrow \mathcal{A}$$

over M whose restriction to N_m is its Albanese morphism. Both constructions are equivariant for Γ/K' . The morphisms $\alpha|_{N_m}$ and α induce isomorphisms on fundamental groups.

Proof. Let $r : \mathcal{Y} \rightarrow \mathcal{X}$ be the simultaneous resolution and let $p^+ : \mathcal{Y} \rightarrow M$ be its smooth proper projective structure map. First, for each m , the map $r_m : Y_m \rightarrow N_m$ is surjective on fundamental groups. Indeed, lift it to the connected covering of N_m corresponding to the image of $\pi_1(Y_m)$. Each connected point inverse of r_m maps to one point of the discrete covering fibre. Since a proper surjection is a quotient map, the lift descends to a continuous section of this covering. A connected covering with a section has one sheet. This proves the assertion.

It follows that

$$W_{\mathbb{Z}} := R^1 p_* \mathbb{Z} \hookrightarrow R^1 p_*^+ \mathbb{Z} =: H_{\mathbb{Z}} \quad (30)$$

is an inclusion of local systems, with saturated image. Here the local-system assertion uses the topological local triviality specified above, and its compatibility with r uses the map of the actual families. Saturation also follows directly: a homomorphism $\pi_1(Y_m) \rightarrow \mathbb{Z}$ whose nonzero integral multiple vanishes on $\ker(r_{m*})$ itself vanishes on that kernel.

We use the normal-projective Albanese theorem in its integral form: if N is normal and projective, then

$$H_1(N, \mathbb{Z}) \longrightarrow H_1(\text{Alb}(N), \mathbb{Z})$$

is surjective with finite kernel [CHK13, Lemma 4.7]. In the present situation $H_1(N_m, \mathbb{Z}) = K'$ is torsion-free, so this map is an isomorphism. In particular its rank is $2a$. Moreover, the image of $H^1(N_m, \mathbb{Q})$ in $H^1(Y_m, \mathbb{Q})$ is a Hodge substructure: it is exactly the pullback from $\text{Alb}(N_m)$ under the algebraic map $Y_m \rightarrow N_m \rightarrow \text{Alb}(N_m)$. This observation also derives the required purity without an assumption about the singularities being rational.

Let $F^1 H_{\mathcal{O}}$ be the holomorphic Hodge subbundle of the weight-one variation of the smooth projective family p^+ . Holomorphicity holds over this analytic base by Griffiths's period-mapping theorem [Gri68, II, Theorem (1.1)]. The intersection

$$W^{1,0} = W_{\mathcal{O}} \cap F^1 H_{\mathcal{O}}$$

is a holomorphic subbundle: it is the kernel of the holomorphic map $W_{\mathcal{O}} \rightarrow H_{\mathcal{O}}/F^1 H_{\mathcal{O}}$ and has the constant rank a . Together with $W_{\mathbb{Z}}$ it defines a polarizable weight-one variation. The lattice dual to $W_{\mathbb{Z}}$ embeds by evaluation in $E = (W^{1,0})^{\vee}$, and the quotient gives a smooth family $E/W_{\mathbb{Z}}^{\vee}$ of abelian varieties, by the integral analytic Jacobian construction [Cla18, §2.1]. Fibrewise the lattice is a full lattice; continuity in a local basis makes the quotient a holomorphic torus family, with no change of lattice index.

On a small base ball, choose a section of the smooth family p^+ . The relative Albanese map of $\mathcal{Y}$ based at that section, followed by the quotient specified by W , is a holomorphic map to this torus family; see the smooth construction in [Fuj83, §1.4] and [Gri68, II, §2(a) and Example (2.15)]. Equivalently, integrate the relative holomorphic one-forms in $W^{1,0}$. Its map on integral first homology is the quotient induced by r_m , so its restriction to every point inverse of r_m has trivial induced fundamental-group map. Each compact connected point inverse therefore lifts to the vector-space cover of the target torus. Holomorphic functions on a compact connected reduced complex space are constant, so this restriction is constant.

Thus the relative map is constant on the fibres of r . Since $\mathcal{X}$ is normal and r is a proper modification, $r_* \mathcal{O}_{\mathcal{Y}} = \mathcal{O}_{\mathcal{X}}$; the map descends holomorphically to $\mathcal{X}$. Changing the chosen local section changes only the origin in the torus family. The resulting local maps glue to the family of Albanese torsors $\mathcal{A}$ and to α , as in the translation-cocycle description of [Cla18, §2.2, Proposition 2.1]. The fibrewise universal property identifies this map with the normal Albanese morphism, which is algebraic on every projective fibre. An element of Γ/K' acts on the intrinsic system $W_{\mathbb{Z}}$ and its Hodge subbundle, and thus induces a holomorphic linear map of the torus families. In charts with origins, the additional translation is obtained by evaluating the relative Albanese map at the image of the chosen local section; it is holomorphic. The fibrewise universal property makes these maps agree on overlaps and obey the group law. This gives the asserted Γ/K' -equivariance of the torsor construction.

Properness of α follows from properness of p : a compact subset of $\mathcal{A}$ projects into a compact subset of M , and its inverse image is closed in the compact inverse image under p . The fibrewise fundamental-group assertion follows from the integral Albanese statement and $\pi_1(N_m) = K'$. The smooth proper torus family q is topologically locally trivial. Since M is contractible, its fibres and those of p induce isomorphisms on fundamental groups. The assertion for α follows from the commutative diagram of these two fibre inclusions. $\square$

7.2 The universal affine base

Proposition 7.3 (The proper map to the affine base). *There is a Γ -equivariant proper flat surjection*

$$j : U \longrightarrow T, \quad T \simeq M \times \mathbb{C}^a,$$

whose fibres are connected, simply connected, normal projective varieties. The action of Γ on T is proper and cocompact and has finite stabilizers. A polarization of the finite cover of X pulls back to a Γ -invariant line bundle L on U , ample on each fibre of j .

Proof. We apply the following precise form of the Kollár–Pardon theorem [KP12, arXiv version 2, Theorem 20]: if N is normal projective, B is smooth projective with contractible universal cover,

and $g : N \rightarrow B$ is a morphism such that the *entire* pullback to that cover is biholomorphic to a semialgebraic open subset of a projective variety, then g is a holomorphic fibre bundle. Surjectivity is part of its conclusion; connected fibres are not a hypothesis or an automatic part of the general statement.

Apply this to $N_m \rightarrow \text{Alb}(N_m)$. The target is an abelian variety with universal cover $\mathbb{C}^a$. By Lemma 7.2, the pullback cover is connected and is precisely the universal cover of N_m . The required semialgebraic presentation is therefore the one supplied by the projection construction. Each Albanese map is a holomorphic fibre bundle and is flat. Its fibres are connected: in the homotopy sequence of a bundle, surjectivity on π_1 implies that π_0 of the fibre is a singleton. Since π_2 of a torus is zero and the map on π_1 is an isomorphism, the same sequence shows that each fibre is simply connected. Local holomorphic products with a ball and normality of N_m show that the fibre is reduced and normal. It is projective, being an algebraic fibre of a morphism of projective varieties. These arguments include $a = 0$, when the Albanese map is the map to a point.

The fibrewise flatness criterion now applies to $\mathcal{X} \rightarrow \mathcal{A} \rightarrow M$: the source is flat over M , the target is smooth over M , and the maps on all fibres over M are flat. More explicitly, for the analytic local rings $R \rightarrow B \rightarrow C$ at m , $\alpha(x)$ and x , respectively, B and C are flat over R , and $C/\mathfrak{m}_R C$ is flat over $B/\mathfrak{m}_R B$. The fibrewise flatness criterion for Noetherian local rings [The26, Tag 00MP] applies with module C , which is finite free over itself, and implies that C is flat over B . Thus α is flat at every point.

Let $T \rightarrow \mathcal{A}$ be its universal covering. The pullback $\mathcal{X} \times_{\mathcal{A}} T$ is connected and simply connected, by the isomorphism on fundamental groups in Lemma 7.2. It is consequently U , and its projection j is proper and flat by base change. It is surjective by the fibrewise surjectivity just proved. Its fibres have all the asserted properties.

We explain the structure of T without requiring the torus family to be isotrivial. On a small contractible ball $B \subset M$, choose an origin for $\mathcal{A}_B$. Relative exponential identifies its fibrewise universal cover with the vector bundle E_B . The inclusion of a fibre in $\mathcal{A}$ induces an isomorphism on π_1 , since M is contractible. Hence the restriction of the global universal cover T to $\mathcal{A}_B$ is connected and is exactly this cover $E_B \rightarrow \mathcal{A}_B$. On overlapping balls, changes of origin lift to translations, while the linear terms are the transition maps of the relative Lie bundle E . Thus $T \rightarrow M$ is an affine bundle with linear part E . Its translation cocycle lies in $H^1(M, \mathcal{O}(E)) = 0$ by Cartan's theorem B [Car52b, Exposé 18, §4]. It follows that $T \simeq E$. The vector bundle E is topologically trivial because M is contractible, and holomorphically trivial by the Oka principle [Gra58, FP00]. Thus $T \simeq M \times \mathbb{C}^a$, which is Stein, contractible and homeomorphic to Euclidean space.

The functorial action on $\mathcal{A}$ lifts to T , since T is its universal cover. Choose a lift γ_T whose value at one point agrees with $j \circ \gamma$. Then $\gamma_T \circ j$ and $j \circ \gamma$ are lifts from the connected space U of the same map to $\mathcal{A}$ and agree at one point, so they agree everywhere. Surjectivity of j then gives uniqueness of γ_T and the group law. If a compact $C \subset T$ meets γC , then the compact set $j^{-1}(C)$ meets its γ -translate. The deck action on U is proper, so only finitely many γ have this property. This proves properness on T , including finite stabilizers and a possible finite kernel. The induced surjection $U/\Gamma \rightarrow T/\Gamma$ shows that the latter quotient is compact. Finally, pull back an ample line bundle from U/Γ . It is Γ -invariant. On a compact fibre its map to U/Γ is finite onto its image, being proper and locally injective, so this pullback is ample on the fibre. $\square$

7.3 Variation of the compact fibres

It remains to show that the compact fibres of j are all polarized-isomorphic. We shall put their parameter space T in a projective Chow space and show that each polarized isomorphism class

is a semialgebraic, Γ -invariant subset. The following topological lemma will force its closure to fill T ; the semialgebraic description will then rule out two distinct dense classes. Once all fibres are polarized-isomorphic, relative Hilbert embeddings and the Oka principle give a holomorphic product.

Lemma 7.4 (Rigidity for invariant closed subsets). *Let a countable discrete group act properly and cocompactly by diffeomorphisms on a contractible smooth manifold T homeomorphic to $\mathbb{R}^b$. If a nonempty closed invariant subset $A \subset T$ has the homotopy type of a finite CW complex, then $A = T$.*

Proof. We give the controlled proper-homotopy argument, including the case of finite stabilizers; compare [KP12, §1, Lemmas 11–12 and Proposition 13]. A proper cocompact smooth action admits an invariant complete Riemannian metric. Its distance d is proper. Fix $o \in T$ and $R > 0$ such that the balls $B(\gamma o, R)$ cover T . Choose continuous functions $\lambda_\gamma : T \rightarrow [0, 1]$ equal to one on these balls and zero outside $B(\gamma o, R + 1)$, obtained by translating one function of distance. Properness makes their supports locally finite. Cocompactness and properness also give a uniform bound N on the number of nonzero terms at any point. Indeed translate that point into a fixed compact set and count orbit centres in its compact $(R + 1)$ -neighbourhood. Finite stabilizers merely give finite repetitions in this count.

Choose a homotopy $H : A \times [0, 1] \rightarrow A$ with $H_0 = \text{id}$ and $H_1(A)$ contained in a compact subset $C \subset A$. Such a homotopy is obtained by factoring a homotopy equivalence through a finite CW complex. Order the countable group once and for all. For $y \in T$, apply, in that order, the finitely many maps

$$a \mapsto \gamma H(\gamma^{-1}a, \lambda_\gamma(y))$$

to a fixed point $a_0 \in A$. Call the resulting point $s(y)$. Local finiteness and $H_0 = \text{id}$ imply that s is continuous, including where a factor enters or leaves its support.

At least one factor has time one. Immediately after that factor the point lies in γC , at uniformly bounded distance from y . Subsequent factors preserve a uniform distance bound. In detail, if $d(a, y) \leq B$ and $\lambda_\delta(y) \neq 0$, then $d(\delta^{-1}a, o) \leq B + R + 1$. The homotopy of the compact set $A \cap \overline{B}(o, B + R + 1)$ has compact image, so the distance of $\delta H(\delta^{-1}a, t)$ from y has a bound depending only on B . Iterating at most N times gives

$$d(s(y), y) \leq B_0$$

for a fixed B_0 . In particular $s : T \rightarrow A$ is proper.

For completeness, bounded distance from the identity gives a proper homotopy in this situation as follows. Choose a contraction $J : T \times [0, 1] \rightarrow T$ from the identity to o . Form $Q(y, x, t)$ by composing the translated contractions

$$x \mapsto \gamma J(\gamma^{-1}x, t\lambda_\gamma(y))$$

in the same fixed order. At $t = 1$ a full contraction occurs, so the output is independent of x . If $d(x, y)$ is bounded, all the intermediate outputs stay at a uniformly bounded distance from y ; the compact-set argument above applies now also before the first full contraction. Join y to $Q(y, y, 1)$ by this construction and return along the construction starting at $s(y)$. The two middle points agree. This is a continuous proper homotopy from id_T to the composite $T \xrightarrow{s} A \hookrightarrow T$.

If $b = 0$ the conclusion is immediate. Otherwise, if $x \notin A$, the latter proper map omits x . Its extension to the one-point compactification $T^+ \simeq S^b$ therefore has degree zero, whereas the proper homotopy to the identity gives degree one. This contradiction proves the lemma. No assertion that $T \rightarrow T/\Gamma$ is a covering has been used. $\square$

Lemma 7.5 (The semialgebraic space of fibres). *Fix a semialgebraic realization $U \subset Z$, where Z is projective, and a projective embedding of Z . Let n be the fibre dimension of j and d the degree of one fibre. In the projective Chow space $\text{Chow}_{n,d}(Z)$, the locus*

$$C = \{c : c \text{ is an integral cycle and } |c| \subset U\}$$

is semialgebraic, and the map $t \mapsto [j^{-1}(t)]$ is a homeomorphism from T onto C .

Proof. A proper flat family of pure-dimensional compact subspaces over a reduced complex base defines an analytic family of fundamental cycles; see [Bar99, p. 27, Remark (iv)]. Its map to the projective Chow space is continuous. Indeed, view cycles in Z as cycles in a fixed smooth projective space. Support and integration continuity [AN67, §2.4(b), Lemma 4 and §2.5, Theorem 4] identify the usual Chow topology with the analytic cycle topology: the natural map is a continuous bijection from the compact fixed-degree Chow variety onto the corresponding fixed-degree locus in the Hausdorff cycle space, hence a homeomorphism. This comparison uses no smoothness of Z . Apply this to the graph of j in $T \times Z$. Every fibre is connected and normal, hence irreducible and reduced, so its fundamental cycle is integral. The degree is locally constant in this family, and hence equals d on the connected base T .

The integral-cycle locus is constructible. More explicitly, its complement is the finite union, over $d_1 + d_2 = d$ with both degrees positive, of the images of the proper addition maps from the corresponding Chow spaces. These images include nonreduced cycles. Containment of the support in U is a semialgebraic condition by the algebraic universal incidence and real quantifier elimination. Thus C is semialgebraic.

Any compact irreducible analytic subvariety of U maps to a point of the Stein manifold T : holomorphic functions are constant on that subvariety, and holomorphic functions on a Stein manifold separate points. An integral n -cycle in U therefore lies in a fibre and, by equality of dimension and irreducibility of that fibre, is its whole reduced cycle. This proves that the displayed cycle map is bijective. Its continuity follows from the analytic-family statement. For inverse continuity, if fibre cycles c_ν converge to the cycle of $j^{-1}(t)$, choose a regular point x of that fibre. Convergence of cycles supplies points $x_\nu \in |c_\nu|$ tending to x ; for example, use a small transverse disk at x , whose local intersection number is one. Then their parameters satisfy $j(x_\nu) \rightarrow j(x) = t$. All spaces involved are metrizable, so this proves inverse continuity. $\square$

Lemma 7.6 (Integral marking of polarized isomorphism classes). *For every fixed polarized fibre (F, L_F) of j , the locus of fibres polarized-isomorphic to it is semialgebraic in C .*

Proof. Write $A_Z = \mathcal{O}_Z(1)|_U$ and choose $k > 0$ for which $H = L_F^k$ is very ample. If $g : F \rightarrow j^{-1}(t)$ is a polarized isomorphism, its graph in $F \times Z$, with the product projective embedding defined by $H \boxtimes \mathcal{O}_Z(1)$, has degree

$$D = \int_{j^{-1}(t)} (kc_1(L) + c_1(A_Z))^n. \quad (31)$$

This integer is independent of t . Here is a direct way to check the constancy, which does not extend L to Z . Choose a smooth Hermitian metric on L on the complex space U , and use its curvature form, together with a Fubini–Study form for A_Z . Near any fixed parameter, properness confines the supports of the fibre cycles to a compact subset of U . Integration of the resulting smooth $2n$ -form over these cycles is continuous in their analytic cycle family. Each value is the integral intersection number in the displayed formula, so it is locally constant. Smooth forms on a complex space here mean forms locally extended from an ambient manifold; a finite partition of unity on the compact set gives the usual continuity-of-integration test. The Chern integral

equals the Cartier intersection number also for a singular fibre: pull the line bundles back to a projective resolution and use its degree-one pushforward of the fundamental cycle. The same argument proves constancy of each mixed intersection separately.

Consider the integral-cycle locus in the projective Chow space of n -cycles of degree D in $F \times Z$, and let P be its locus of cycles G such that

$$|G| \subset F \times U, \quad p_F : |G| \rightarrow F \text{ is bijective,} \quad p_Z : |G| \rightarrow U \text{ is injective.}$$

All three conditions are first-order conditions on the universal incidence, and hence are semialgebraic. The first projection is a proper quasi-finite algebraic map, hence finite. It is birational, since in characteristic zero a finite map bijective on complex points has degree one. Its target F is normal, so it is an isomorphism. The second projection has an irreducible compact image of dimension n , which by the preceding lemma is a whole fibre of j . This image is normal, and the same finite birational argument shows that the second projection is an isomorphism onto that fibre. Consequently P parametrizes exactly the isomorphisms from F to fibres whose graphs have the specified degree.

The universal incidence over P maps bijectively and closedly to $F \times P$: closedness follows by projection along the compact factor Z . It is therefore homeomorphic to $F \times P$. In particular it gives a continuous evaluation map

$$e : F \times P \longrightarrow U.$$

Connected semialgebraic sets are path connected, and P has only finitely many connected components. A path in one such component gives a homotopy of the maps $e_p : F \rightarrow U$. Hence $c_1(e_p^*L) \in H^2(F, \mathbb{Z})$ is constant on each component. This is the integral marking test; neither definability of L nor an extension of L to the compactification has been assumed.

For a simply connected normal projective variety F , the map

$$c_1 : \text{Pic}(F) \longrightarrow H^2(F, \mathbb{Z})$$

is injective. Indeed $\text{Pic}^0(F)$ is projective for normal projective F and is smooth in characteristic zero [Kle05, Theorem 5.4 and Remark 5.6]. GAGA identifies algebraic and analytic line bundles on F [Ser56, Proposition 18]. The exponential sequence and $H^1(F, \mathbb{Z}) = 0$ identify $\text{Pic}^0(F)$ analytically with the vector group $H^1(F, \mathcal{O}_F)$. A compact complex vector group is zero. The kernel of c_1 , which is the image of this vector group in the exponential sequence, is therefore zero.

It follows that the condition $e_p^*L \simeq L_F$ selects a union P_L of connected components of P , so P_L is semialgebraic. Taking the second image of a graph gives a semialgebraic map to C . One can see this directly by expressing equality of supports as

$$z \in |c| \iff (\exists x \in F) (x, z) \in |G|;$$

integrality makes this support determine the cycle. The image of P_L is exactly the requested polarized isomorphism class, including all its members by the graph-degree calculation. Quantifier elimination proves the assertion. $\square$

Proposition 7.7 (Triviality of the compact factor). *There is a simply connected normal projective variety F such that*

$$\tilde{X} \simeq M \times \mathbb{C}^a \times F.$$

Proof. By Lemma 7.6, every polarized isomorphism class is a semialgebraic subset of the semialgebraic model C of T . It is Γ -invariant, since the polarization is pulled back from U/Γ . Its closure

in C is a nonempty closed invariant semialgebraic subset. Compatible semialgebraic triangulation [Lo64, §3, Theorem 4] gives that closure the homotopy type of a finite CW complex. Under the homeomorphism of Lemma 7.5, Lemma 7.4 therefore forces it to be all of T . Thus every class is dense. Two disjoint semialgebraic subsets cannot both be dense in the same semialgebraic space: finite cell decomposition shows that a dense semialgebraic subset contains a dense relatively open subset. All fibres are consequently isomorphic as polarized varieties.

We next justify local holomorphic triviality for these possibly singular fibres. Choose r so large that L_F^r is very ample and $H^i(F, L_F^r) = 0$ for $i > 0$. The same holds on every fibre. The constant-cohomology base-change theorem [Gra60, §7, Satz 5] makes $j_* L^r$ locally free, with its formation commuting with fibres: L^r is flat over the base and all its fibre cohomology dimensions are constant. After trivializing it on a small open set $B \subset T$, the evaluation map embeds $j^{-1}(B)$ in $B \times \mathbb{P}^q$ and gives a holomorphic Hilbert map $h : B \rightarrow \text{Hilb}(\mathbb{P}^q)$. Here one uses the analytic-base formulation: Douady's universal proper-flat family [Dou66, §§9.7–9.8, Theorems 1–2] agrees with the analytification of the projective Hilbert scheme [Gro61, §3, Theorem 3.2]. The natural map between them has the same points and the same local deformation functors, since GAGA applies on every Artinian complex base [Gro03, Exposé XII, Theorem 4.4]; thus it is an isomorphism of analytic germs. This supplies the holomorphic classifying map also for singular fibres. All its points lie in the single orbit of the embedded F under $G_0 = \text{PGL}_{q+1}(\mathbb{C})$.

This orbit is a locally closed smooth complex subvariety. Since B is reduced, a holomorphic map with image in it factors holomorphically through it. For its closed algebraic stabilizer $G \subset G_0$, the quotient map $G_0 \rightarrow G_0/G$ has local holomorphic sections, by the holomorphic submersion theorem. Locally lifting h and applying the inverse projective transformations gives a holomorphic product $B \times F$. The resulting transition maps take values in the complex linear algebraic group G . Thus j is the bundle associated to a holomorphic principal G -bundle on T .

Since T is contractible and paracompact, that principal bundle is topologically trivial. The Oka principle for complex Lie groups over a Stein base makes it holomorphically trivial; one may use [FP00, Theorem 1.3 and Example (A)] on its continuous trivializing section. If necessary, first reduce to the identity component of G using the triviality of the discrete covering G/G° over the simply connected space T . We obtain $U \simeq T \times F$. Proposition 7.3 gives the required product, with all the stated properties of F . $\square$

Proposition 7.8 (The converse). *If $\tilde{X} \simeq D \times \mathbb{C}^m \times F$, with D a bounded symmetric domain and F projective, then $\tilde{X}$ is biholomorphic to a semialgebraic open subset of a projective variety.*

Proof. In the Borel embedding, D is an open orbit in its projective compact dual $D^\vee$ [Bor54, §4, Theorem 2 and §6, Proposition 2]; see also [FGW25, §2.1.1, Borel embedding theorem]. The acting adjoint real semisimple group is the identity component of the real points of a real algebraic group [Mil05, Proposition 1.7]; it is semialgebraic, and its action on $D^\vee$ is algebraic. The image of its orbit map is semialgebraic by real quantifier elimination. Thus $D \subset D^\vee$ is a semialgebraic open subset. Taking the usual affine chart $\mathbb{C}^m \subset \mathbb{P}^m$ gives

$$D \times \mathbb{C}^m \times F \subset D^\vee \times \mathbb{P}^m \times F$$

as a semialgebraic open subset of a projective variety. Point factors are allowed. This construction concerns the biholomorphism type of the covering space and imposes no algebraicity condition on deck transformations. $\square$

8 Assembly of the classification

Starting from condition (i) of Theorem 1.1, take a functorial projective resolution of the normal uniformizing model. The quotient construction of Theorem 5.9 gives the intrinsic transverse domain and its equivariant projection, including the descended normal-source family. Theorem 6.6 proves that the transverse deck action is discrete, makes it free after the indicated finite-index passage, and identifies its simply connected domain with a bounded symmetric domain. The fiber kernel is virtually abelian by Theorem 6.8. Proposition 7.7 then separates the affine and compact factors and proves condition (ii), with $D = M$ and $m = a$ in its notation. The converse is Proposition 7.8. Every passage to finite index preserves the universal cover of X .

Corollary 8.1 (Remmert reduction of semialgebraic covers). *Let X be a connected normal projective complex variety whose ordinary universal cover U has a semialgebraic projective-open presentation. Write $U \simeq B \times F$, where $B = D \times \mathbb{C}^m$, as in Theorem 1.1. Then U is holomorphically convex and the projection $p : U \rightarrow B$ is its Remmert reduction: B is Stein, p is proper with connected compact normal projective fibres, and the canonical map $\mathcal{O}_B \rightarrow p_*\mathcal{O}_U$ is an isomorphism. Moreover, U is Stein if and only if F is a point.*

Proof. The base B is the Stein base T in the proof of Proposition 7.3. The factor F is connected, normal and projective, possibly singular. Since F is compact, p is proper and surjective. Holomorphic functions on F are constant by [Dem12, Corollary II.5.8]. Fix $z_0 \in F$. For every open $V \subset B$ and $h \in \mathcal{O}_U(V \times F)$, fibrewise constancy gives $h(b, z) = h(b, z_0)$. Restriction along the holomorphic section $b \mapsto (b, z_0)$ therefore inverts pullback, proving $\mathcal{O}_B \simeq p_*\mathcal{O}_U$; equality on points suffices because U is reduced. For every compact $K \subset U$, this gives

$$\widehat{K}_{\mathcal{O}(U)} = p^{-1}(\widehat{p(K)}_{\mathcal{O}(B)}).$$

The right side is compact since B is Stein and p is proper. Thus U is holomorphically convex, and the proper Stein-target map with this sheaf identity is its Remmert reduction. If F is a point, then $U \simeq B$ is Stein; if F has two distinct points, global holomorphic functions on U cannot separate the corresponding points of a fibre, so U is not Stein. A connected zero-dimensional normal projective variety is a single reduced point, so this criterion includes the zero-dimensional case. $\square$

Corollary 8.2 (Singer vanishing for semialgebraic covers). *Let X be a smooth connected closed aspherical projective complex n -fold whose ordinary universal cover has a semialgebraic projective-open presentation. Its L^2 -Betti numbers, computed from the reduced L^2 -cohomology of the universal cover, satisfy*

$$b_j^{(2)}(X) = 0 \quad (j \geq 0, j \neq n).$$

If $m > 0$ for the affine factor in the product constructed in the proof of Theorem 1.1, then $b_j^{(2)}(X) = 0$ for every $j \geq 0$.

Proof. The case $n = 0$ is immediate. Since $\widetilde{X}$ is contractible, so is F : the other two factors are contractible. A positive-dimensional normal projective variety has a nonzero top fundamental class. Hence F is a point and $\dim_{\mathbb{C}} D = n - m$.

Put $\Gamma = \pi_1(X')$ for the finite cover $X' \rightarrow X$ used in Section 7. Lemmas 7.1 and 7.2, with $m = a$ as above, give $K' \triangleleft \Gamma$ with $K' \simeq \mathbb{Z}^{2m}$. If $m > 0$, this is an infinite normal amenable subgroup, so the Cheeger–Gromov theorem [CG86, Theorem 0.3(3) and the proof of Corollary 0.6]

makes every group L^2 -Betti number of Γ vanish. Asphericity identifies these with $b_j^{(2)}(X')$, and finite-cover proportionality gives $b_j^{(2)}(X') = [\pi_1(X) : \Gamma] b_j^{(2)}(X)$, proving the claim.

If $m = 0$, then $\tilde{X} \simeq D$ has real dimension $2n$. As in the proof of Theorem 6.6, its Bergman metric is complete and symmetric of noncompact type. The deck group acts freely and cocompactly by Bergman isometries. Olbrich's theorem on the universal symmetric space [Olb02, Proposition 1.2 and §4] gives $\ker \Delta_j(D) = 0$ for $j \neq n$. The cocompact L^2 -de Rham identification of $b_j^{(2)}(X)$ with the von Neumann dimension of this harmonic kernel completes the proof. $\square$

9 Quasi-projective covering spaces

For a quasi-projective universal cover, one can also describe a finite cover of the projective quotient. Claudon, Höring and Kollár proved this refinement, and a structure theorem for more general covers, from smooth abundance. We now apply the log-abundance theorem to supply that premise. In covering statements we use the same symbol for an algebraic variety and its analytification.

Corollary 9.1 (Quasi-projective covers). *Let X be a connected normal projective variety over $\mathbb{C}$, with ordinary universal cover U . The following are equivalent:*

- (i) *U is biholomorphic to a quasi-projective variety.*
- (ii) *There are a finite étale Galois cover $X' \rightarrow X$, an abelian variety A , and a morphism $\alpha : X' \rightarrow A$ that is a locally trivial holomorphic fibre bundle with simply connected projective fibre.*
- (iii) *For some integer $m \geq 0$ and some simply connected projective variety F , there is a biholomorphism*

$$U \simeq \mathbb{C}^m \times F.$$

A point is allowed as the abelian variety or the fibre.

More generally, let $V \rightarrow X$ be a connected infinite étale Galois covering of complex spaces, with V biholomorphic to a quasi-projective variety. There are an intermediate finite étale Galois cover $X' \rightarrow X$, a morphism $\alpha : X' \rightarrow A$ to an abelian variety that is a locally trivial holomorphic fibre bundle, and an étale cover $\tilde{A} \rightarrow A$, where $\tilde{A}$ has no positive-dimensional compact analytic subvariety, such that $V \rightarrow X'$ is biholomorphic as a covering to

$$X' \times_A \tilde{A} \longrightarrow X'.$$

No algebraicity of the deck transformations is assumed.

Proof. For every smooth projective complex variety Y with nef K_Y , the pair $(Y, 0)$ satisfies the hypotheses of [Ope26a, Theorem 1.1]. Thus K_Y is semiample. This is exactly the premise of [CHK13, Conjecture 1.2], so [CHK13, Theorem 1.1 and Corollary 1.5] give the two assertions. $\square$

The finite cover X' is described as a bundle over A , which need not be a product bundle. The statement for general covers asserts only the displayed pullback direction.

Corollary 9.2 (Projective affine-space uniformization). *Let X be a connected smooth projective complex n -fold, where $n \geq 0$. If its ordinary universal cover is biholomorphic to $\mathbb{C}^n$, then there is a finite étale Galois cover $A \rightarrow X$ with A an abelian variety.*

Proof. The simply connected projective fibre F of the bundle in Corollary 9.1(ii) lifts to a holomorphic embedding in the universal cover $\mathbb{C}^n$. Every coordinate function is constant on the connected compact complex space F , so F is a point. The bundle morphism $X' \rightarrow A$ is therefore an isomorphism. $\square$

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